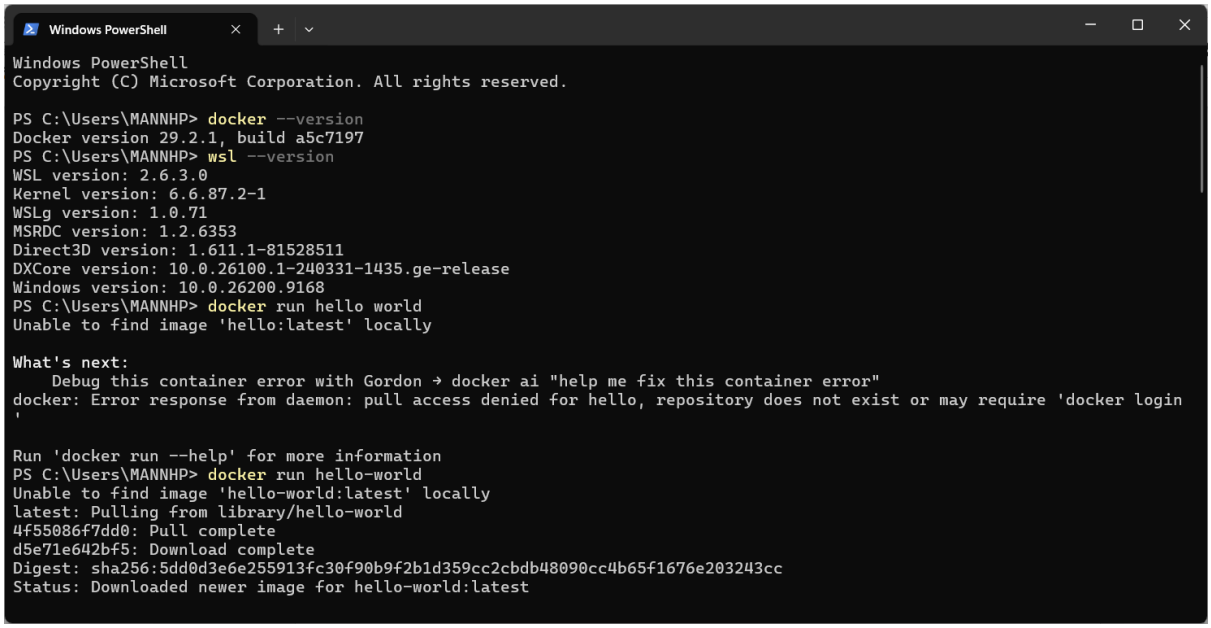
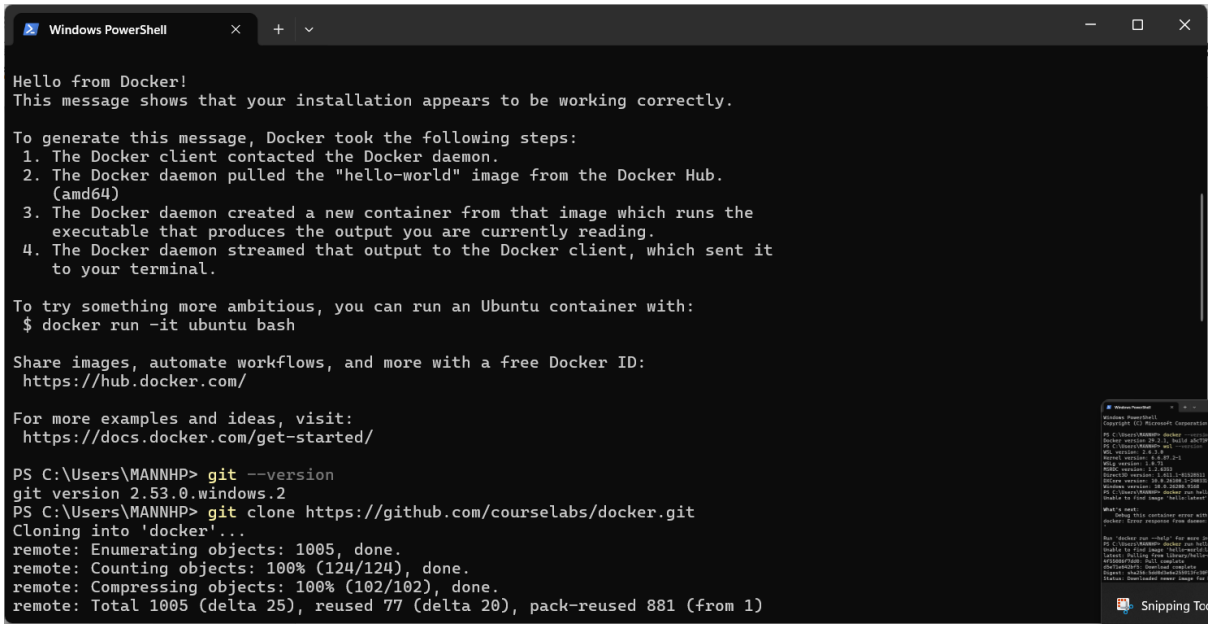
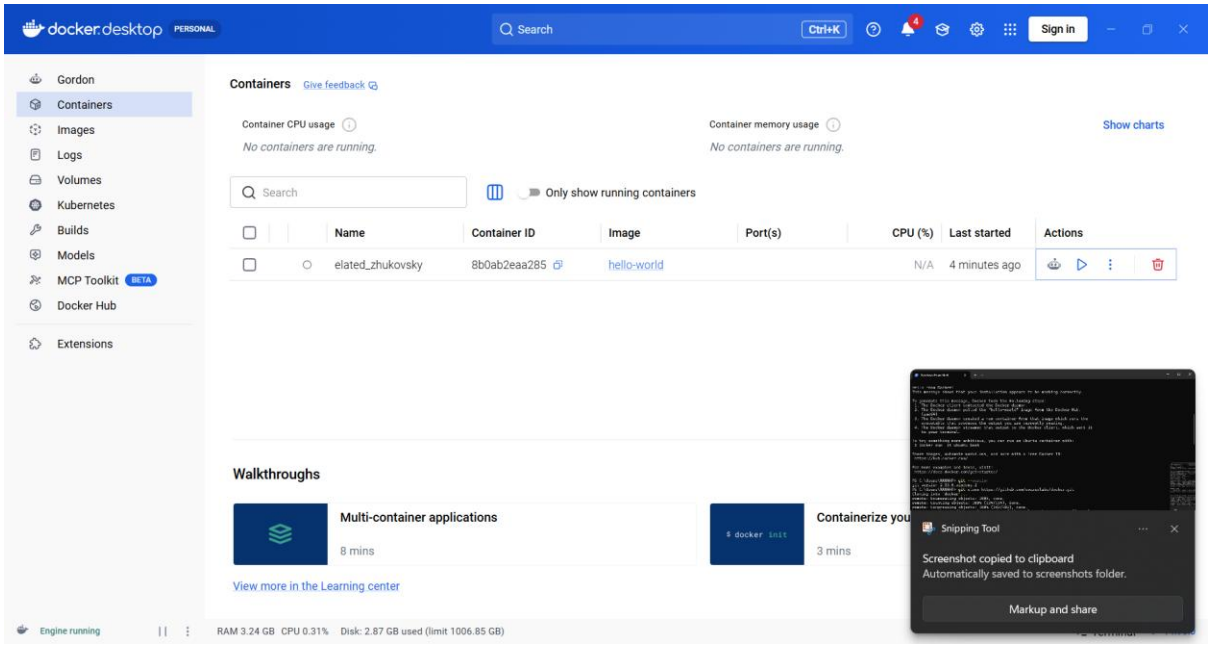
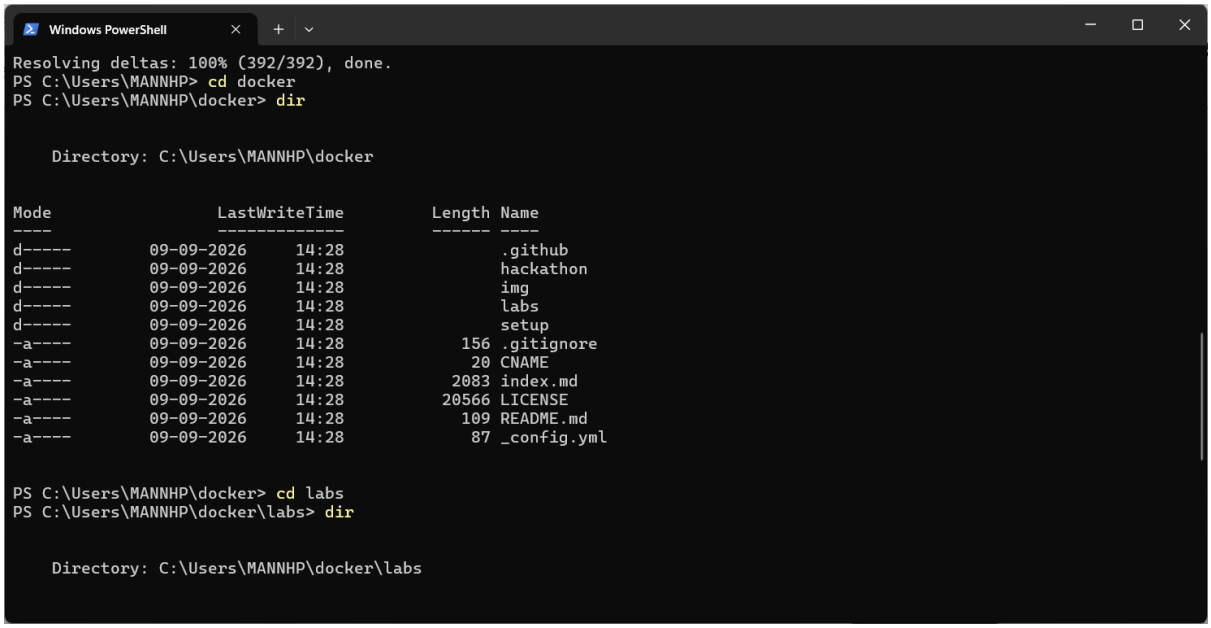
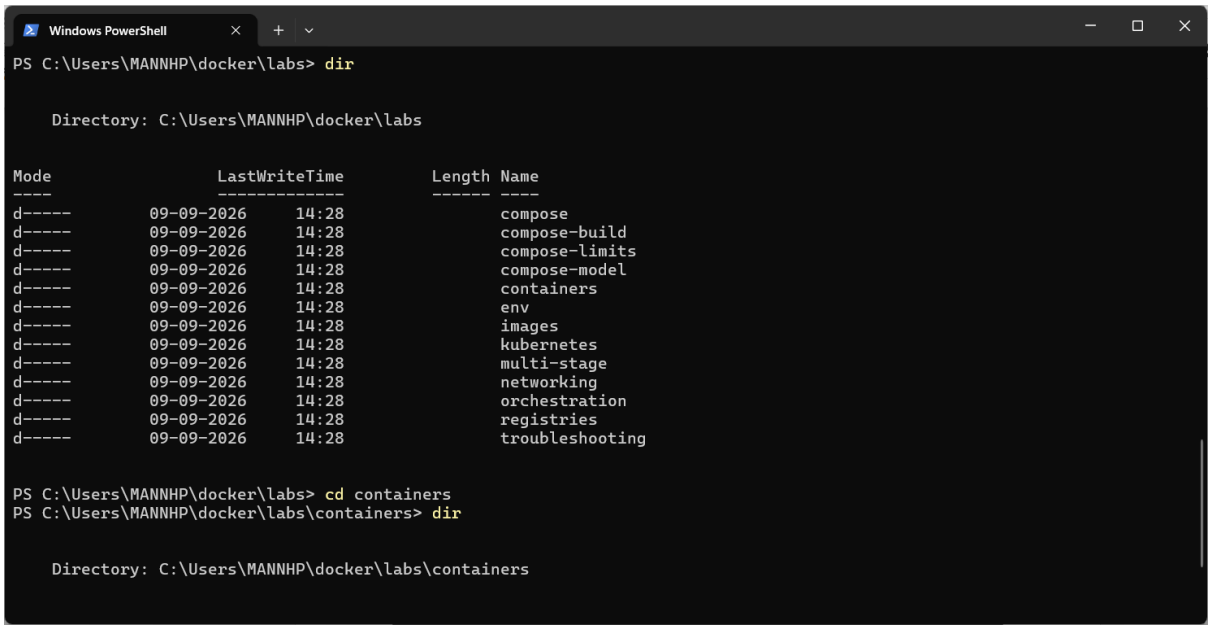
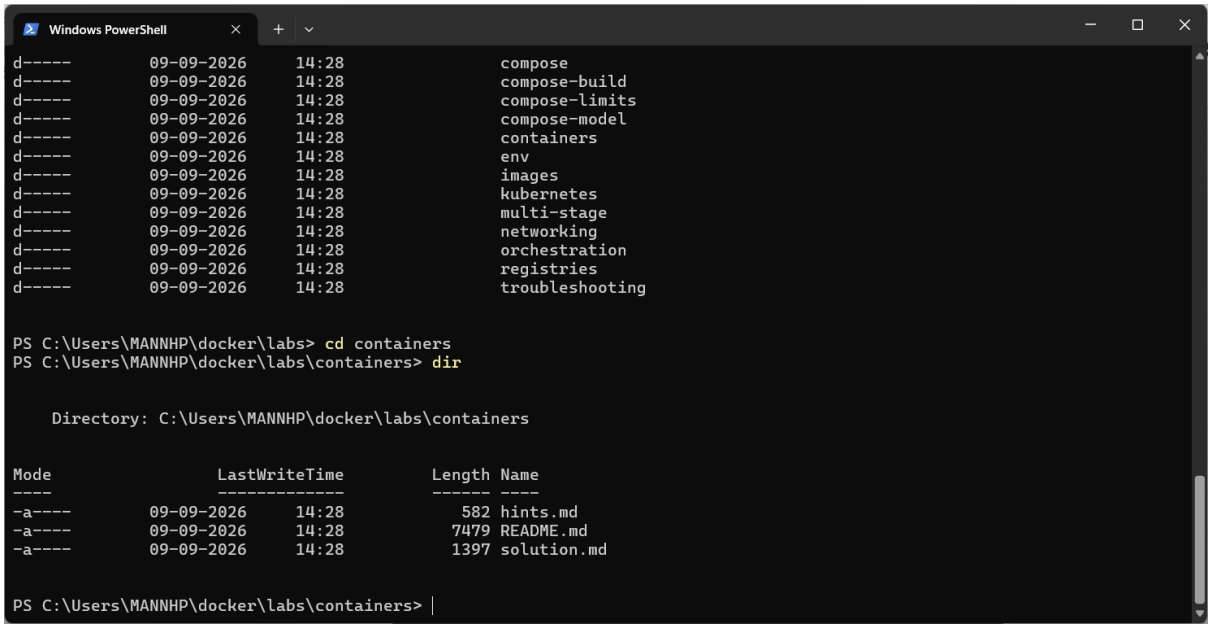

Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\MANNHP> docker --version
Docker version 29.2.1, build a5c7197
PS C:\Users\MANNHP> wsl --version
WSL version: 2.6.3.0
Kernel version: 6.6.87.2-1
WSLg version: 1.0.71
MSRDC version: 1.2.6353
Direct3D version: 1.611.1-81528511
DXCore version: 10.0.26100.1-240331-1435.ge-release
Windows version: 10.0.26200.9168
PS C:\Users\MANNHP> docker run hello world
Unable to find image 'hello:latest' locally
What's next:
Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: pull access denied for hello, repository does not exist or may require 'docker login
'
Run 'docker run --help' for more information
PS C:\Users\MANNHP> docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

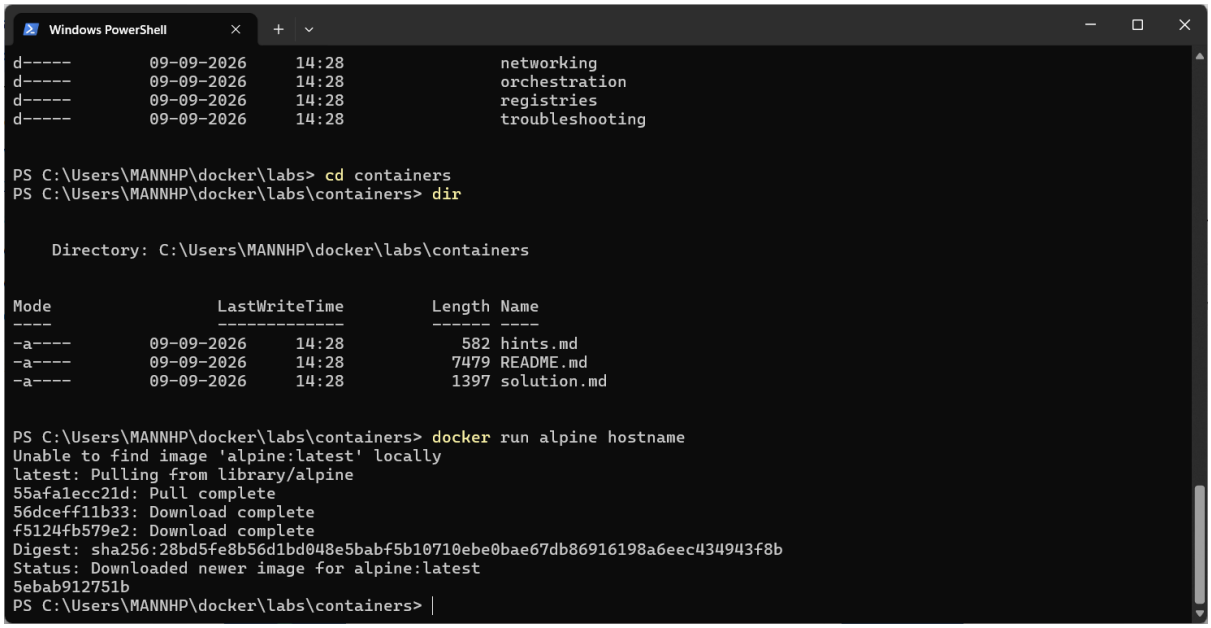

Windows PowerShell
Hello from Docker!
This message shows that your installation appears to be working correctly.
To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the
executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
to your terminal.
To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash
Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/
For more examples and ideas, visit:
https://docs.docker.com/get-started/
PS C:\Users\MANNHP> git --version
git version 2.53.0.windows.2
PS C:\Users\MANNHP> git clone https://github.com/courselabs/docker.git
Cloning into 'docker'...
remote: Enumerating objects: 1005, done.
remote: Counting objects: 100% (124/124), done.
remote: Compressing objects: 100% (102/102), done.
remote: Total 1005 (delta 25), reused 77 (delta 20), pack-reused 881 (from 1)

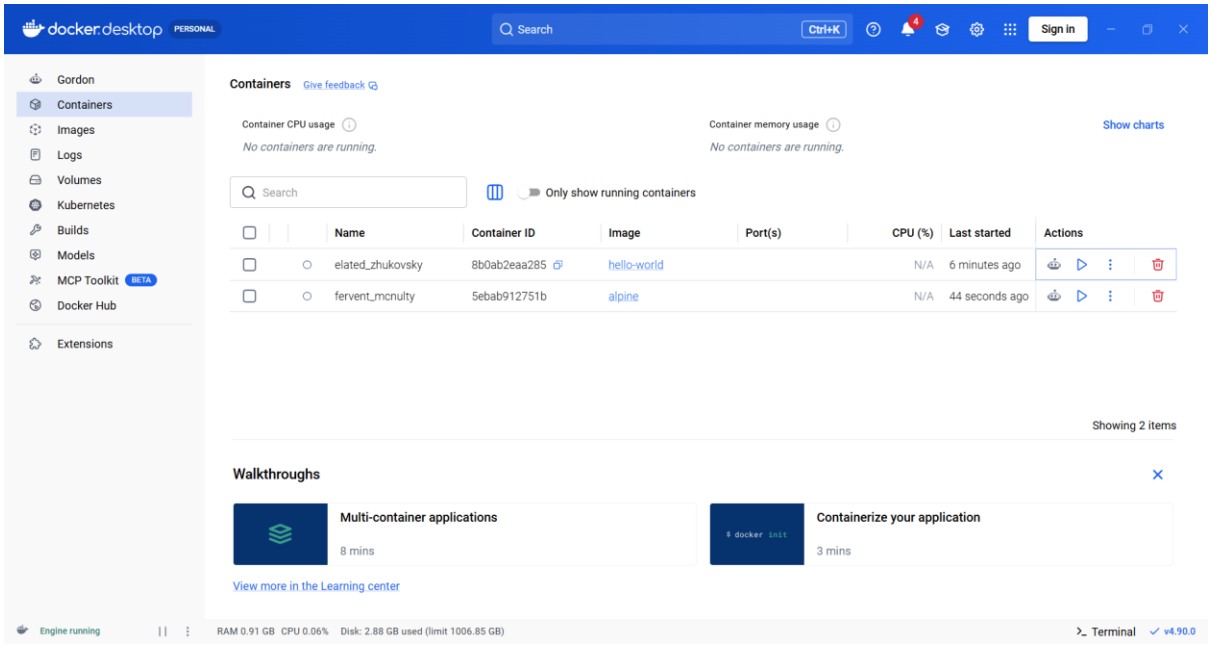

docker desktop
PERSONAL
Search
Ctrl+K
Sign in
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Containers
Give feedback
Container CPU usage
No containers are running.
Container memory usage
No containers are running.
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
elated_zhukovsky
8b0ab2eaa285
hello-world
N/A
4 minutes ago
Walkthroughs
Multi-container applications
8 mins
$ docker init
Containerize you
3 mins
View more in the Learning center
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Markup and share
Engine running
RAM 3.24 GB CPU 0.31% Disk: 2.87 GB used (limit 1006.85 GB)

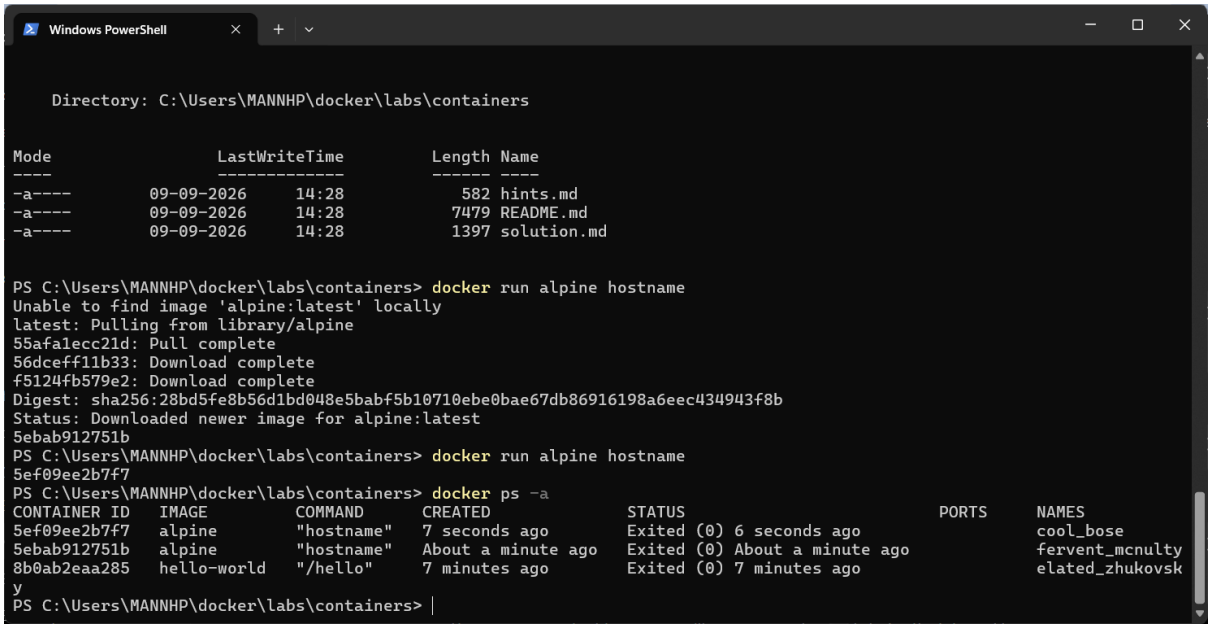

Windows PowerShell
Resolving deltas: 100% (392/392), done.
PS C:\Users\MANNHP> cd docker
PS C:\Users\MANNHP\docker> dir
Directory: C:\Users\MANNHP\docker
Mode LastWriteTime Length Name
d----- 09-09-2026 14:28 .github
d----- 09-09-2026 14:28 hackathon
d----- 09-09-2026 14:28 img
d----- 09-09-2026 14:28 labs
d----- 09-09-2026 14:28 setup
-a---- 09-09-2026 14:28 156 .gitignore
-a---- 09-09-2026 14:28 20 CNAME
-a---- 09-09-2026 14:28 2083 index.md
-a---- 09-09-2026 14:28 20566 LICENSE
-a---- 09-09-2026 14:28 109 README.md
-a---- 09-09-2026 14:28 87 _config.yml
PS C:\Users\MANNHP\docker> cd labs
PS C:\Users\MANNHP\docker\labs> dir
Directory: C:\Users\MANNHP\docker\labs

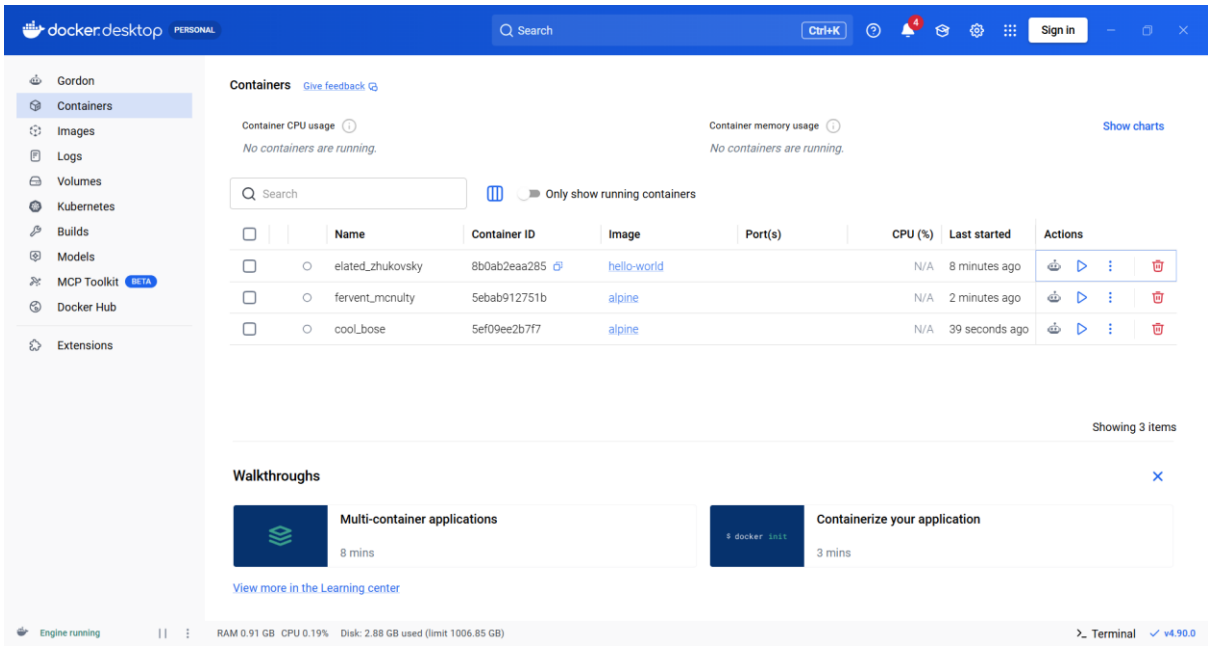

Windows PowerShell
PS C:\Users\MANNHP\docker\labs> dir
Directory: C:\Users\MANNHP\docker\labs
Mode LastWriteTime Length Name
d----- 09-09-2026 14:28 compose
d----- 09-09-2026 14:28 compose-build
d----- 09-09-2026 14:28 compose-limits
d----- 09-09-2026 14:28 compose-model
d----- 09-09-2026 14:28 containers
d----- 09-09-2026 14:28 env
d----- 09-09-2026 14:28 images
d----- 09-09-2026 14:28 kubernetes
d----- 09-09-2026 14:28 multi-stage
d----- 09-09-2026 14:28 networking
d----- 09-09-2026 14:28 orchestration
d----- 09-09-2026 14:28 registries
d----- 09-09-2026 14:28 troubleshooting
PS C:\Users\MANNHP\docker\labs> cd containers
PS C:\Users\MANNHP\docker\labs\containers> dir
Directory: C:\Users\MANNHP\docker\labs\containers

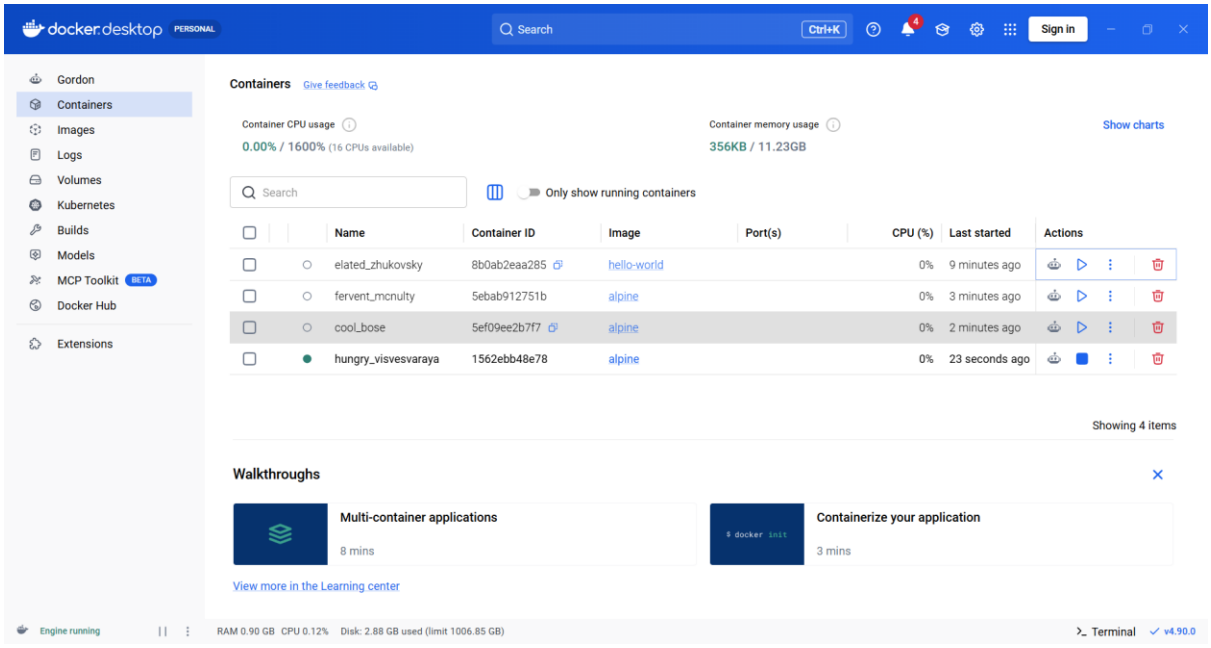

Windows PowerShell
d----- 09-09-2026 14:28 compose
d----- 09-09-2026 14:28 compose-build
d----- 09-09-2026 14:28 compose-limits
d----- 09-09-2026 14:28 compose-model
d----- 09-09-2026 14:28 containers
d----- 09-09-2026 14:28 env
d----- 09-09-2026 14:28 images
d----- 09-09-2026 14:28 kubernetes
d----- 09-09-2026 14:28 multi-stage
d----- 09-09-2026 14:28 networking
d----- 09-09-2026 14:28 orchestration
d----- 09-09-2026 14:28 registries
d----- 09-09-2026 14:28 troubleshooting
PS C:\Users\MANNHP\docker\labs> cd containers
PS C:\Users\MANNHP\docker\labs\containers> dir
Directory: C:\Users\MANNHP\docker\labs\containers
Mode LastWriteTime Length Name
-a---- 09-09-2026 14:28 582 hints.md
-a---- 09-09-2026 14:28 7479 README.md
-a---- 09-09-2026 14:28 1397 solution.md
PS C:\Users\MANNHP\docker\labs\containers>

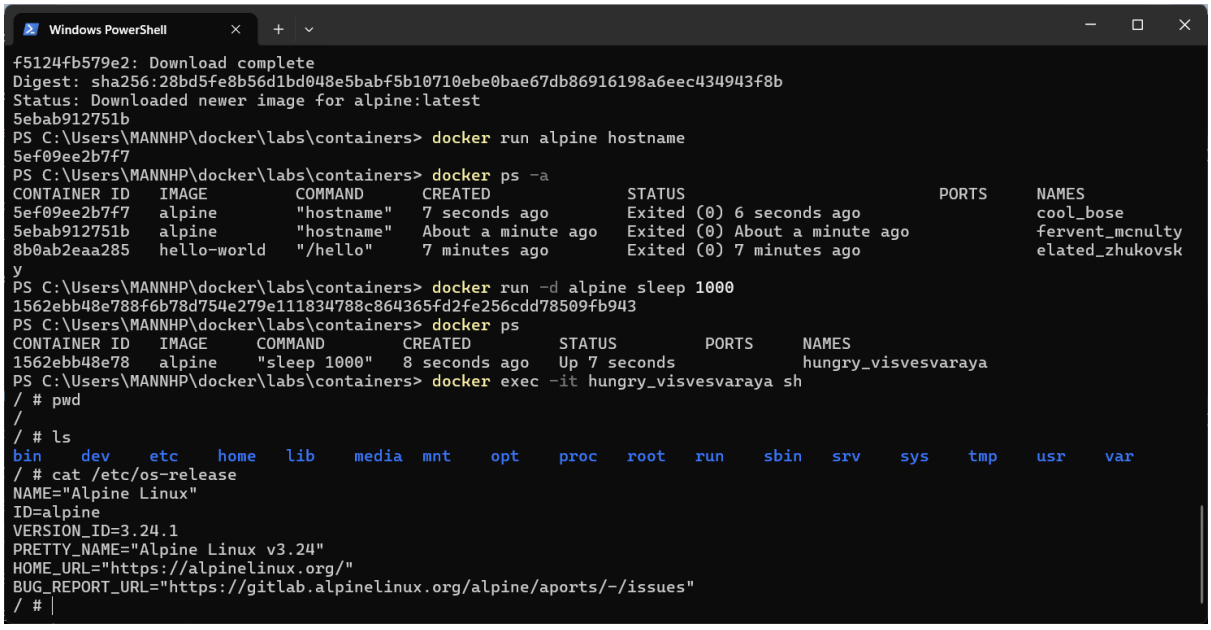

Windows PowerShell
d----- 09-09-2026 14:28 networking
d----- 09-09-2026 14:28 orchestration
d----- 09-09-2026 14:28 registries
d----- 09-09-2026 14:28 troubleshooting
PS C:\Users\MANNHP\docker\labs> cd containers
PS C:\Users\MANNHP\docker\labs\containers> dir
Directory: C:\Users\MANNHP\docker\labs\containers
Mode LastWriteTime Length Name
-a---- 09-09-2026 14:28 582 hints.md
-a---- 09-09-2026 14:28 7479 README.md
-a---- 09-09-2026 14:28 1397 solution.md
PS C:\Users\MANNHP\docker\labs\containers> docker run alpine hostname
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
55afa1ecc21d: Pull complete
56dceff11b33: Download complete
f5124fb579e2: Download complete
Digest: sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
Status: Downloaded newer image for alpine:latest
5ebab912751b
PS C:\Users\MANNHP\docker\labs\containers>


docker desktop
PERSONAL
Search
Ctrl+K
Sign in
Gordon
Containers
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Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Containers
Give feedback
Container CPU usage
No containers are running.
Container memory usage
No containers are running.
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
elated_zhukovsky
8b0ab2eaa285
hello-world
N/A
6 minutes ago
fervent_mcnulty
5ebab912751b
alpine
N/A
44 seconds ago
Showing 2 items
Walkthroughs
Multi-container applications
8 mins
$ docker init
Containerize your application
3 mins
View more in the Learning center
Engine running
RAM 0.91 GB CPU 0.06% Disk: 2.88 GB used (limit 1006.85 GB)
Terminal
v4.90.0


Windows PowerShell
Directory: C:\Users\MANNHP\docker\labs\containers
Mode LastWriteTime Length Name
-a---- 09-09-2026 14:28 582 hints.md
-a---- 09-09-2026 14:28 7479 README.md
-a---- 09-09-2026 14:28 1397 solution.md
PS C:\Users\MANNHP\docker\labs\containers> docker run alpine hostname
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
55afa1ecc21d: Pull complete
56dceff11b33: Download complete
f5124fb579e2: Download complete
Digest: sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
Status: Downloaded newer image for alpine:latest
5ebab912751b
PS C:\Users\MANNHP\docker\labs\containers> docker run alpine hostname
5ef09ee2b7f7
PS C:\Users\MANNHP\docker\labs\containers> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
5ef09ee2b7f7 alpine "hostname" 7 seconds ago Exited (0) 6 seconds ago cool_bose
5ebab912751b alpine "hostname" About a minute ago Exited (0) About a minute ago fervent_mcnulty
8b0ab2eaa285 hello-world "/hello" 7 minutes ago Exited (0) 7 minutes ago elated_zhukovsky
PS C:\Users\MANNHP\docker\labs\containers>

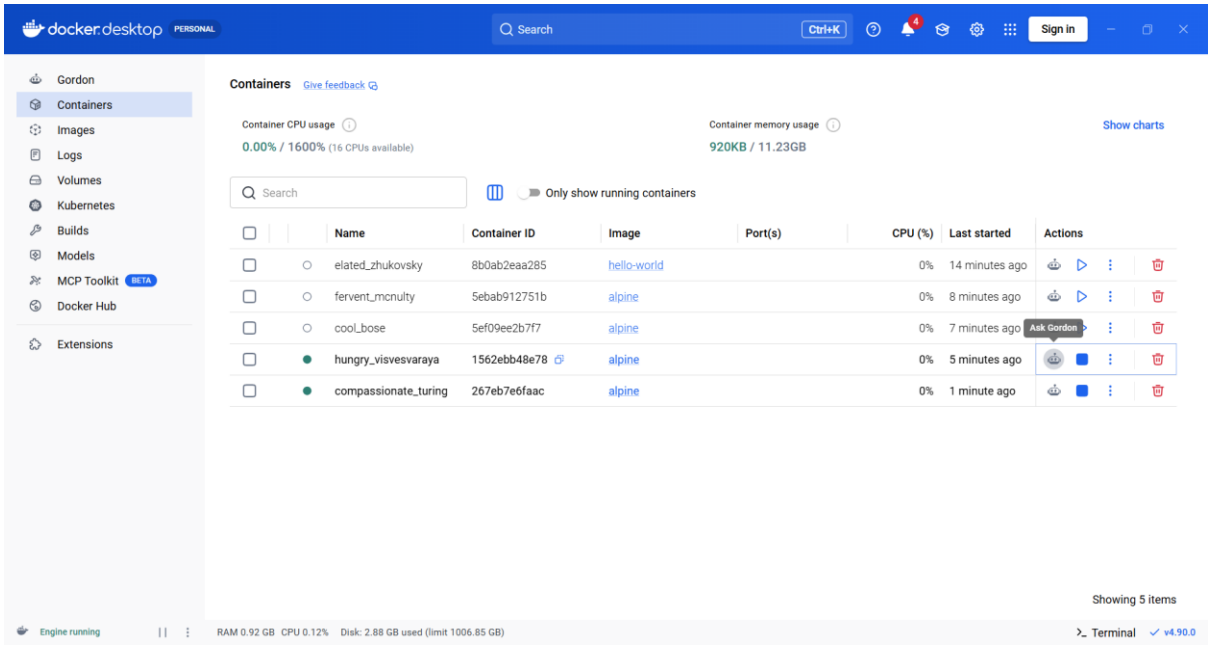

docker desktop
PERSONAL
Search
Ctrl+K
Sign in
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Containers
Give feedback
Container CPU usage
No containers are running.
Container memory usage
No containers are running.
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
elated_zhukovsky
8b0ab2eaa285
hello-world
N/A
8 minutes ago
fervent_mcnulty
5ebab912751b
alpine
N/A
2 minutes ago
cool_bose
5ef09ee2b7f7
alpine
N/A
39 seconds ago
Showing 3 items
Walkthroughs
Multi-container applications
8 mins
Containerize your application
3 mins
View more in the Learning center
Engine running
RAM 0.91 GB CPU 0.19% Disk: 2.88 GB used (limit 1006.85 GB)
Terminal
v4.90.0

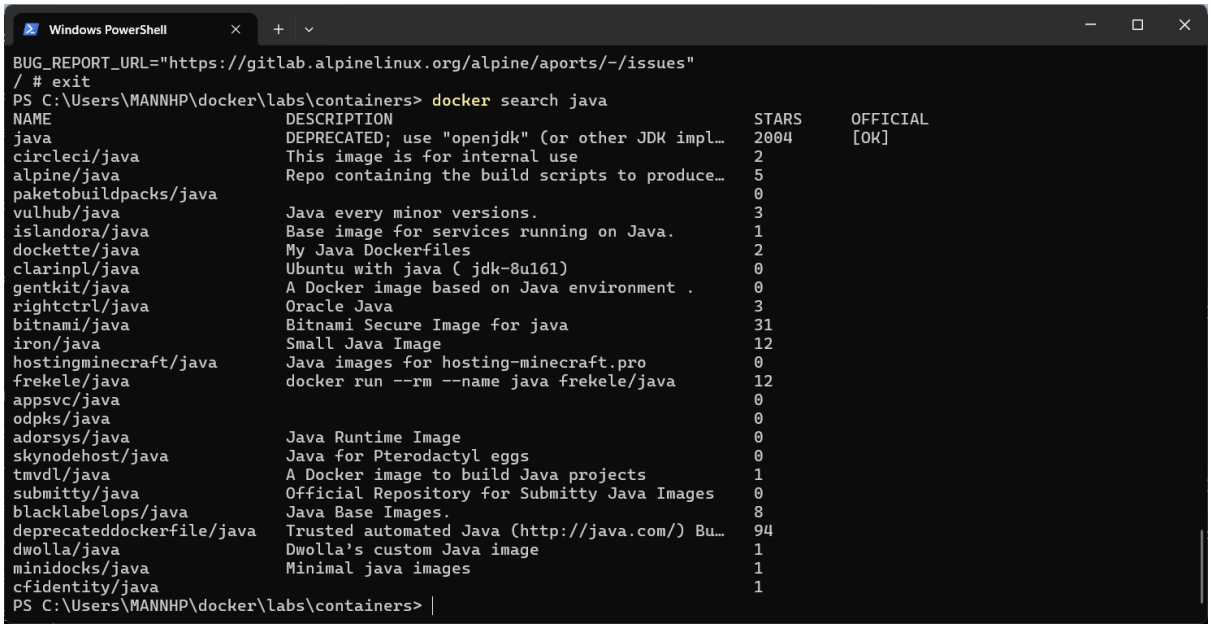

docker desktop
PERSONAL
Search
Ctrl+K
Sign in
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Containers
Give feedback
Container CPU usage
0.00% / 1600% (16 CPUs available)
Container memory usage
356KB / 11.23GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
elated_zhukovsky
8b0ab2eaa285
hello-world
0%
9 minutes ago
fervent_mcnulty
5ebab912751b
alpine
0%
3 minutes ago
cool_bose
5ef09ee2b7f7
alpine
0%
2 minutes ago
hungry_visvesvaraya
1562ebb48e78
alpine
0%
23 seconds ago
Showing 4 items
Walkthroughs
Multi-container applications
8 mins
Containerize your application
3 mins
View more in the Learning center
Engine running
RAM 0.90 GB CPU 0.12% Disk: 2.88 GB used (limit 1006.85 GB)
Terminal
v4.90.0

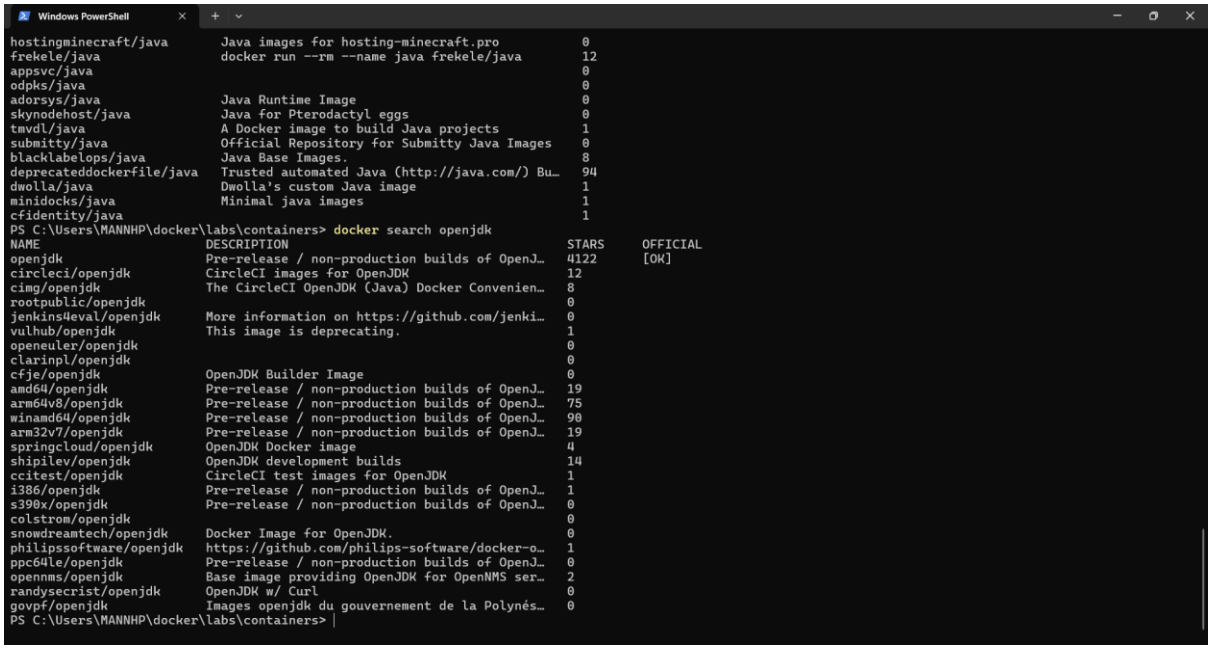

Windows PowerShell
f5124fb579e2: Download complete
Digest: sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
Status: Downloaded newer image for alpine:latest
5ebab912751b
PS C:\Users\MANNHP\docker\labs\containers> docker run alpine hostname
5ef09ee2b7f7
PS C:\Users\MANNHP\docker\labs\containers> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
5ef09ee2b7f7 alpine "hostname" 7 seconds ago Exited (0) 6 seconds ago cool_bose
5ebab912751b alpine "hostname" About a minute ago Exited (0) About a minute ago fervent_mcnulty
8b0ab2eaa285 hello-world "/hello" 7 minutes ago Exited (0) 7 minutes ago elated_zhukovsky
PS C:\Users\MANNHP\docker\labs\containers> docker run -d alpine sleep 1000
1562ebb48e788f6b78d754e279e111834788c864365fd2fe256cdd78509fb943
PS C:\Users\MANNHP\docker\labs\containers> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
1562ebb48e78 alpine "sleep 1000" 8 seconds ago Up 7 seconds hungry_visvesvaraya
PS C:\Users\MANNHP\docker\labs\containers> docker exec -it hungry_visvesvaraya sh
/ # pwd
/
/ # ls
bin dev etc home lib media mnt opt proc root run sbin srv sys tmp usr var
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.24.1
PRETTY_NAME="Alpine Linux v3.24"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ #

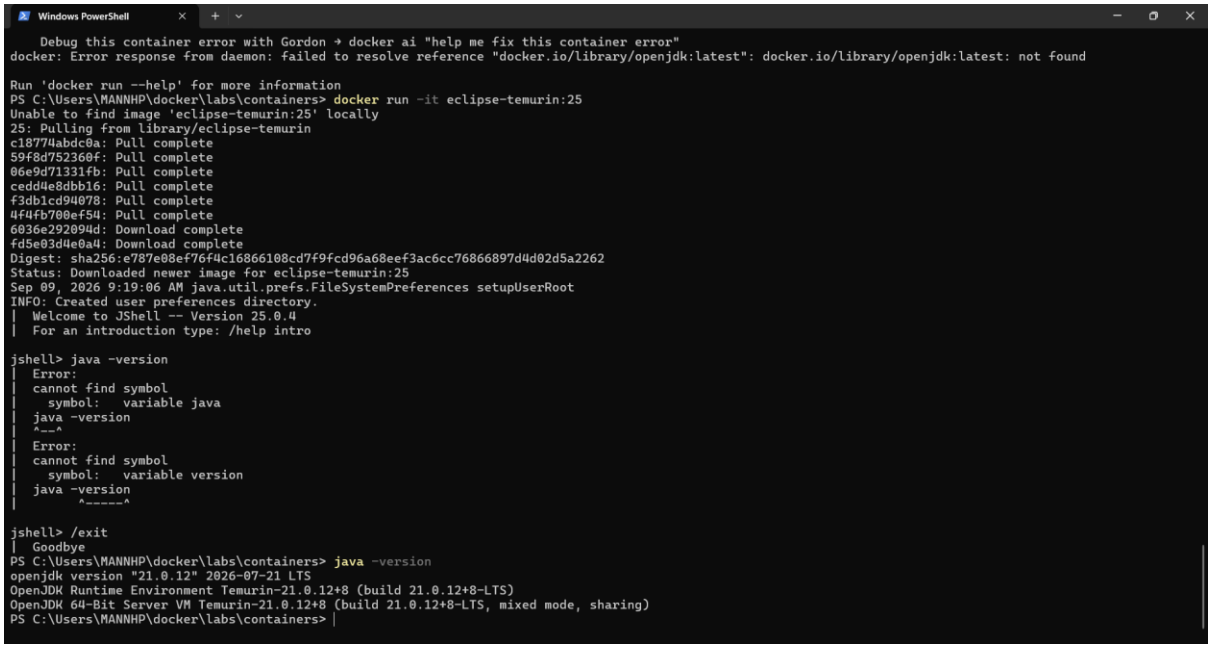
```
Windows PowerShell
/ # ls
bin    dev    etc    home   lib    media  mnt    opt    proc   root   run    sbin   srv    sys    tmp    usr    var
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.24.1
PRETTY_NAME="Alpine Linux v3.24"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit

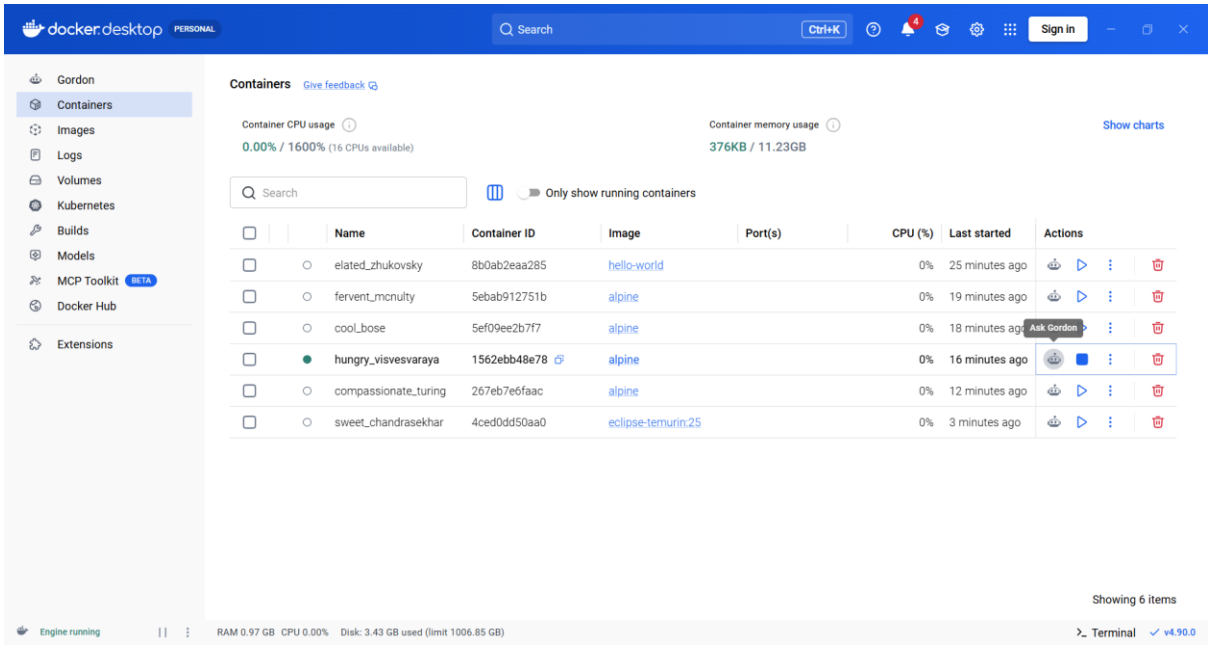
What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug hungry_visvesvara
ya
    Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\MANNHP\docker\labs\containers> docker run -it alpine
/ # ls /
bin    dev    etc    home   lib    media  mnt    opt    proc   root   run    sbin   srv    sys    tmp    usr    var
/ # whoami
root
/ # hostname
267eb7e6faac
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.24.1
PRETTY_NAME="Alpine Linux v3.24"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ #
```

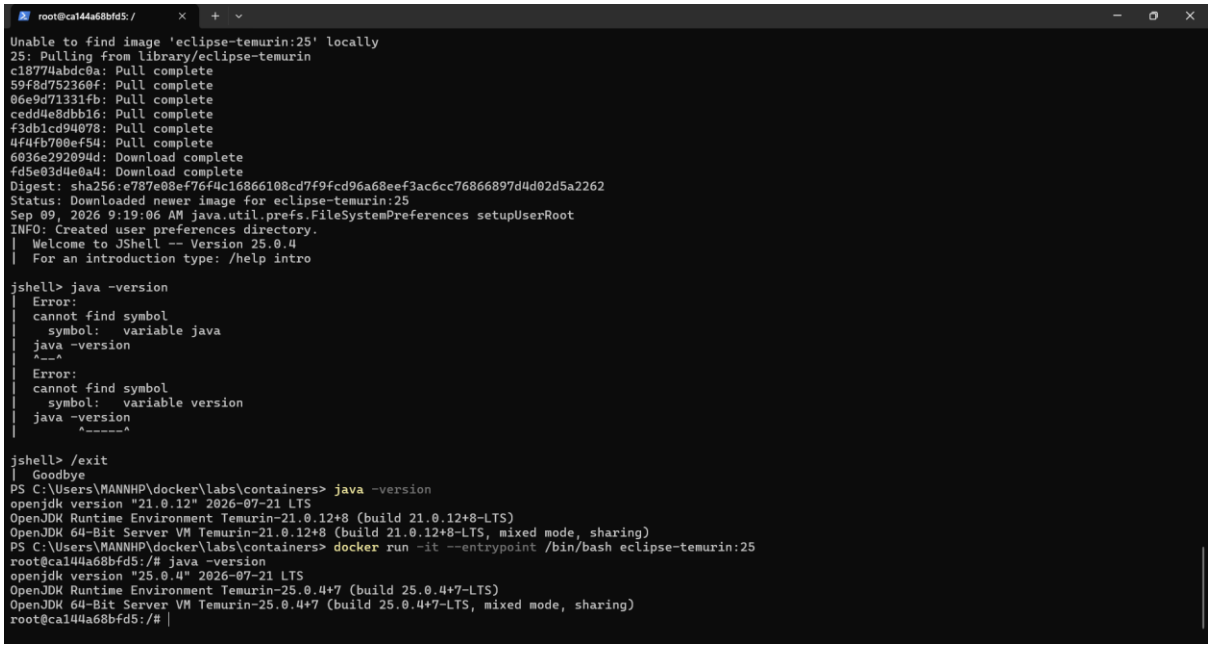


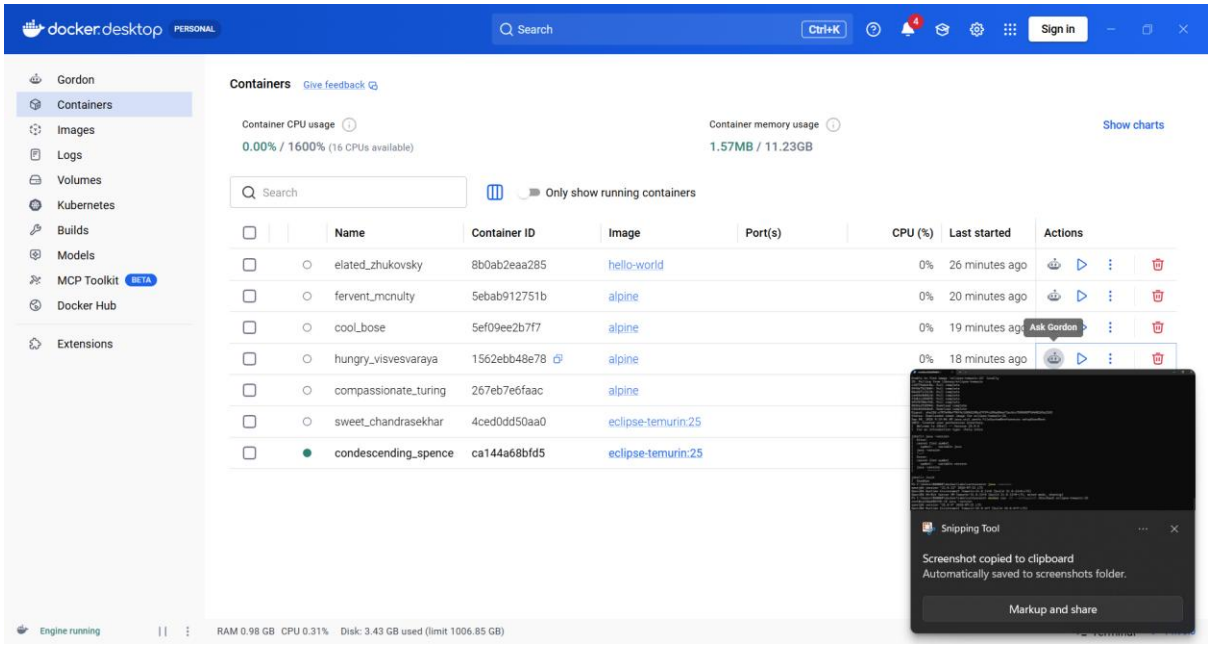

Windows PowerShell
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit
PS C:\Users\MANNHP\docker\labs\containers> docker search java
NAME DESCRIPTION STARS OFFICIAL
java DEPRECATED; use "openjdk" (or other JDK impl… 2004 [OK]
circleci/java This image is for internal use 2
alpine/java Repo containing the build scripts to produce… 5
paketobuildpacks/java 0
vulhub/java Java every minor versions. 3
islandora/java Base image for services running on Java. 1
dockette/java My Java Dockerfiles 2
clarinpl/java Ubuntu with java (jdk-8u161) 0
gentkit/java A Docker image based on Java environment . 0
rightctrl/java Oracle Java 3
bitnami/java Bitnami Secure Image for java 31
iron/java Small Java Image 12
hostingminecraft/java Java images for hosting-minecraft.pro 0
frekele/java docker run --rm --name java frekele/java 12
appsvc/java 0
odpks/java 0
adorsys/java Java Runtime Image 0
skynodehost/java Java for Pterodactyl eggs 0
tmvdl/java A Docker image to build Java projects 1
submitty/java Official Repository for Submitty Java Images 0
blacklabelops/java Java Base Images. 8
deprecateddockerfile/java Trusted automated Java (http://java.com/) Bu… 94
dwolla/java Dwolla's custom Java image 1
minidocks/java Minimal java images 1
cfidentity/java 1
PS C:\Users\MANNHP\docker\labs\containers>


Windows PowerShell
hostingminecraft/java Java images for hosting-minecraft.pro 0
frekele/java docker run --rm --name java frekele/java 12
appsvc/java 0
odpks/java 0
adorsys/java Java Runtime Image 0
skynodehost/java Java for Pterodactyl eggs 0
tmvdl/java A Docker image to build Java projects 1
submitty/java Official Repository for Submitty Java Images 0
blacklabelops/java Java Base Images. 8
deprecateddockerfile/java Trusted automated Java (http://java.com/) Bu… 94
dwolla/java Dwolla's custom Java image 1
minidocks/java Minimal java images 1
cfidentity/java 1
PS C:\Users\MANNHP\docker\labs\containers> docker search openjdk
NAME DESCRIPTION STARS OFFICIAL
openjdk Pre-release / non-production builds of OpenJ… 4122 [OK]
circleci/openjdk CircleCI images for OpenJDK 12
cimg/openjdk The CircleCI OpenJDK (Java) Docker Convenien… 8
rootpublic/openjdk 0
jenkins4eval/openjdk More information on https://github.com/jenki… 0
vulhub/openjdk This image is deprecating. 1
openeuler/openjdk 0
clarinpl/openjdk 0
cfje/openjdk OpenJDK Builder Image 0
amd64/openjdk Pre-release / non-production builds of OpenJ… 19
arm64v8/openjdk Pre-release / non-production builds of OpenJ… 75
winamd64/openjdk Pre-release / non-production builds of OpenJ… 90
arm32v7/openjdk Pre-release / non-production builds of OpenJ… 19
springcloud/openjdk OpenJDK Docker image 4
shipilev/openjdk OpenJDK development builds 14
ccitest/openjdk CircleCI test images for OpenJDK 1
i386/openjdk Pre-release / non-production builds of OpenJ… 1
s390x/openjdk Pre-release / non-production builds of OpenJ… 0
colstrom/openjdk 0
snowdreamtech/openjdk Docker Image for OpenJDK. 0
philipssoftware/openjdk https://github.com/philips-software/docker-o… 1
ppc64le/openjdk Pre-release / non-production builds of OpenJ… 0
opennms/openjdk Base image providing OpenJDK for OpenNMS ser… 2
randysecrist/openjdk OpenJDK w/ Curl 0
govpf/openjdk Images openjdk du gouvernement de la Polynés… 0
PS C:\Users\MANNHP\docker\labs\containers>


Windows PowerShell
Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: failed to resolve reference "docker.io/library/openjdk:latest": docker.io/library/openjdk:latest: not found
Run 'docker run --help' for more information
PS C:\Users\MANNHP\docker\labs\containers> docker run -it eclipse-temurin:25
Unable to find image 'eclipse-temurin:25' locally
25: Pulling from library/eclipse-temurin
c18774abdc0a: Pull complete
59f8d752360f: Pull complete
06e9d71331fb: Pull complete
cedd4e8dbb16: Pull complete
f3db1cd94078: Pull complete
4f4fb700ef54: Pull complete
6036e292094d: Download complete
fd5e03d4e0a4: Download complete
Digest: sha256:e787e08ef76f4c16866108cd7f9fcd96a68eef3ac6cc76866897d4d02d5a2262
Status: Downloaded newer image for eclipse-temurin:25
Sep 09, 2026 9:19:06 AM java.util.prefs.FileSystemPreferences setupUserRoot
INFO: Created user preferences directory.
| Welcome to JShell -- Version 25.0.4
| For an introduction type: /help intro
jshell> java -version
| Error:
| cannot find symbol
| symbol: variable java
| java -version
| ^--^
| Error:
| cannot find symbol
| symbol: variable version
| java -version
| ^-----^
jshell> /exit
| Goodbye
PS C:\Users\MANNHP\docker\labs\containers> java -version
openjdk version "21.0.12" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-21.0.12+8 (build 21.0.12+8-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.12+8 (build 21.0.12+8-LTS, mixed mode, sharing)
PS C:\Users\MANNHP\docker\labs\containers>


docker desktop
PERSONAL
Search
Ctrl+K
Sign in
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Containers
Give feedback
Container CPU usage
0.00% / 1600% (16 CPUs available)
Container memory usage
376KB / 11.23GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
elated_zhukovsky
8b0ab2eaa285
hello-world
0%
25 minutes ago
fervent_mcnulty
5ebab912751b
alpine
0%
19 minutes ago
cool_bose
5ef09ee2b7f7
alpine
0%
18 minutes ago
Ask Gordon
hungry_visvesvaraya
1562ebb48e78
alpine
0%
16 minutes ago
compassionate_turing
267eb7e6faac
alpine
0%
12 minutes ago
sweet_chandrasekhar
4ced0dd50aa0
eclipse-temurin:25
0%
3 minutes ago
Showing 6 items
Engine running
RAM 0.97 GB CPU 0.00% Disk: 3.43 GB used (limit 1006.85 GB)
Terminal
v4.90.0

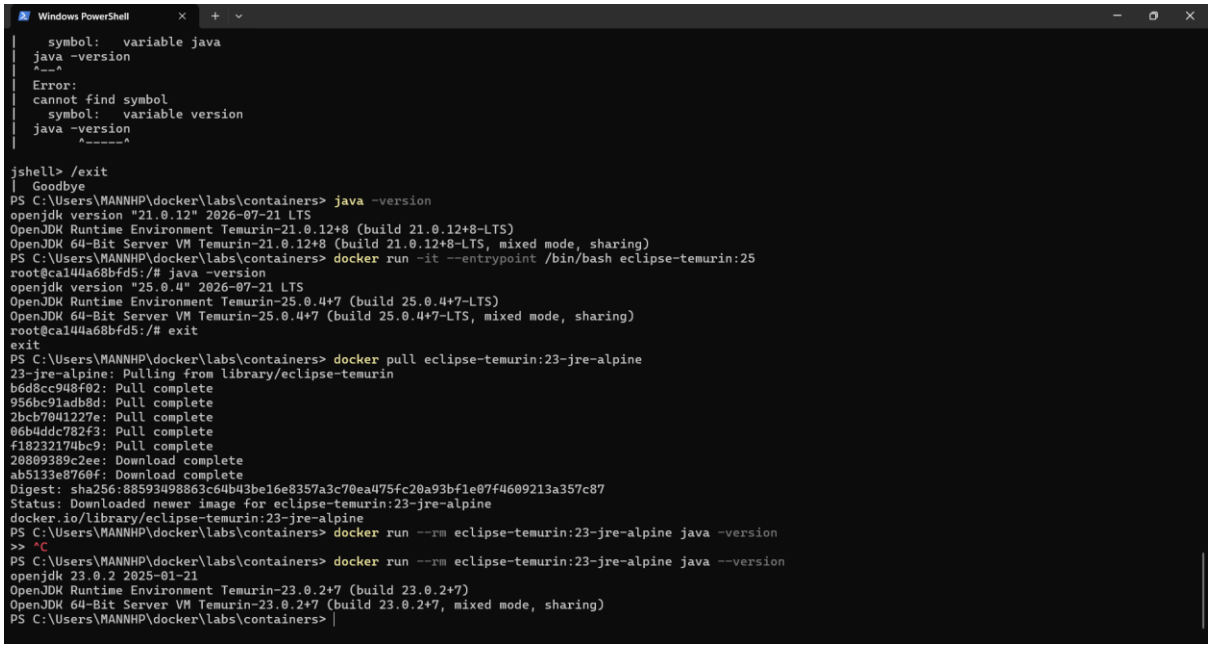

root@ca144a68bfd5: /
Unable to find image 'eclipse-temurin:25' locally
25: Pulling from library/eclipse-temurin
c18774abdc0a: Pull complete
59f8d752360f: Pull complete
06e9d71331fb: Pull complete
cedd4e8dbb16: Pull complete
f3db1cd94078: Pull complete
4f4fb700ef54: Pull complete
6036e292094d: Download complete
fd5e03d4e0a4: Download complete
Digest: sha256:e787e08ef76f4c16866108cd7f9fcd96a68eef3ac6cc76866897d4d02d5a2262
Status: Downloaded newer image for eclipse-temurin:25
Sep 09, 2026 9:19:06 AM java.util.prefs.FileSystemPreferences setupUserRoot
INFO: Created user preferences directory.
| Welcome to JShell -- Version 25.0.4
| For an introduction type: /help intro
jshell> java -version
| Error:
| cannot find symbol
| symbol: variable java
| java -version
| ^--^
| Error:
| cannot find symbol
| symbol: variable version
| java -version
| ^-----^
jshell> /exit
| Goodbye
PS C:\Users\MANNHP\docker\labs\containers> java -version
openjdk version "21.0.12" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-21.0.12+8 (build 21.0.12+8-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.12+8 (build 21.0.12+8-LTS, mixed mode, sharing)
PS C:\Users\MANNHP\docker\labs\containers> docker run -it --entrypoint /bin/bash eclipse-temurin:25
root@ca144a68bfd5:/# java -version
openjdk version "25.0.4" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-25.0.4+7 (build 25.0.4+7-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4+7 (build 25.0.4+7-LTS, mixed mode, sharing)
root@ca144a68bfd5:/#

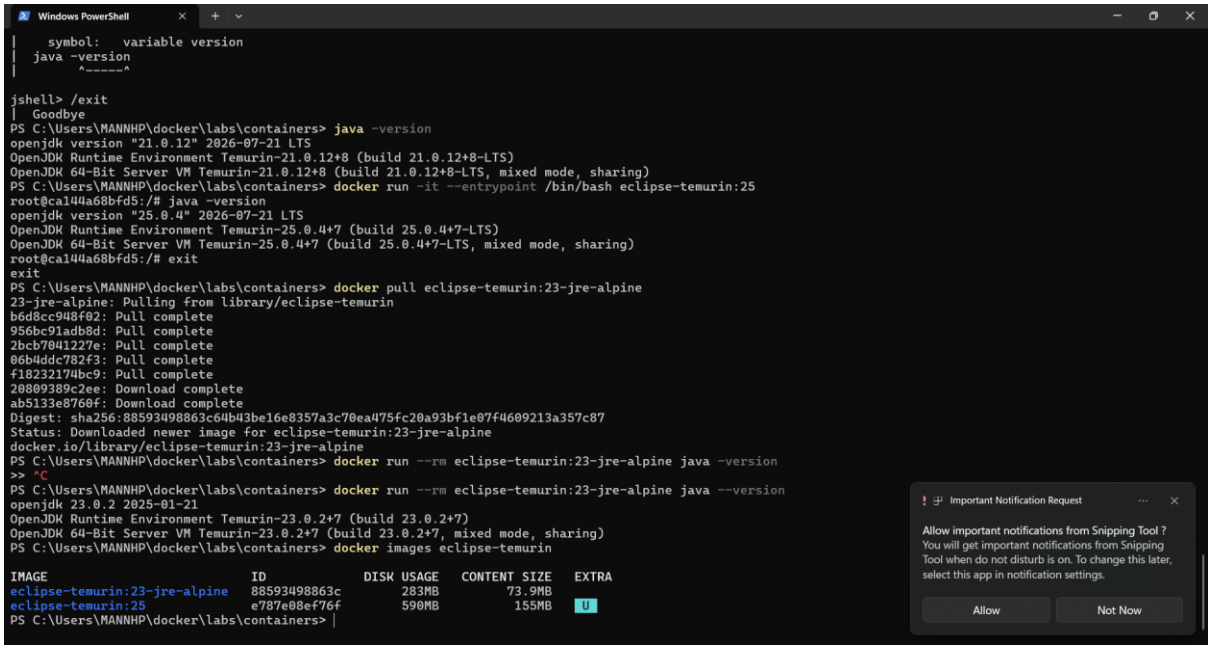


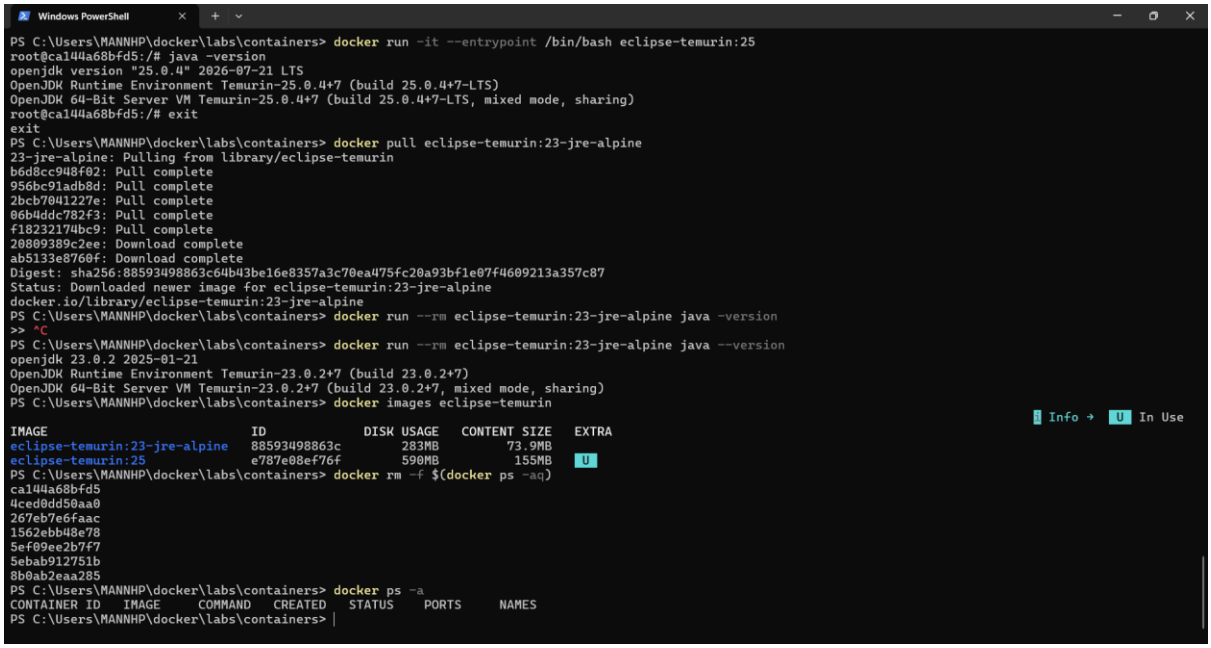

```
Windows PowerShell
|  Welcome to JShell -- Version 25.0.4
|  For an introduction type: /help intro

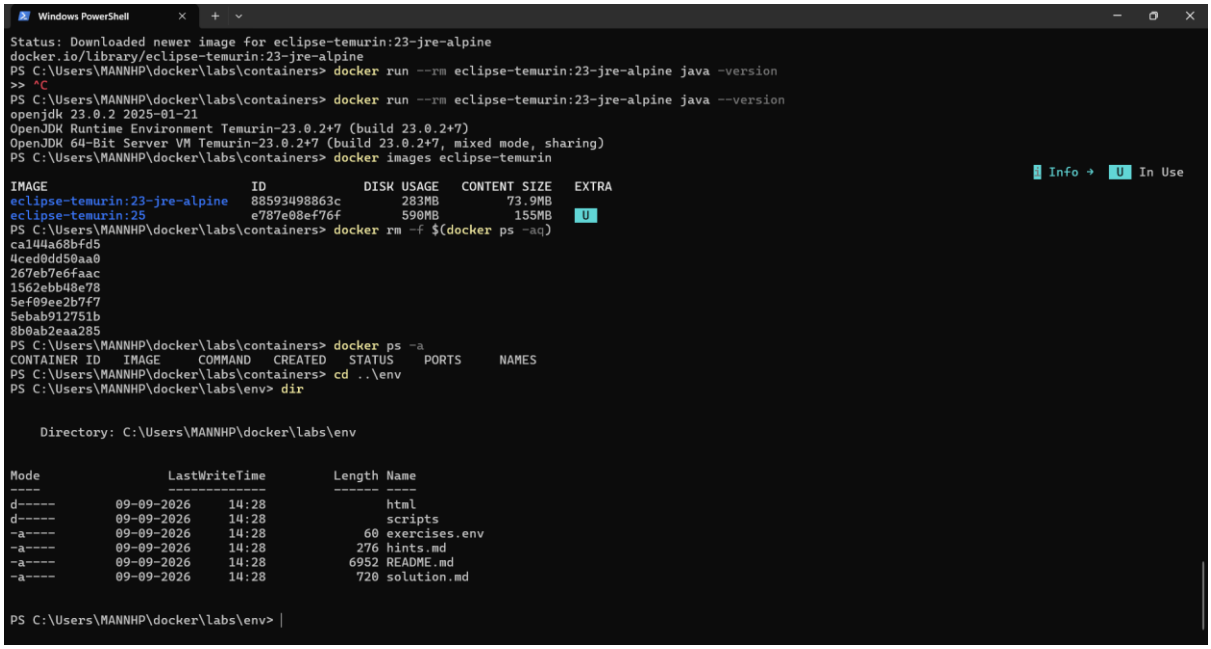
jshell> java -version
|  Error:
|  cannot find symbol
|    symbol:   variable java
|  java -version
|  ^--^
|  Error:
|  cannot find symbol
|    symbol:   variable version
|  java -version
|       ^-----^

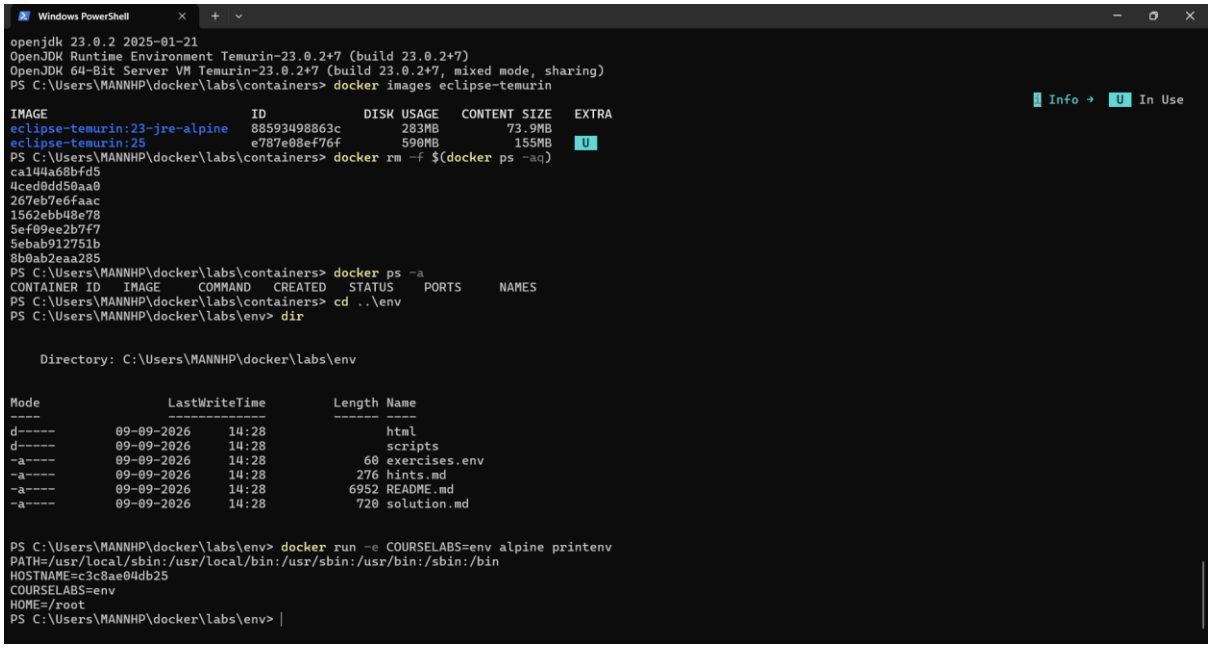
jshell> /exit
|  Goodbye
PS C:\Users\MANNHP\docker\labs\containers> java -version
openjdk version "21.0.12" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-21.0.12+8 (build 21.0.12+8-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.12+8 (build 21.0.12+8-LTS, mixed mode, sharing)
PS C:\Users\MANNHP\docker\labs\containers> docker run -it --entrypoint /bin/bash eclipse-temurin:25
root@ca144a68bfd5:/# java -version
openjdk version "25.0.4" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-25.0.4+7 (build 25.0.4+7-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4+7 (build 25.0.4+7-LTS, mixed mode, sharing)
root@ca144a68bfd5:/# exit
exit
PS C:\Users\MANNHP\docker\labs\containers> docker pull eclipse-temurin:23-jre-alpine
23-jre-alpine: Pulling from library/eclipse-temurin
b6d8cc948f02: Pull complete
956bc91adb8d: Pull complete
2bcb7041227e: Pull complete
06b4ddc782f3: Pull complete
f18232174bc9: Pull complete
20809389c2ee: Download complete
ab5133e8760f: Download complete
Digest: sha256:88593498863c64b43be16e8357a3c70ea475fc20a93bf1e07f4609213a357c87
Status: Downloaded newer image for eclipse-temurin:23-jre-alpine
docker.io/library/eclipse-temurin:23-jre-alpine
PS C:\Users\MANNHP\docker\labs\containers>
```

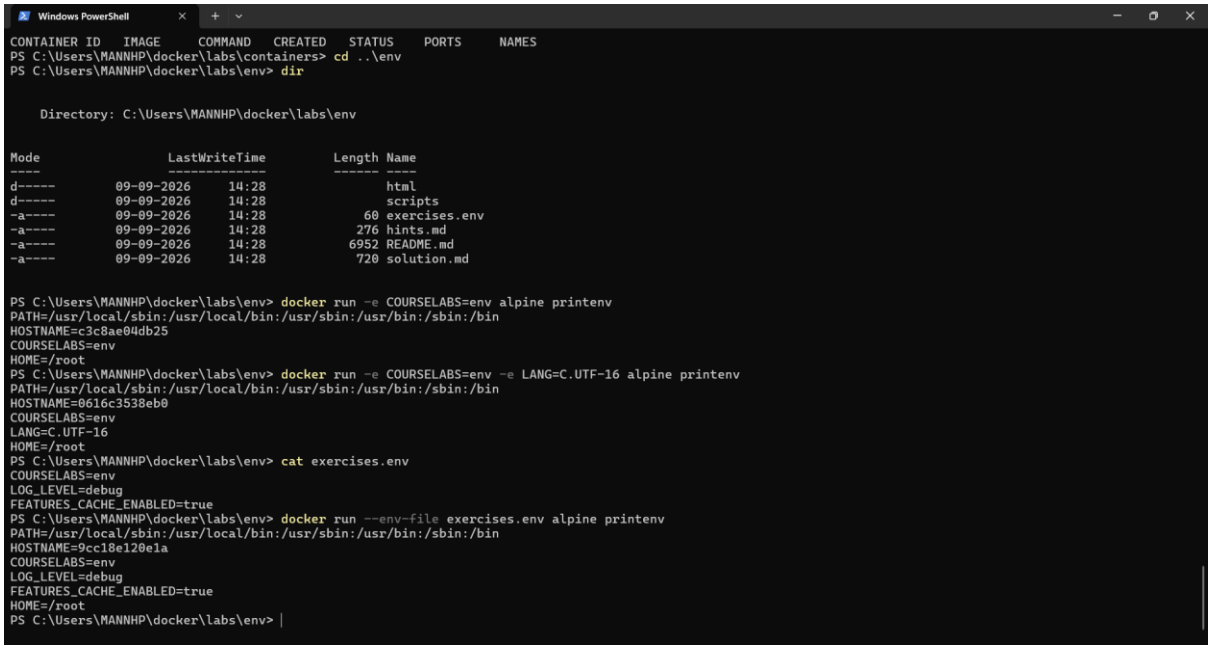

Windows PowerShell
| symbol: variable java
| java -version
| ^--^
| Error:
| cannot find symbol
| symbol: variable version
| java -version
| ^-----^
jshell> /exit
| Goodbye
PS C:\Users\MANNHP\docker\labs\containers> java -version
openjdk version "21.0.12" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-21.0.12+8 (build 21.0.12+8-LTS)
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PS C:\Users\MANNHP\docker\labs\containers> docker run -it --entrypoint /bin/bash eclipse-temurin:25
root@ca144a68bfd5:/# java -version
openjdk version "25.0.4" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-25.0.4+7 (build 25.0.4+7-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4+7 (build 25.0.4+7-LTS, mixed mode, sharing)
root@ca144a68bfd5:/# exit
exit
PS C:\Users\MANNHP\docker\labs\containers> docker pull eclipse-temurin:23-jre-alpine
23-jre-alpine: Pulling from library/eclipse-temurin
b6d8cc948f02: Pull complete
956bc91adb8d: Pull complete
2bcb7041227e: Pull complete
06b4ddc782f3: Pull complete
f18232174bc9: Pull complete
20809389c2ee: Download complete
ab5133e8760f: Download complete
Digest: sha256:88593498863c64b43be16e8357a3c70ea475fc20a93bf1e07f4609213a357c87
Status: Downloaded newer image for eclipse-temurin:23-jre-alpine
docker.io/library/eclipse-temurin:23-jre-alpine
PS C:\Users\MANNHP\docker\labs\containers> docker run --rm eclipse-temurin:23-jre-alpine java -version
>> ^C
PS C:\Users\MANNHP\docker\labs\containers> docker run --rm eclipse-temurin:23-jre-alpine java --version
openjdk 23.0.2 2025-01-21
OpenJDK Runtime Environment Temurin-23.0.2+7 (build 23.0.2+7)
OpenJDK 64-Bit Server VM Temurin-23.0.2+7 (build 23.0.2+7, mixed mode, sharing)
PS C:\Users\MANNHP\docker\labs\containers>

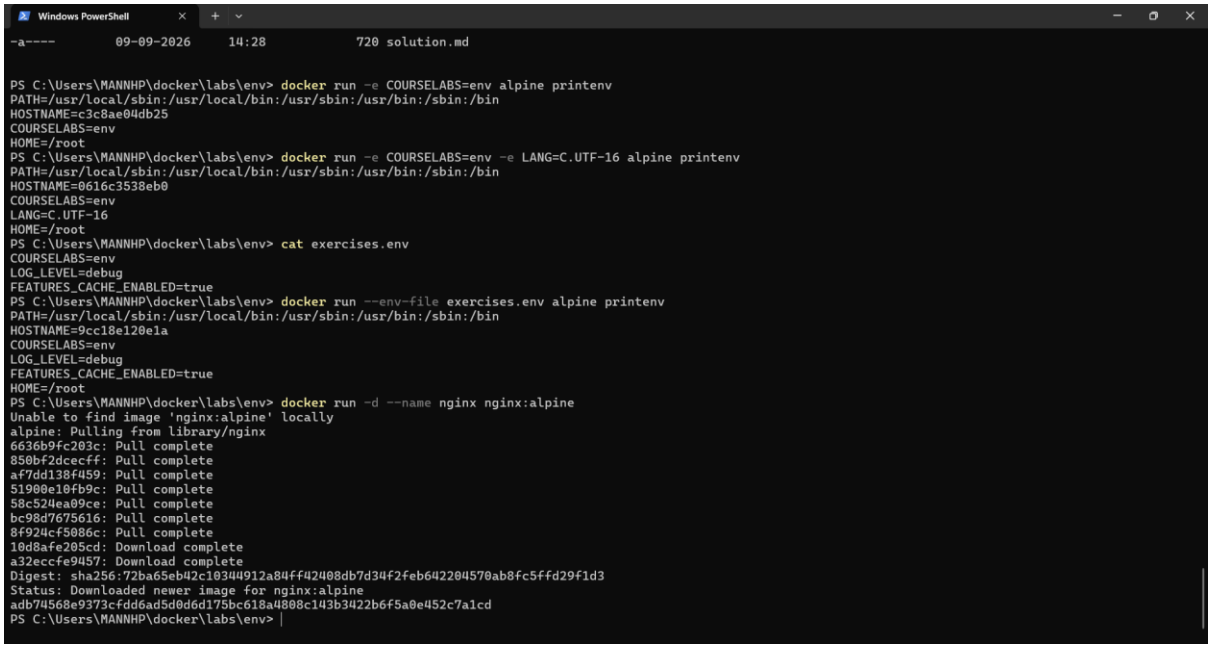

Windows PowerShell
| symbol: variable version
| java -version
| ^-----^
jshell> /exit
| Goodbye
PS C:\Users\MANNHP\docker\labs\containers> java -version
openjdk version "21.0.12" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-21.0.12+8 (build 21.0.12+8-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.12+8 (build 21.0.12+8-LTS, mixed mode, sharing)
PS C:\Users\MANNHP\docker\labs\containers> docker run -it --entrypoint /bin/bash eclipse-temurin:25
root@ca144a68bfd5:/# java -version
openjdk version "25.0.4" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-25.0.4+7 (build 25.0.4+7-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4+7 (build 25.0.4+7-LTS, mixed mode, sharing)
root@ca144a68bfd5:/# exit
exit
PS C:\Users\MANNHP\docker\labs\containers> docker pull eclipse-temurin:23-jre-alpine
23-jre-alpine: Pulling from library/eclipse-temurin
b6d8cc948f02: Pull complete
956bc91adb8d: Pull complete
2bcb7041227e: Pull complete
06b4ddc782f3: Pull complete
f18232174bc9: Pull complete
20809389c2ee: Download complete
ab5133e8760f: Download complete
Digest: sha256:88593498863c64b43be16e8357a3c70ea475fc20a93bf1e07f4609213a357c87
Status: Downloaded newer image for eclipse-temurin:23-jre-alpine
docker.io/library/eclipse-temurin:23-jre-alpine
PS C:\Users\MANNHP\docker\labs\containers> docker run --rm eclipse-temurin:23-jre-alpine java -version
>> ^C
PS C:\Users\MANNHP\docker\labs\containers> docker run --rm eclipse-temurin:23-jre-alpine java --version
openjdk 23.0.2 2025-01-21
OpenJDK Runtime Environment Temurin-23.0.2+7 (build 23.0.2+7)
OpenJDK 64-Bit Server VM Temurin-23.0.2+7 (build 23.0.2+7, mixed mode, sharing)
PS C:\Users\MANNHP\docker\labs\containers> docker images eclipse-temurin
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
eclipse-temurin:23-jre-alpine 88593498863c 283MB 73.9MB
eclipse-temurin:25 e787e08ef76f 590MB 155MB U
PS C:\Users\MANNHP\docker\labs\containers>
Important Notification Request
Allow important notifications from Snipping Tool ?
You will get important notifications from Snipping Tool when do not disturb is on. To change this later, select this app in notification settings.
Allow
Not Now

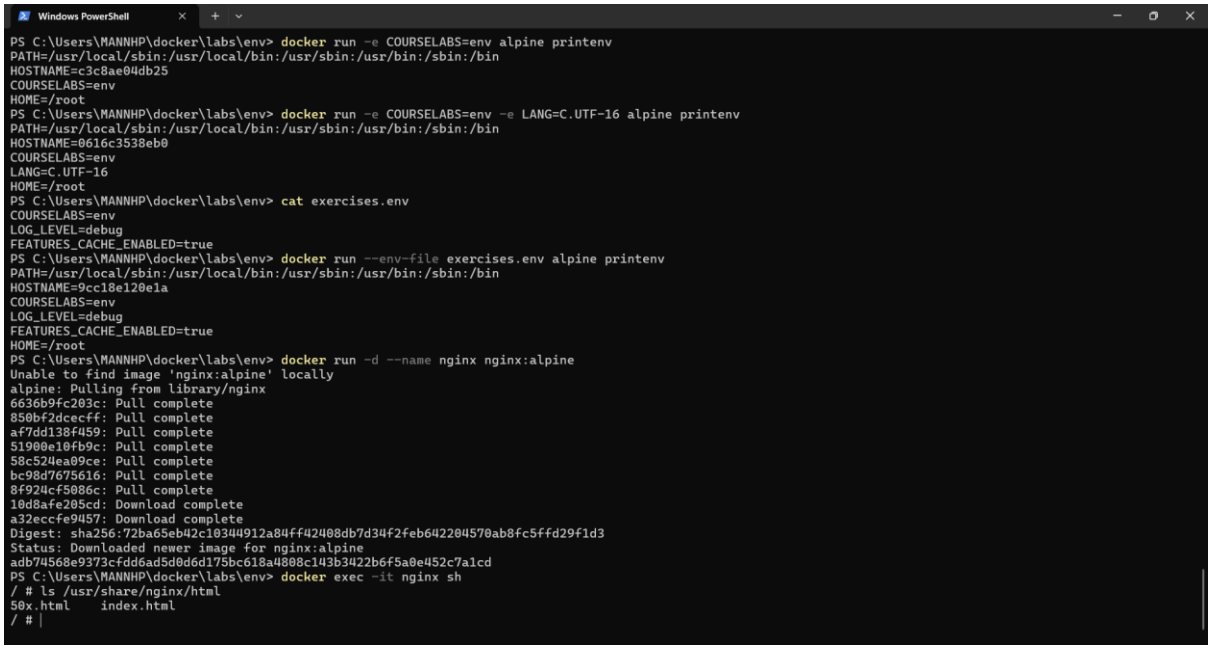

Windows PowerShell
PS C:\Users\MANNHP\docker\labs\containers> docker run -it --entrypoint /bin/bash eclipse-temurin:25
root@ca144a68bfd5:/# java -version
openjdk version "25.0.4" 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-25.0.4+7 (build 25.0.4+7-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4+7 (build 25.0.4+7-LTS, mixed mode, sharing)
root@ca144a68bfd5:/# exit
exit
PS C:\Users\MANNHP\docker\labs\containers> docker pull eclipse-temurin:23-jre-alpine
23-jre-alpine: Pulling from library/eclipse-temurin
b6d8cc948f02: Pull complete
956bc91adb8d: Pull complete
2bcb7041227e: Pull complete
06b4ddc782f3: Pull complete
f18232174bc9: Pull complete
20809389c2ee: Download complete
ab5133e8760f: Download complete
Digest: sha256:88593498863c64b43be16e8357a3c70ea475fc20a93bf1e07f4609213a357c87
Status: Downloaded newer image for eclipse-temurin:23-jre-alpine
docker.io/library/eclipse-temurin:23-jre-alpine
PS C:\Users\MANNHP\docker\labs\containers> docker run --rm eclipse-temurin:23-jre-alpine java -version
>> ^C
PS C:\Users\MANNHP\docker\labs\containers> docker run --rm eclipse-temurin:23-jre-alpine java --version
openjdk 23.0.2 2025-01-21
OpenJDK Runtime Environment Temurin-23.0.2+7 (build 23.0.2+7)
OpenJDK 64-Bit Server VM Temurin-23.0.2+7 (build 23.0.2+7, mixed mode, sharing)
PS C:\Users\MANNHP\docker\labs\containers> docker images eclipse-temurin
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
eclipse-temurin:23-jre-alpine 88593498863c 283MB 73.9MB
eclipse-temurin:25 e787e08ef76f 590MB 155MB U
PS C:\Users\MANNHP\docker\labs\containers> docker rm -f $(docker ps -aq)
ca144a68bfd5
4ced0dd50aa0
267eb7e6faac
1562ebb48e78
5ef09ee2b7f7
5ebab912751b
8b0ab2eaa285
PS C:\Users\MANNHP\docker\labs\containers> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
PS C:\Users\MANNHP\docker\labs\containers>

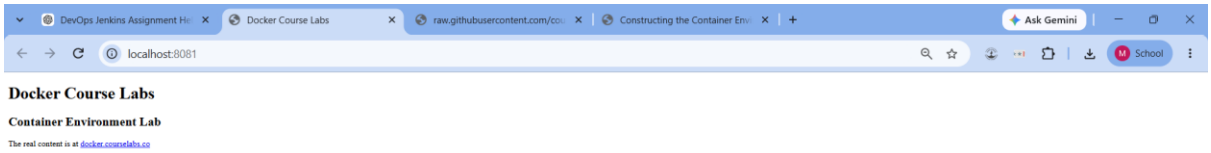

Windows PowerShell
Status: Downloaded newer image for eclipse-temurin:23-jre-alpine
docker.io/library/eclipse-temurin:23-jre-alpine
PS C:\Users\MANNHP\docker\labs\containers> docker run --rm eclipse-temurin:23-jre-alpine java -version
>> ^C
PS C:\Users\MANNHP\docker\labs\containers> docker run --rm eclipse-temurin:23-jre-alpine java --version
openjdk 23.0.2 2025-01-21
OpenJDK Runtime Environment Temurin-23.0.2+7 (build 23.0.2+7)
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PS C:\Users\MANNHP\docker\labs\containers> docker images eclipse-temurin
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
eclipse-temurin:23-jre-alpine 88593498863c 283MB 73.9MB
eclipse-temurin:25 e787e08ef76f 590MB 155MB U
PS C:\Users\MANNHP\docker\labs\containers> docker rm -f $(docker ps -aq)
ca144a68bfd5
4ced0dd50aa0
267eb7e6faac
1562ebb48e78
5ef09ee2b7f7
5ebab912751b
8b0ab2eaa285
PS C:\Users\MANNHP\docker\labs\containers> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
PS C:\Users\MANNHP\docker\labs\containers> cd ..\env
PS C:\Users\MANNHP\docker\labs\env> dir
Directory: C:\Users\MANNHP\docker\labs\env
Mode LastWriteTime Length Name
d----- 09-09-2026 14:28 html
d----- 09-09-2026 14:28 scripts
-a---- 09-09-2026 14:28 60 exercises.env
-a---- 09-09-2026 14:28 276 hints.md
-a---- 09-09-2026 14:28 6952 README.md
-a---- 09-09-2026 14:28 720 solution.md
PS C:\Users\MANNHP\docker\labs\env>

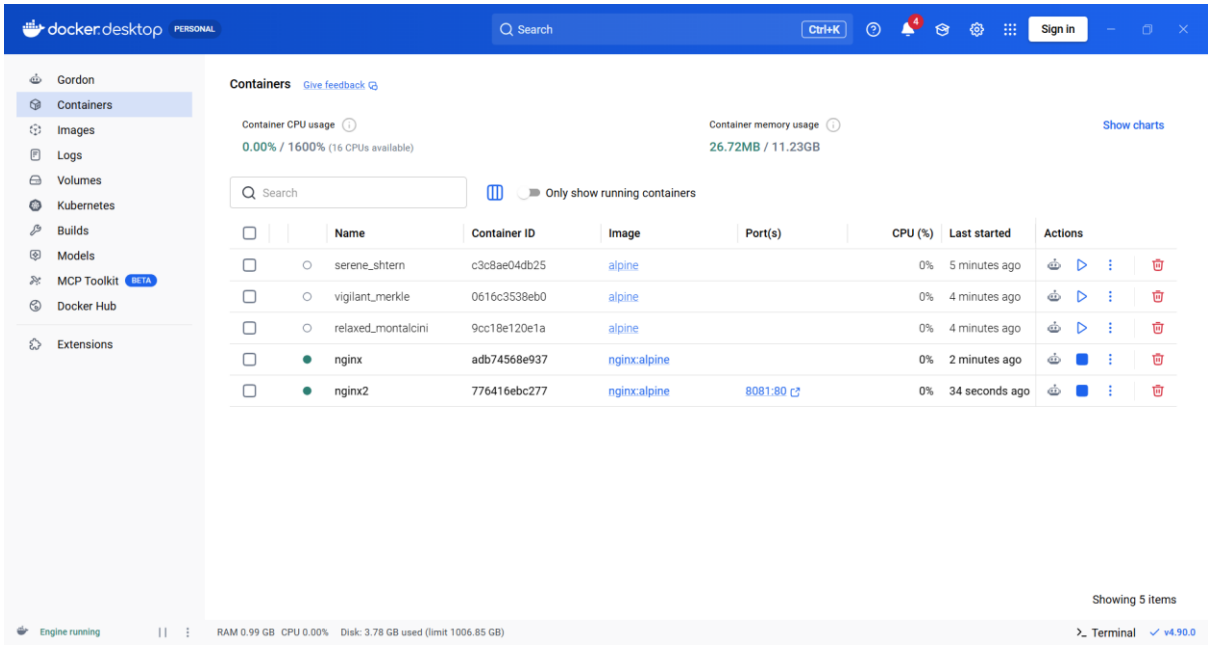

Windows PowerShell
openjdk 23.0.2 2025-01-21
OpenJDK Runtime Environment Temurin-23.0.2+7 (build 23.0.2+7)
OpenJDK 64-Bit Server VM Temurin-23.0.2+7 (build 23.0.2+7, mixed mode, sharing)
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eclipse-temurin:25 e787e08ef76f 590MB 155MB U
PS C:\Users\MANNHP\docker\labs\containers> docker rm -f $(docker ps -aq)
ca144a68bfd5
4ced0dd50aa0
267eb7e6faac
1562ebb48e78
5ef09ee2b7f7
5ebab912751b
8b0ab2eaa285
PS C:\Users\MANNHP\docker\labs\containers> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
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Directory: C:\Users\MANNHP\docker\labs\env
Mode LastWriteTime Length Name
d----- 09-09-2026 14:28 html
d----- 09-09-2026 14:28 scripts
-a---- 09-09-2026 14:28 60 exercises.env
-a---- 09-09-2026 14:28 276 hints.md
-a---- 09-09-2026 14:28 6952 README.md
-a---- 09-09-2026 14:28 720 solution.md
PS C:\Users\MANNHP\docker\labs\env> docker run -e COURSELABS=env alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=c3c8ae04db25
COURSELABS=env
HOME=/root
PS C:\Users\MANNHP\docker\labs\env>

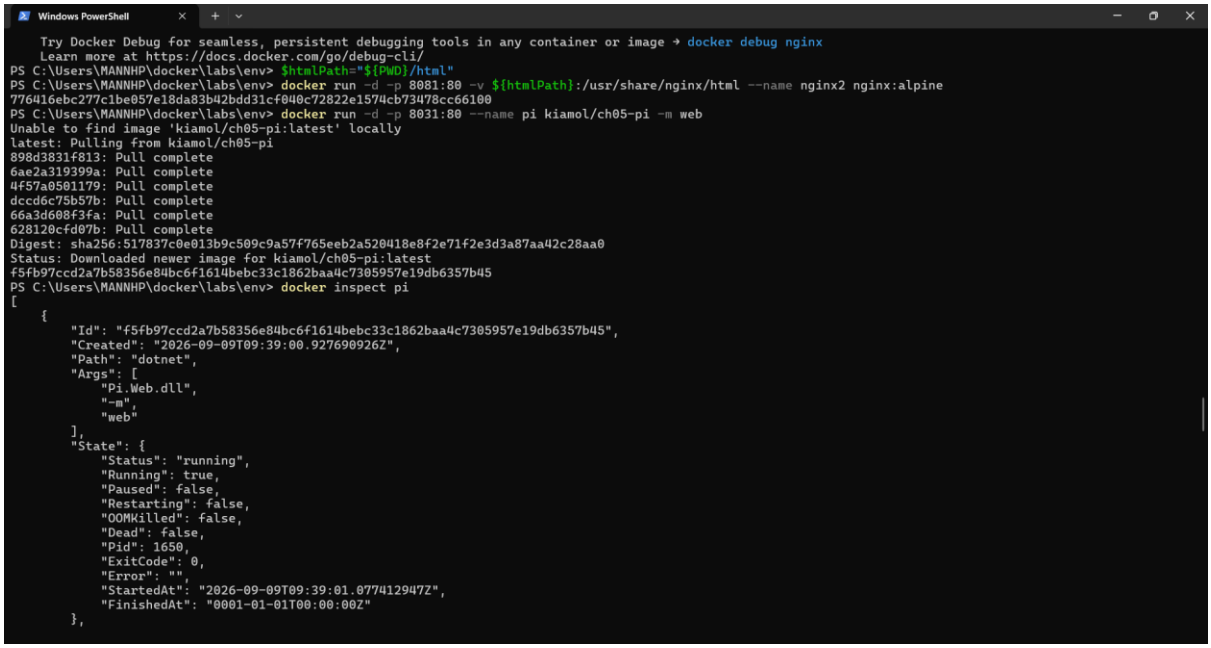

Windows PowerShell
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
PS C:\Users\MANNHP\docker\labs\containers> cd ..\env
PS C:\Users\MANNHP\docker\labs\env> dir
Directory: C:\Users\MANNHP\docker\labs\env
Mode LastWriteTime Length Name
d----- 09-09-2026 14:28 html
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-a---- 09-09-2026 14:28 276 hints.md
-a---- 09-09-2026 14:28 6952 README.md
-a---- 09-09-2026 14:28 720 solution.md
PS C:\Users\MANNHP\docker\labs\env> docker run -e COURSELABS=env alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=c3c8ae04db25
COURSELABS=env
HOME=/root
PS C:\Users\MANNHP\docker\labs\env> docker run -e COURSELABS=env -e LANG=C.UTF-16 alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=0616c3538eb0
COURSELABS=env
LANG=C.UTF-16
HOME=/root
PS C:\Users\MANNHP\docker\labs\env> cat exercises.env
COURSELABS=env
LOG_LEVEL=debug
FEATURES_CACHE_ENABLED=true
PS C:\Users\MANNHP\docker\labs\env> docker run --env-file exercises.env alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=9cc18e120e1a
COURSELABS=env
LOG_LEVEL=debug
FEATURES_CACHE_ENABLED=true
HOME=/root
PS C:\Users\MANNHP\docker\labs\env>

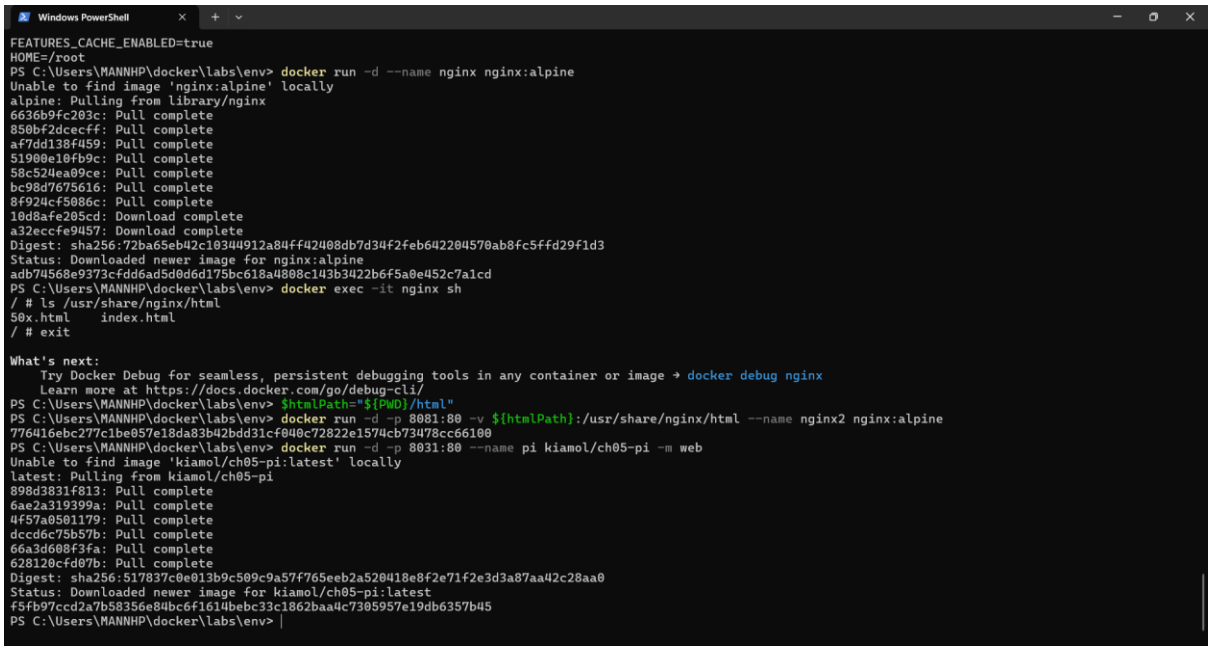

Windows PowerShell
-a---- 09-09-2026 14:28 720 solution.md
PS C:\Users\MANNHP\docker\labs\env> docker run -e COURSELABS=env alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=c3c8ae04db25
COURSELABS=env
HOME=/root
PS C:\Users\MANNHP\docker\labs\env> docker run -e COURSELABS=env -e LANG=C.UTF-16 alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=0616c3538eb0
COURSELABS=env
LANG=C.UTF-16
HOME=/root
PS C:\Users\MANNHP\docker\labs\env> cat exercises.env
COURSELABS=env
LOG_LEVEL=debug
FEATURES_CACHE_ENABLED=true
PS C:\Users\MANNHP\docker\labs\env> docker run --env-file exercises.env alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=9cc18e120e1a
COURSELABS=env
LOG_LEVEL=debug
FEATURES_CACHE_ENABLED=true
HOME=/root
PS C:\Users\MANNHP\docker\labs\env> docker run -d --name nginx nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
6636b9fc203c: Pull complete
850bf2dcecff: Pull complete
af7dd138f459: Pull complete
51900e10fb9c: Pull complete
58c524ea09ce: Pull complete
bc98d7675616: Pull complete
8f924cf5086c: Pull complete
10d8afe205cd: Download complete
a32eccfe9457: Download complete
Digest: sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5ffd29f1d3
Status: Downloaded newer image for nginx:alpine
adb74568e9373cfdd6ad5d0d6d175bc618a4808c143b3422b6f5a0e452c7a1cd
PS C:\Users\MANNHP\docker\labs\env>

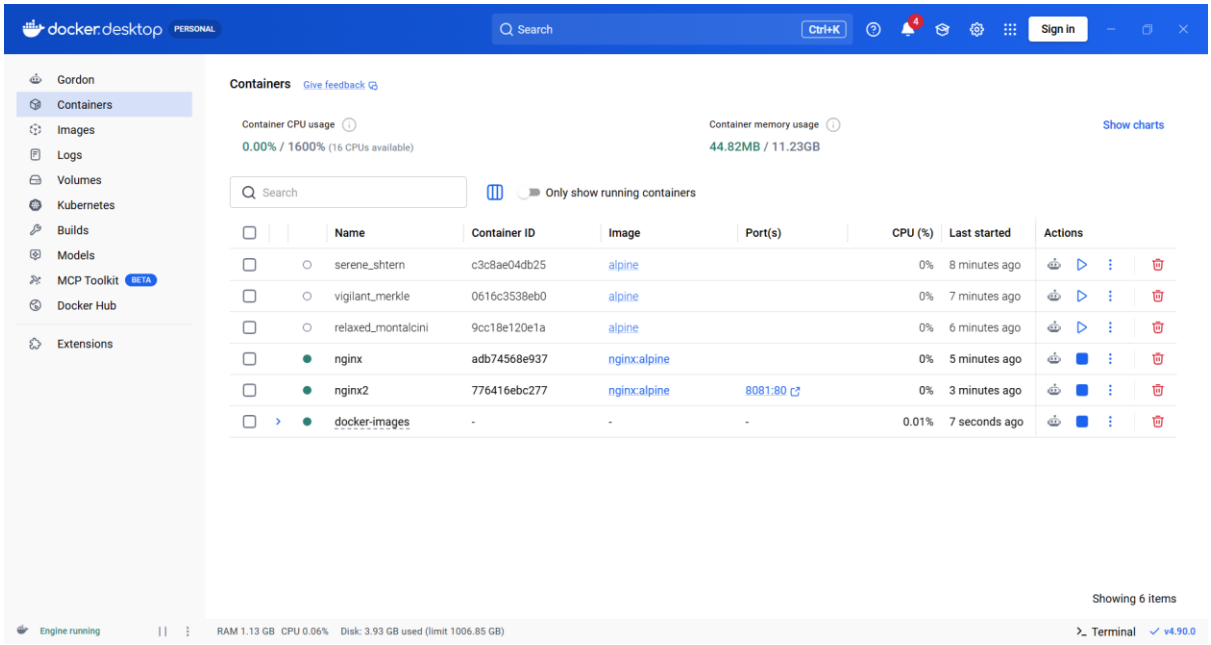

Windows PowerShell
PS C:\Users\MANNHP\docker\labs\env> docker run -e COURSELABS=env alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=c3c8ae04db25
COURSELABS=env
HOME=/root
PS C:\Users\MANNHP\docker\labs\env> docker run -e COURSELABS=env -e LANG=C.UTF-16 alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=0616c3538eb0
COURSELABS=env
LANG=C.UTF-16
HOME=/root
PS C:\Users\MANNHP\docker\labs\env> cat exercises.env
COURSELABS=env
LOG_LEVEL=debug
FEATURES_CACHE_ENABLED=true
PS C:\Users\MANNHP\docker\labs\env> docker run --env-file exercises.env alpine printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=9cc18e120e1a
COURSELABS=env
LOG_LEVEL=debug
FEATURES_CACHE_ENABLED=true
HOME=/root
PS C:\Users\MANNHP\docker\labs\env> docker run -d --name nginx nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
6636b9fc203c: Pull complete
850bf2dcecff: Pull complete
af7dd138f459: Pull complete
51900e10fb9c: Pull complete
58c524ea09ce: Pull complete
bc98d7675616: Pull complete
8f924cf5086c: Pull complete
10d8afe205cd: Download complete
a32eccfe9457: Download complete
Digest: sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5ffd29f1d3
Status: Downloaded newer image for nginx:alpine
adb74568e9373cfdd6ad5d0d6d175bc618a4808c143b3422b6f5a0e452c7a1cd
PS C:\Users\MANNHP\docker\labs\env> docker exec -it nginx sh
/ # ls /usr/share/nginx/html
50x.html   index.html
/ #

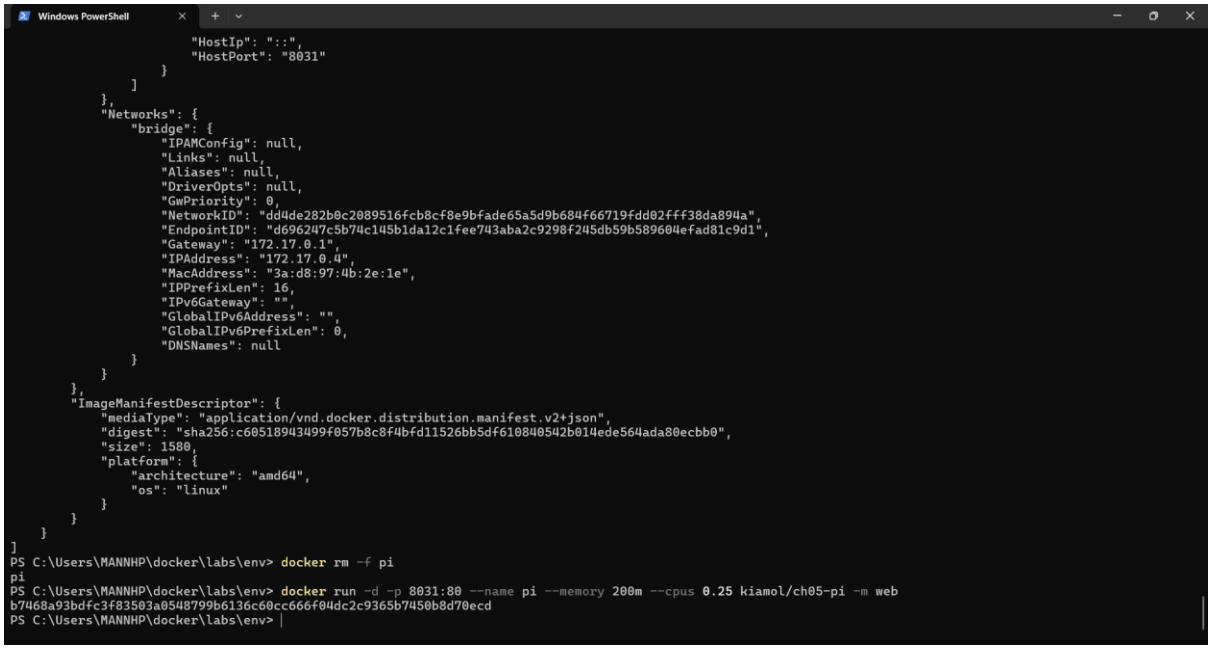

DevOps Jenkins Assignment He
Docker Course Labs
raw.githubusercontent.com/cou
Constructing the Container Env
Ask Gemini
localhost:8081
School
Docker Course Labs
Container Environment Lab
The real content is at docker.courselabs.co

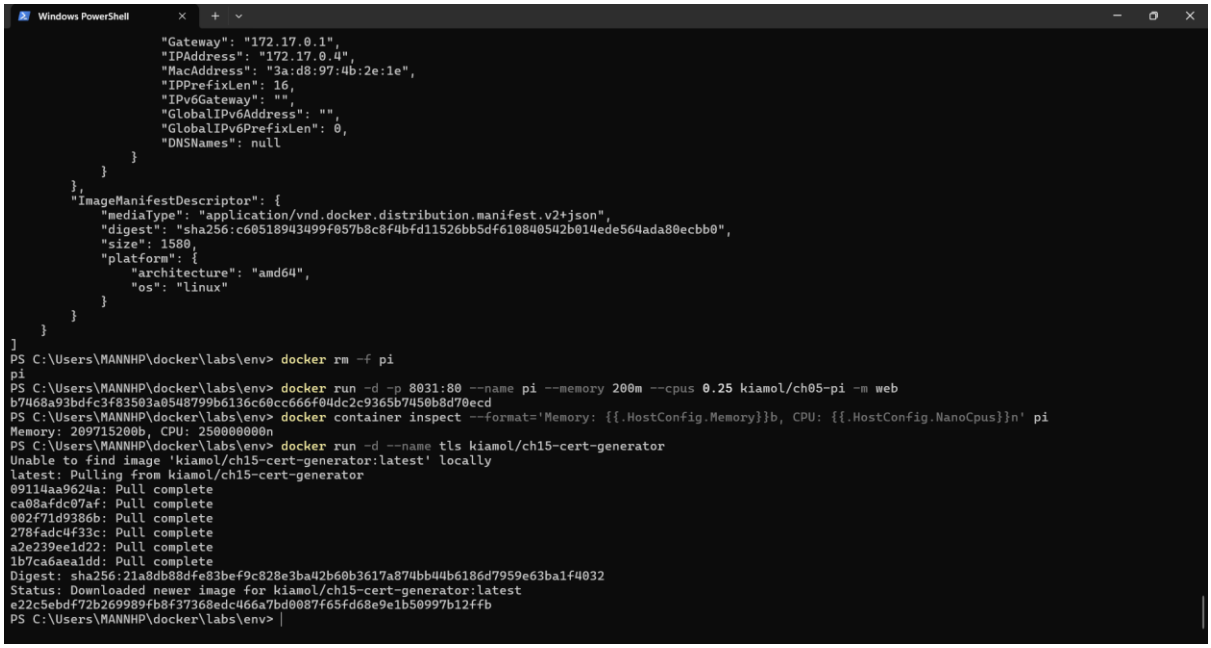

docker desktop
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Extensions
Containers
Give feedback
Container CPU usage
0.00% / 1600% (16 CPUs available)
Container memory usage
26.72MB / 11.23GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
serene_shtern c3c8ae04db25 alpine 0% 5 minutes ago
vigilant_merkle 0616c3538eb0 alpine 0% 4 minutes ago
relaxed_montalcini 9cc18e120e1a alpine 0% 4 minutes ago
nginx adb74568e937 nginx:alpine 0% 2 minutes ago
nginx2 776416ebc277 nginx:alpine 8081:80 0% 34 seconds ago
Showing 5 items
Engine running
RAM 0.99 GB CPU 0.00% Disk: 3.78 GB used (limit 1006.85 GB)
Terminal
v4.90.0

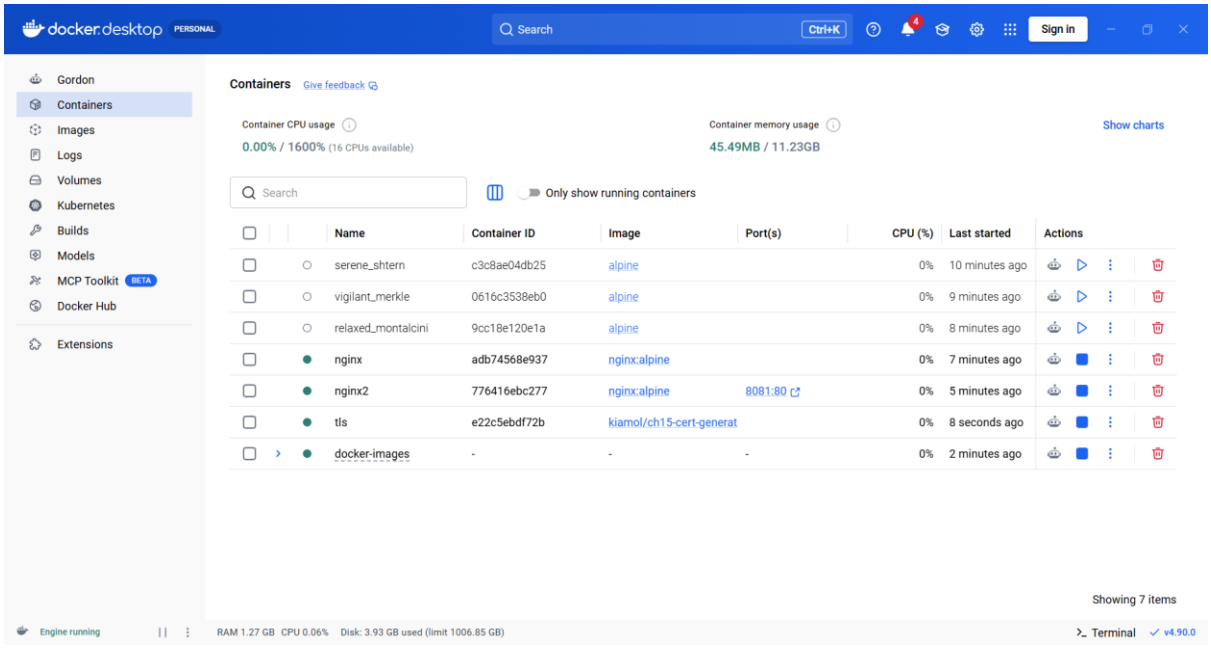

Windows PowerShell
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug nginx
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\MANNHP\docker\labs\env> $htmlPath="${PWD}/html"
PS C:\Users\MANNHP\docker\labs\env> docker run -d -p 8081:80 -v ${htmlPath}:/usr/share/nginx/html --name nginx2 nginx:alpine
776416ebc277c1be057e18da83b42bdd31cf040c72822e1574cb73478cc66100
PS C:\Users\MANNHP\docker\labs\env> docker run -d -p 8031:80 --name pi kiamol/ch05-pi -m web
Unable to find image 'kiamol/ch05-pi:latest' locally
latest: Pulling from kiamol/ch05-pi
898d3831f813: Pull complete
6ae2a319399a: Pull complete
4f57a0501179: Pull complete
dccd6c75b57b: Pull complete
66a3d608f3fa: Pull complete
628120cfd07b: Pull complete
Digest: sha256:517837c0e013b9c509c9a57f765eeb2a520418e8f2e71f2e3d3a87aa42c28aa0
Status: Downloaded newer image for kiamol/ch05-pi:latest
f5fb97ccd2a7b58356e84bc6f1614bebc33c1862baa4c7305957e19db6357b45
PS C:\Users\MANNHP\docker\labs\env> docker inspect pi
[
{
"Id": "f5fb97ccd2a7b58356e84bc6f1614bebc33c1862baa4c7305957e19db6357b45",
"Created": "2026-09-09T09:39:00.927690926Z",
"Path": "dotnet",
"Args": [
"Pi.Web.dll",
"-m",
"web"
],
"State": {
"Status": "running",
"Running": true,
"Paused": false,
"Restarting": false,
"OOMKilled": false,
"Dead": false,
"Pid": 1650,
"ExitCode": 0,
"Error": "",
"StartedAt": "2026-09-09T09:39:01.0774129477Z",
"FinishedAt": "0001-01-01T00:00:00Z"
},

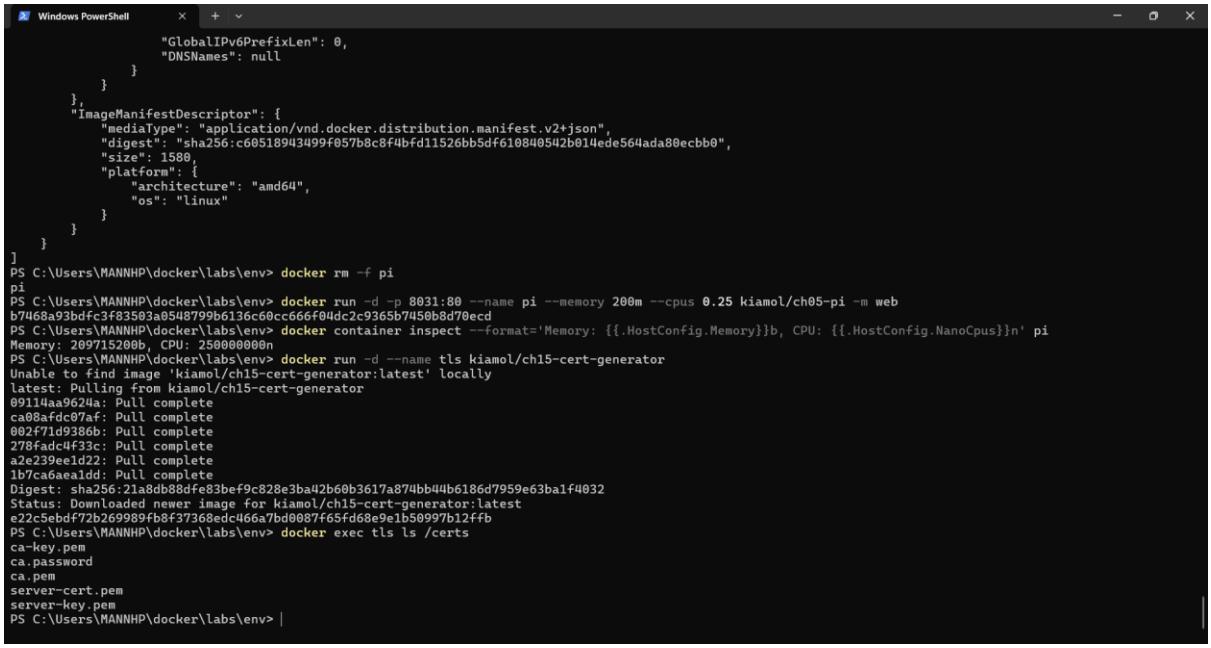

Windows PowerShell
FEATURES_CACHE_ENABLED=true
HOME=/root
PS C:\Users\MANNHP\docker\labs\env> docker run -d --name nginx nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
6636b9fc203c: Pull complete
850bf2dcecff: Pull complete
af7dd138f459: Pull complete
51900e10fb9c: Pull complete
58c524ea09ce: Pull complete
bc98d7675616: Pull complete
8f924cf5086c: Pull complete
10d8afe205cd: Download complete
a32eccfe9457: Download complete
Digest: sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5ffd29f1d3
Status: Downloaded newer image for nginx:alpine
adb74568e9373cfdd6ad5d0d6d175bc618a4808c143b3422b6f5a0e452c7a1cd
PS C:\Users\MANNHP\docker\labs\env> docker exec -it nginx sh
/ # ls /usr/share/nginx/html
50x.html index.html
/ # exit
What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug nginx
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\MANNHP\docker\labs\env> $htmlPath="${PWD}/html"
PS C:\Users\MANNHP\docker\labs\env> docker run -d -p 8081:80 -v ${htmlPath}:/usr/share/nginx/html --name nginx2 nginx:alpine
776416ebc277c1be057e18da83b42bdd31cf040c72822e1574cb73478cc66100
PS C:\Users\MANNHP\docker\labs\env> docker run -d -p 8031:80 --name pi kiamol/ch05-pi -m web
Unable to find image 'kiamol/ch05-pi:latest' locally
latest: Pulling from kiamol/ch05-pi
898d3831f813: Pull complete
6ae2a319399a: Pull complete
4f57a0501179: Pull complete
dccd6c75b57b: Pull complete
66a3d608f3fa: Pull complete
628120cfd07b: Pull complete
Digest: sha256:517837c0e013b9c509c9a57f765eeb2a520418e8f2e71f2e3d3a87aa42c28aa0
Status: Downloaded newer image for kiamol/ch05-pi:latest
f5fb97ccd2a7b58356e84bc6f1614bebc33c1862baa4c7305957e19db6357b45
PS C:\Users\MANNHP\docker\labs\env>

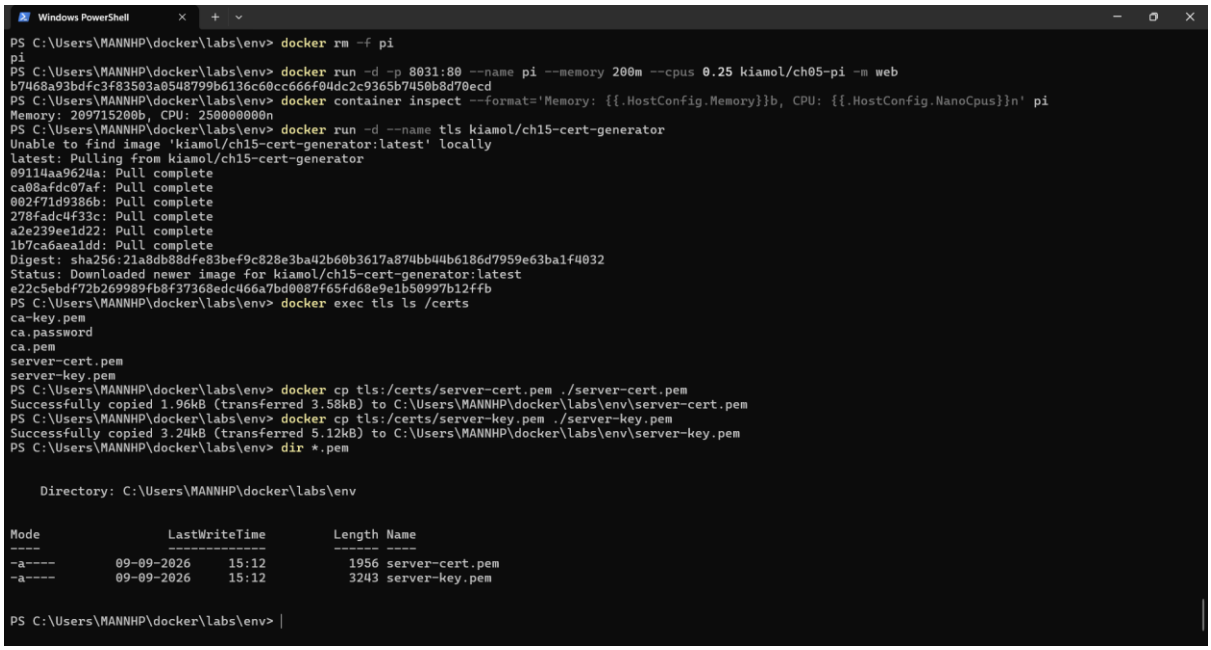

docker desktop
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Container CPU usage
0.00% / 1600% (16 CPUs available)
Container memory usage
44.82MB / 11.23GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
serene_shtern c3c8ae04db25 alpine 0% 8 minutes ago
vigilant_merkle 0616c3538eb0 alpine 0% 7 minutes ago
relaxed_montalcini 9cc18e120e1a alpine 0% 6 minutes ago
nginx adb74568e937 nginx:alpine 0% 5 minutes ago
nginx2 776416ebc277 nginx:alpine 8081:80 0% 3 minutes ago
docker-images - - - 0.01% 7 seconds ago
Showing 6 items
Engine running
RAM 1.13 GB CPU 0.06% Disk: 3.93 GB used (limit 1006.85 GB)
Terminal
v4.90.0

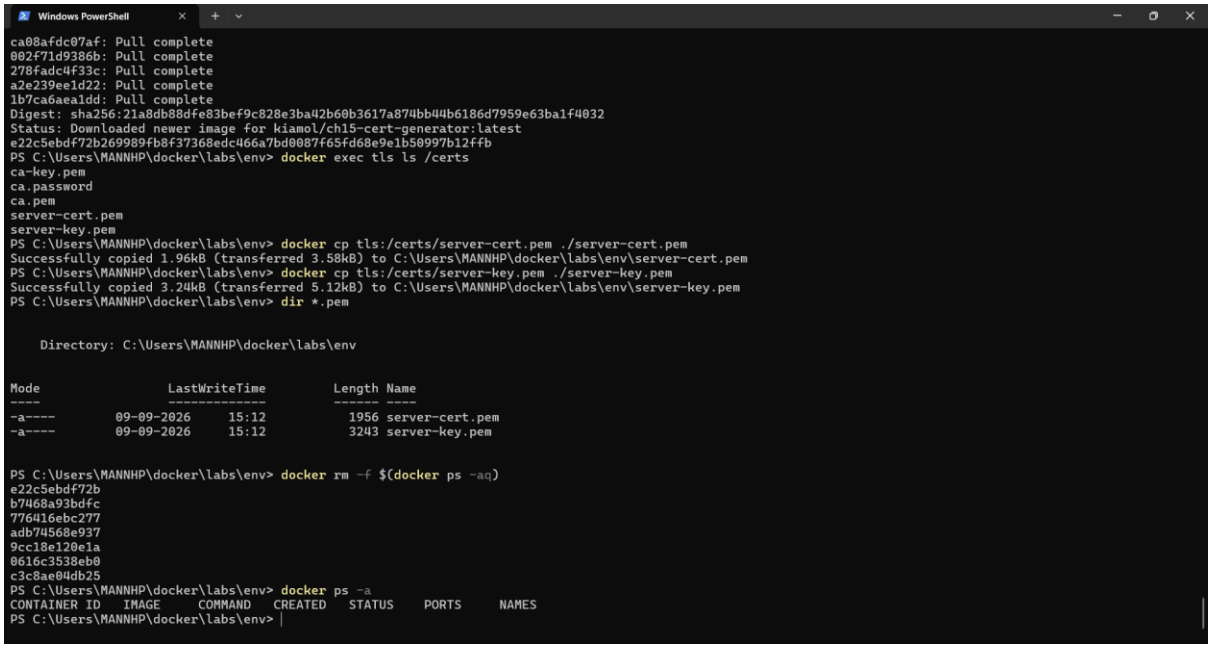

Windows PowerShell
"HostIp": "::",
"HostPort": "8031"
"Networks": {
"bridge": {
"IPAMConfig": null,
"Links": null,
"Aliases": null,
"DriverOpts": null,
"GwPriority": 0,
"NetworkID": "dd4de282b0c2089516fcb8cf8e9bfade65a5d9b684f66719fdd02fff38da894a",
"EndpointID": "d696247c5b74c145b1da12c1fee743aba2c9298f245db59b589604efad81c9d1",
"Gateway": "172.17.0.1",
"IPAddress": "172.17.0.4",
"MacAddress": "3a:d8:97:4b:2e:1e",
"IPPrefixLen": 16,
"IPv6Gateway": "",
"GlobalIPv6Address": "",
"GlobalIPv6PrefixLen": 0,
"DNSNames": null
"ImageManifestDescriptor": {
"mediaType": "application/vnd.docker.distribution.manifest.v2+json",
"digest": "sha256:c605189434 99f057b8c8f4bfd11526bb5df610840542b014ede564ada80ecbb0",
"size": 1580,
"platform": {
"architecture": "amd64",
"os": "linux"
PS C:\Users\MANNHP\docker\labs\env> docker rm -f pi
pi
PS C:\Users\MANNHP\docker\labs\env> docker run -d -p 8031:80 --name pi --memory 200m --cpus 0.25 kiamol/ch05-pi -m web
b7468a93bdfc3f83503a0548799b6136c60cc666f04dc2c9365b7450b8d70ecd
PS C:\Users\MANNHP\docker\labs\env>


Windows PowerShell
"Gateway": "172.17.0.1",
"IPAddress": "172.17.0.4",
"MacAddress": "3a:d8:97:4b:2e:1e",
"IPPrefixLen": 16,
"IPv6Gateway": "",
"GlobalIPv6Address": "",
"GlobalIPv6PrefixLen": 0,
"DNSNames": null
"ImageManifestDescriptor": {
"mediaType": "application/vnd.docker.distribution.manifest.v2+json",
"digest": "sha256:c60518943499f057b8c8f4bfd11526bb5df610840542b014ede564ada80ecbb0",
"size": 1580,
"platform": {
"architecture": "amd64",
"os": "linux"
PS C:\Users\MANNHP\docker\labs\env> docker rm -f pi
pi
PS C:\Users\MANNHP\docker\labs\env> docker run -d -p 8031:80 --name pi --memory 200m --cpus 0.25 kiamol/ch05-pi -m web
b7468a93bdfc3f83503a0548799b6136c60cc666f04dc2c9365b7450b8d70ecd
PS C:\Users\MANNHP\docker\labs\env> docker container inspect --format='Memory: {{.HostConfig.Memory}}b, CPU: {{.HostConfig.NanoCpus}}n' pi
Memory: 209715200b, CPU: 250000000n
PS C:\Users\MANNHP\docker\labs\env> docker run -d --name tls kiamol/ch15-cert-generator
Unable to find image 'kiamol/ch15-cert-generator:latest' locally
latest: Pulling from kiamol/ch15-cert-generator
09114aa9624a: Pull complete
ca08afdc07af: Pull complete
002f71d9386b: Pull complete
278fadc4f33c: Pull complete
a2e239ee1d22: Pull complete
1b7ca6aea1dd: Pull complete
Digest: sha256:21a8db88dfe83bef9c828e3ba42b60b3617a874bb44b6186d7959e63ba1f4032
Status: Downloaded newer image for kiamol/ch15-cert-generator:latest
e22c5ebdf72b269989fb8f37368edc466a7bd0087f65fd68e9e1b50997b12ffb
PS C:\Users\MANNHP\docker\labs\env>

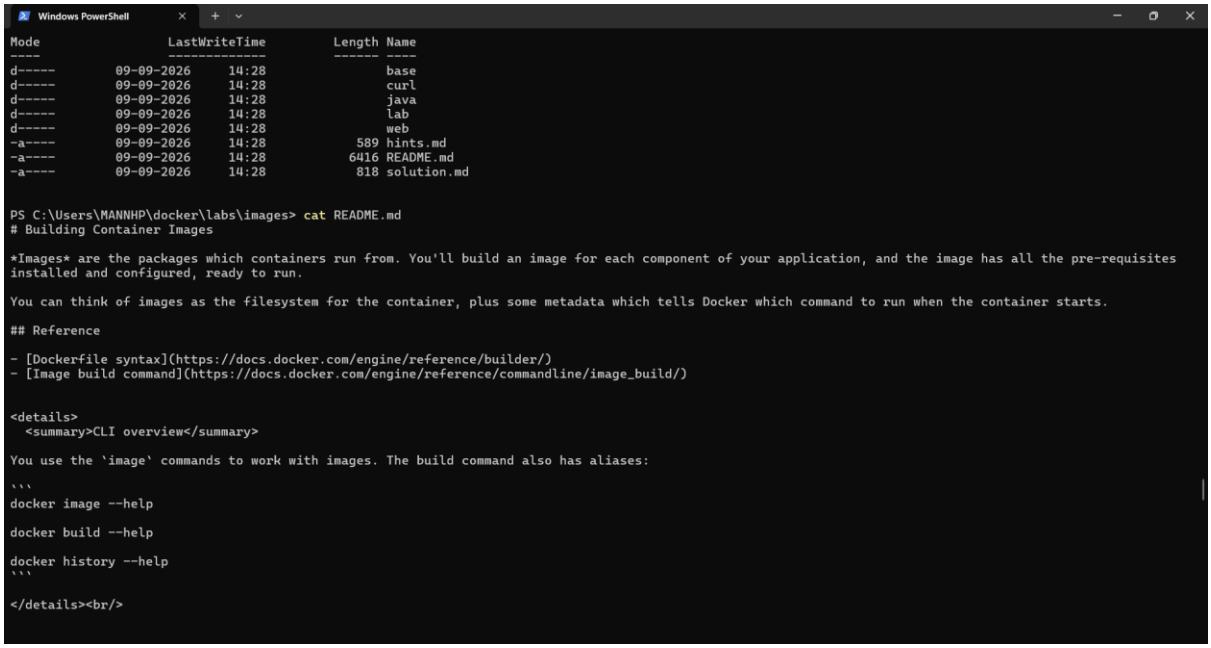

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Container ID
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Port(s)
CPU (%)
Last started
Actions
serene_shtern c3c8ae04db25 alpine 0% 10 minutes ago
vigilant_merkle 0616c3538eb0 alpine 0% 9 minutes ago
relaxed_montalcini 9cc18e120e1a alpine 0% 8 minutes ago
nginx adb74568e937 nginx:alpine 0% 7 minutes ago
nginx2 776416ebc277 nginx:alpine 8081:80 0% 5 minutes ago
tls e22c5ebdf72b kiamol/ch15-cert-generat 0% 8 seconds ago
docker-images - - - 0% 2 minutes ago
Showing 7 items
Engine running
RAM 1.27 GB CPU 0.06% Disk: 3.93 GB used (limit 1006.85 GB)
Terminal
v4.90.0

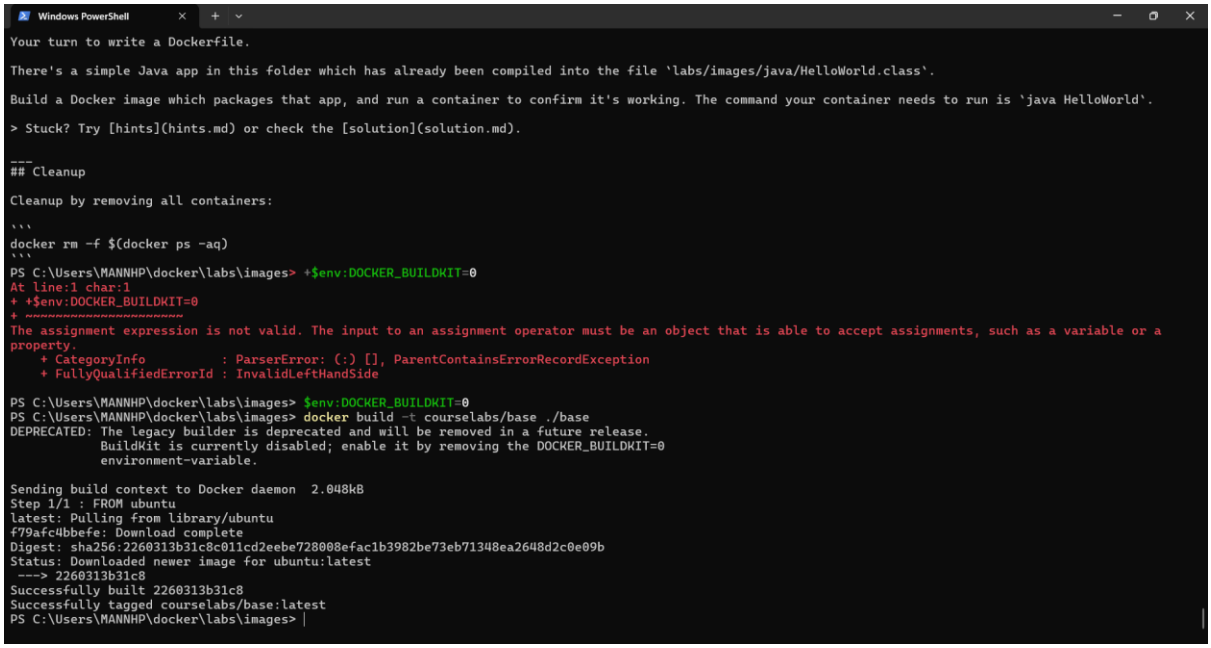

Windows PowerShell
"GlobalIPv6PrefixLen": 0,
"DNSNames": null
}
}
},
"ImageManifestDescriptor": {
"mediaType": "application/vnd.docker.distribution.manifest.v2+json",
"digest": "sha256:c60518943499f057b8c8f4bfd11526bb5df610840542b014ede564ada80ecbb0",
"size": 1580,
"platform": {
"architecture": "amd64",
"os": "linux"
}
}
}
]
PS C:\Users\MANNHP\docker\labs\env> docker rm -f pi
pi
PS C:\Users\MANNHP\docker\labs\env> docker run -d -p 8031:80 --name pi --memory 200m --cpus 0.25 kiamol/ch05-pi -m web
b7468a93bdfc3f83503a0548799b6136c60cc666f04dc2c9365b7450b8d70ecd
PS C:\Users\MANNHP\docker\labs\env> docker container inspect --format='Memory: {{.HostConfig.Memory}}b, CPU: {{.HostConfig.NanoCpus}}n' pi
Memory: 209715200b, CPU: 250000000n
PS C:\Users\MANNHP\docker\labs\env> docker run -d --name tls kiamol/ch15-cert-generator
Unable to find image 'kiamol/ch15-cert-generator:latest' locally
latest: Pulling from kiamol/ch15-cert-generator
09114aa9624a: Pull complete
ca08afdc07af: Pull complete
002f71d9386b: Pull complete
278fadc4f33c: Pull complete
a2e239ee1d22: Pull complete
1b7ca6aea1dd: Pull complete
Digest: sha256:21a8db88dfe83bef9c828e3ba42b60b3617a874bb44b6186d7959e63ba1f4032
Status: Downloaded newer image for kiamol/ch15-cert-generator:latest
e22c5ebdf72b269989fb8f37368edc466a7bd0087f65fd68e9e1b50997b12ffb
PS C:\Users\MANNHP\docker\labs\env> docker exec tls ls /certs
ca-key.pem
ca.password
ca.pem
server-cert.pem
server-key.pem
PS C:\Users\MANNHP\docker\labs\env>


Windows PowerShell
PS C:\Users\MANNHP\docker\labs\env> docker rm -f pi
pi
PS C:\Users\MANNHP\docker\labs\env> docker run -d -p 8031:80 --name pi --memory 200m --cpus 0.25 kiamol/ch05-pi -m web
b7468a93bdfc3f83503a0548799b6136c60cc666f04dc2c9365b7450b8d70ecd
PS C:\Users\MANNHP\docker\labs\env> docker container inspect --format='Memory: {{.HostConfig.Memory}}b, CPU: {{.HostConfig.NanoCpus}}n' pi
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Unable to find image 'kiamol/ch15-cert-generator:latest' locally
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ca08afdc07af: Pull complete
002f71d9386b: Pull complete
278fadc4f33c: Pull complete
a2e239ee1d22: Pull complete
1b7ca6aea1dd: Pull complete
Digest: sha256:21a8db88dfe83bef9c828e3ba42b60b3617a874bb44b6186d7959e63ba1f4032
Status: Downloaded newer image for kiamol/ch15-cert-generator:latest
e22c5ebdf72b269989fb8f37368edc466a7bd0087f65fd68e9e1b50997b12ffb
PS C:\Users\MANNHP\docker\labs\env> docker exec tls ls /certs
ca-key.pem
ca.password
ca.pem
server-cert.pem
server-key.pem
PS C:\Users\MANNHP\docker\labs\env> docker cp tls:/certs/server-cert.pem ./server-cert.pem
Successfully copied 1.96kB (transferred 3.58kB) to C:\Users\MANNHP\docker\labs\env\server-cert.pem
PS C:\Users\MANNHP\docker\labs\env> docker cp tls:/certs/server-key.pem ./server-key.pem
Successfully copied 3.24kB (transferred 5.12kB) to C:\Users\MANNHP\docker\labs\env\server-key.pem
PS C:\Users\MANNHP\docker\labs\env> dir *.pem
Directory: C:\Users\MANNHP\docker\labs\env
Mode LastWriteTime Length Name
---- ------------- ------ ----
-a---- 09-09-2026 15:12 1956 server-cert.pem
-a---- 09-09-2026 15:12 3243 server-key.pem
PS C:\Users\MANNHP\docker\labs\env>

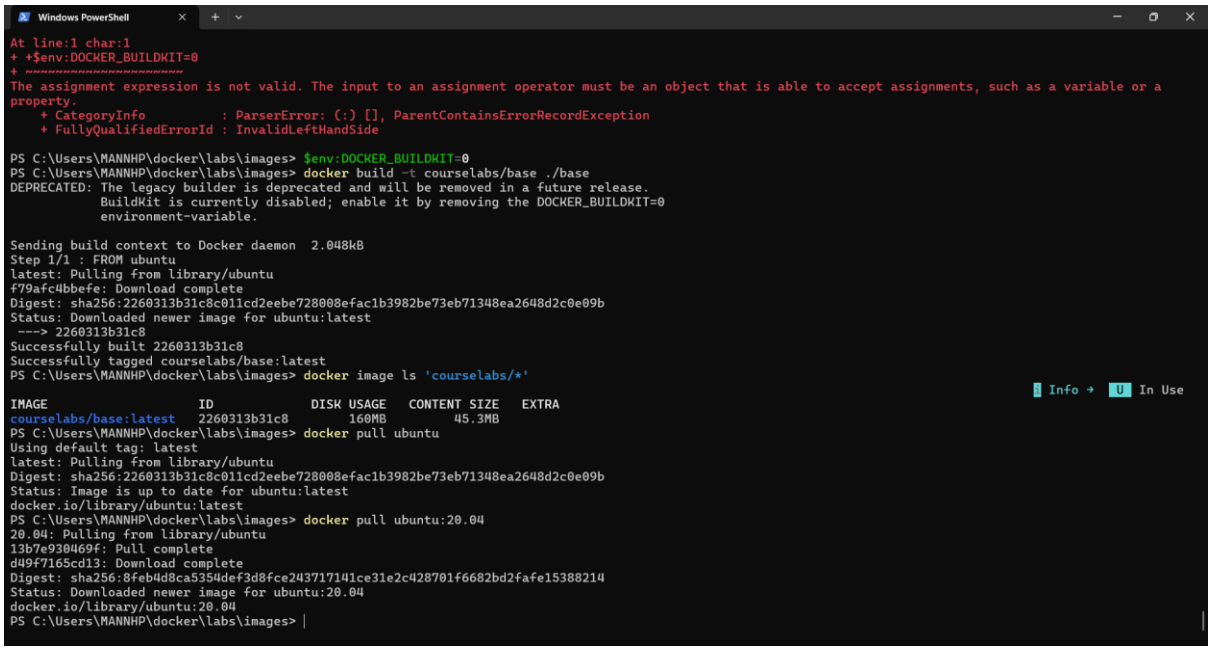

Windows PowerShell
ca08afdc07af: Pull complete
002f71d9386b: Pull complete
278fadc4f33c: Pull complete
a2e239ee1d22: Pull complete
1b7ca6aea1dd: Pull complete
Digest: sha256:21a8db88dfe83bef9c828e3ba42b60b3617a874bb44b6186d7959e63ba1f4032
Status: Downloaded newer image for kiamol/ch15-cert-generator:latest
e22c5ebdf72b269989fb8f37368edc466a7bd0087f65fd68e9e1b50997b12ffb
PS C:\Users\MANNHP\docker\labs\env> docker exec tls ls /certs
ca-key.pem
ca.password
ca.pem
server-cert.pem
server-key.pem
PS C:\Users\MANNHP\docker\labs\env> docker cp tls:/certs/server-cert.pem ./server-cert.pem
Successfully copied 1.96kB (transferred 3.58kB) to C:\Users\MANNHP\docker\labs\env\server-cert.pem
PS C:\Users\MANNHP\docker\labs\env> docker cp tls:/certs/server-key.pem ./server-key.pem
Successfully copied 3.24kB (transferred 5.12kB) to C:\Users\MANNHP\docker\labs\env\server-key.pem
PS C:\Users\MANNHP\docker\labs\env> dir *.pem
Directory: C:\Users\MANNHP\docker\labs\env
Mode LastWriteTime Length Name
-a---- 09-09-2026 15:12 1956 server-cert.pem
-a---- 09-09-2026 15:12 3243 server-key.pem
PS C:\Users\MANNHP\docker\labs\env> docker rm -f $(docker ps -aq)
e22c5ebdf72b
b7468a93bdfc
776416ebc277
adb74568e937
9cc18e120e1a
0616c3538eb0
c3c8ae04db25
PS C:\Users\MANNHP\docker\labs\env> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
PS C:\Users\MANNHP\docker\labs\env>

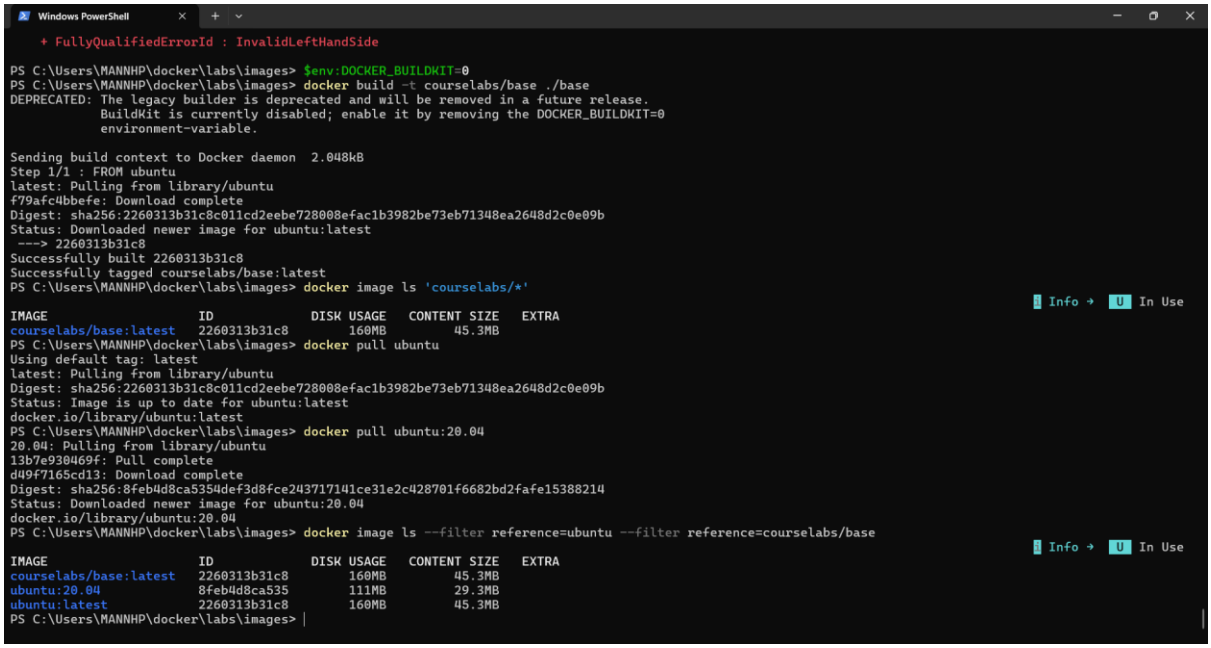

Windows PowerShell
Directory: C:\Users\MANNHP\docker\labs\env
Mode LastWriteTime Length Name
-a---- 09-09-2026 15:12 1956 server-cert.pem
-a---- 09-09-2026 15:12 3243 server-key.pem
PS C:\Users\MANNHP\docker\labs\env> docker rm -f $(docker ps -aq)
e22c5ebdf72b
b7468a93bdfc
776416ebc277
adb74568e937
9cc18e120e1a
0616c3538eb0
c3c8ae04db25
PS C:\Users\MANNHP\docker\labs\env> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
PS C:\Users\MANNHP\docker\labs\env> cd ..\images
PS C:\Users\MANNHP\docker\labs\images> dir
Directory: C:\Users\MANNHP\docker\labs\images
Mode LastWriteTime Length Name
d----- 09-09-2026 14:28 base
d----- 09-09-2026 14:28 curl
d----- 09-09-2026 14:28 java
d----- 09-09-2026 14:28 lab
d----- 09-09-2026 14:28 web
-a---- 09-09-2026 14:28 589 hints.md
-a---- 09-09-2026 14:28 6416 README.md
-a---- 09-09-2026 14:28 818 solution.md
PS C:\Users\MANNHP\docker\labs\images>

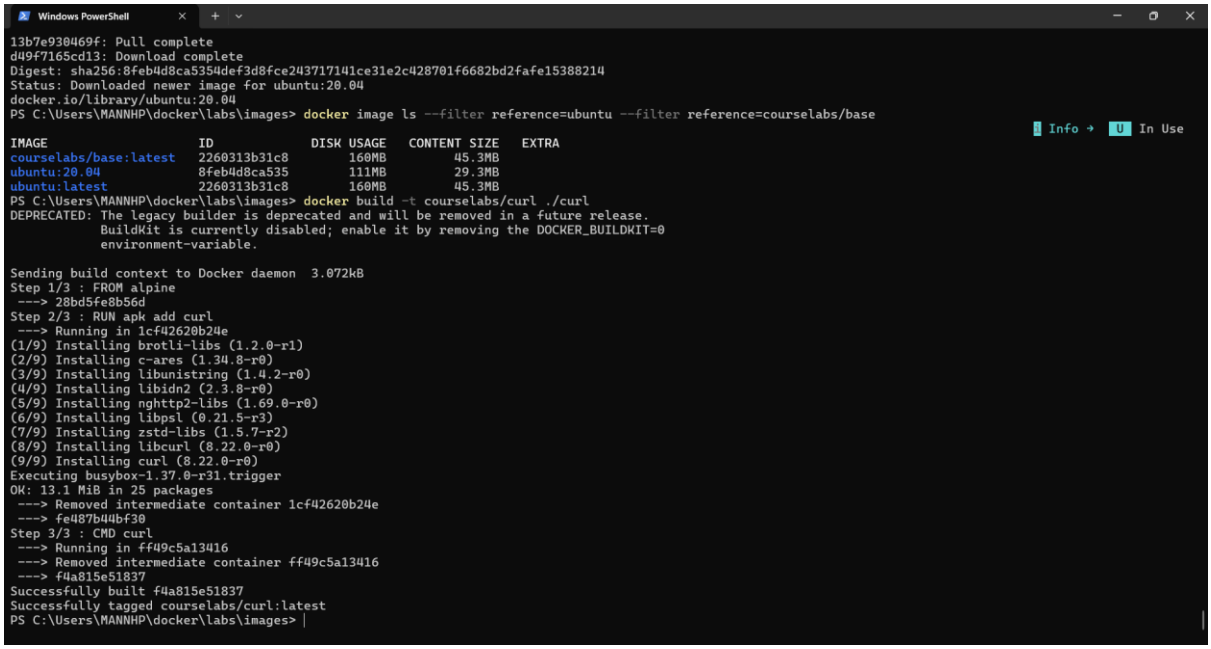

Windows PowerShell
Mode LastWriteTime Length Name
d----- 09-09-2026 14:28 base
d----- 09-09-2026 14:28 curl
d----- 09-09-2026 14:28 java
d----- 09-09-2026 14:28 lab
d----- 09-09-2026 14:28 web
-a---- 09-09-2026 14:28 589 hints.md
-a---- 09-09-2026 14:28 6416 README.md
-a---- 09-09-2026 14:28 818 solution.md
PS C:\Users\MANNHP\docker\labs\images> cat README.md
Building Container Images
Images are the packages which containers run from. You'll build an image for each component of your application, and the image has all the pre-requisites installed and configured, ready to run.
You can think of images as the filesystem for the container, plus some metadata which tells Docker which command to run when the container starts.
Reference
- [Dockerfile syntax](https://docs.docker.com/engine/reference/builder/)
- [Image build command](https://docs.docker.com/engine/reference/commandline/image_build/)
<details>
<summary>CLI overview</summary>
You use the `image` commands to work with images. The build command also has aliases:
```
docker image --help
docker build --help
docker history --help
```
</details>

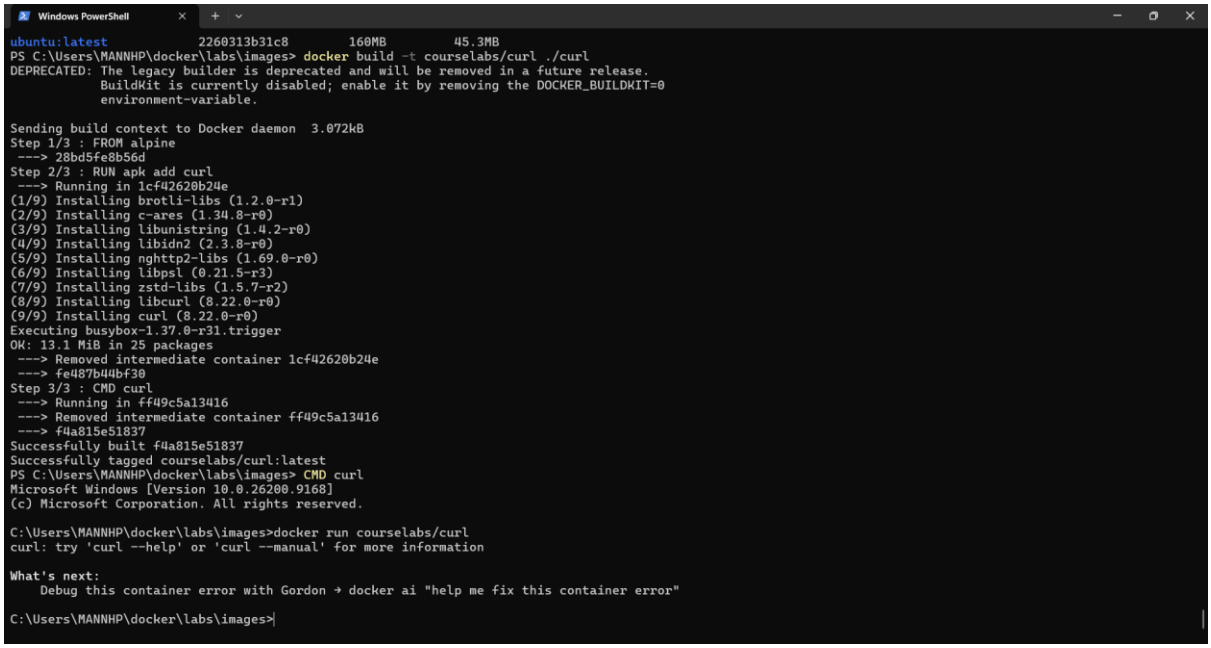

Windows PowerShell
Your turn to write a Dockerfile.
There's a simple Java app in this folder which has already been compiled into the file `labs/images/java/HelloWorld.class`.
Build a Docker image which packages that app, and run a container to confirm it's working. The command your container needs to run is `java HelloWorld`.
> Stuck? Try [hints](hints.md) or check the [solution](solution.md).

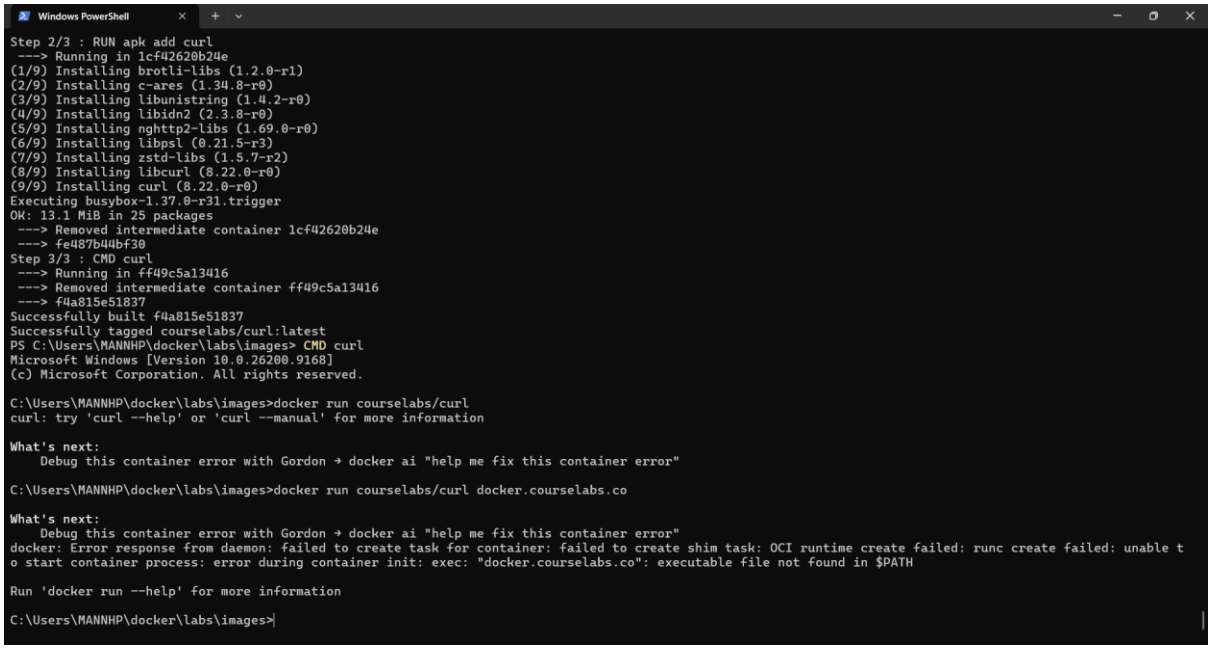
Cleanup
Cleanup by removing all containers:
```
docker rm -f $(docker ps -aq)
```
PS C:\Users\MANNHP\docker\labs\images> +$env:DOCKER_BUILDKIT=0
At line:1 char:1
+ +$env:DOCKER_BUILDKIT=0
+ ~~~~~~~~~~~~~~~~~~~~~~
The assignment expression is not valid. The input to an assignment operator must be an object that is able to accept assignments, such as a variable or a property.
+ CategoryInfo : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : InvalidLeftHandSide
PS C:\Users\MANNHP\docker\labs\images> $env:DOCKER_BUILDKIT=0
PS C:\Users\MANNHP\docker\labs\images> docker build -t courselabs/base ./base
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
BuildKit is currently disabled; enable it by removing the DOCKER_BUILDKIT=0
environment-variable.
Sending build context to Docker daemon 2.048kB
Step 1/1 : FROM ubuntu
latest: Pulling from library/ubuntu
f79afc4bbefe: Download complete
Digest: sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
Status: Downloaded newer image for ubuntu:latest
---> 2260313b31c8
Successfully built 2260313b31c8
Successfully tagged courselabs/base:latest
PS C:\Users\MANNHP\docker\labs\images>

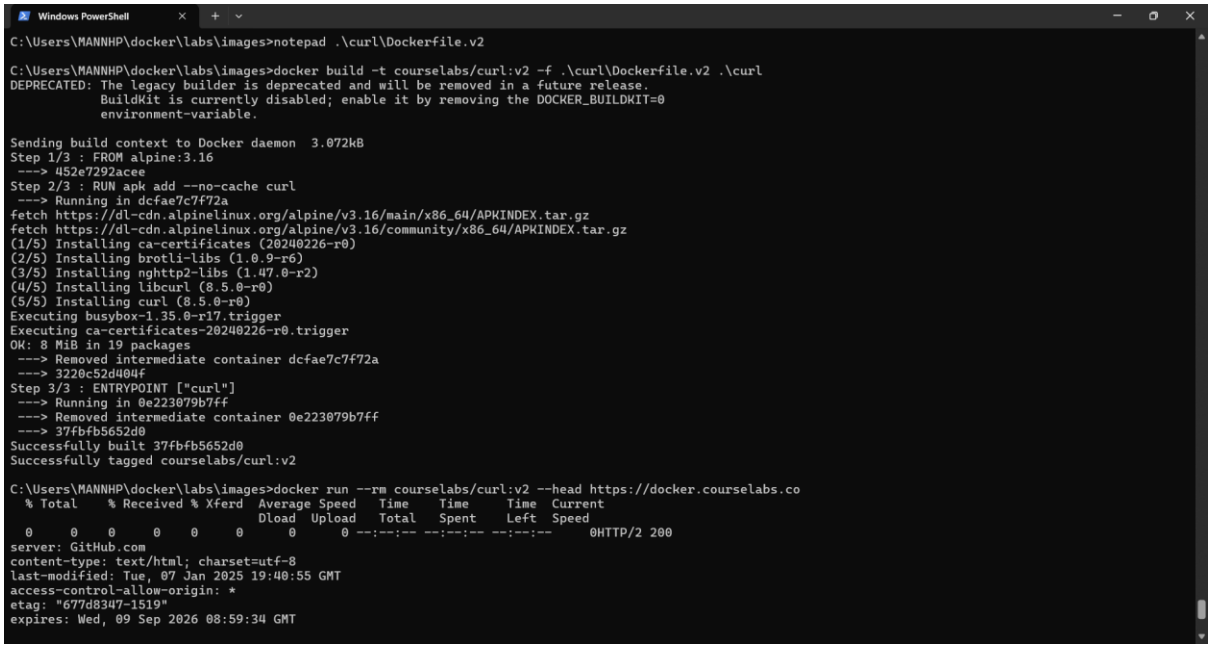

Windows PowerShell
At line:1 char:1
+ +$env:DOCKER_BUILDKIT=0
+ ~~~~~~~~~~~~~~~~~~~~~~
The assignment expression is not valid. The input to an assignment operator must be an object that is able to accept assignments, such as a variable or a property.
+ CategoryInfo : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : InvalidLeftHandSide
PS C:\Users\MANNHP\docker\labs\images> $env:DOCKER_BUILDKIT=0
PS C:\Users\MANNHP\docker\labs\images> docker build -t courselabs/base ./base
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
BuildKit is currently disabled; enable it by removing the DOCKER_BUILDKIT=0
environment-variable.
Sending build context to Docker daemon 2.048kB
Step 1/1 : FROM ubuntu
latest: Pulling from library/ubuntu
f79afc4bbefe: Download complete
Digest: sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
Status: Downloaded newer image for ubuntu:latest
---> 2260313b31c8
Successfully built 2260313b31c8
Successfully tagged courselabs/base:latest
PS C:\Users\MANNHP\docker\labs\images> docker image ls 'courselabs/*'
I Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
courselabs/base:latest 2260313b31c8 160MB 45.3MB
PS C:\Users\MANNHP\docker\labs\images> docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
PS C:\Users\MANNHP\docker\labs\images> docker pull ubuntu:20.04
20.04: Pulling from library/ubuntu
13b7e930469f: Pull complete
d49f7165cd13: Download complete
Digest: sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafe15388214
Status: Downloaded newer image for ubuntu:20.04
docker.io/library/ubuntu:20.04
PS C:\Users\MANNHP\docker\labs\images>


Windows PowerShell
+ FullyQualifiedErrorId : InvalidLeftHandSide
PS C:\Users\MANNHP\docker\labs\images> $env:DOCKER_BUILDKIT=0
PS C:\Users\MANNHP\docker\labs\images> docker build -t courselabs/base ./base
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
BuildKit is currently disabled; enable it by removing the DOCKER_BUILDKIT=0
environment-variable.
Sending build context to Docker daemon 2.048kB
Step 1/1 : FROM ubuntu
latest: Pulling from library/ubuntu
f79afc4bbefe: Download complete
Digest: sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
Status: Downloaded newer image for ubuntu:latest
---> 2260313b31c8
Successfully built 2260313b31c8
Successfully tagged courselabs/base:latest
PS C:\Users\MANNHP\docker\labs\images> docker image ls 'courselabs/*'
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
courselabs/base:latest 2260313b31c8 160MB 45.3MB
PS C:\Users\MANNHP\docker\labs\images> docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
PS C:\Users\MANNHP\docker\labs\images> docker pull ubuntu:20.04
20.04: Pulling from library/ubuntu
13b7e930469f: Pull complete
d49f7165cd13: Download complete
Digest: sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafe15388214
Status: Downloaded newer image for ubuntu:20.04
docker.io/library/ubuntu:20.04
PS C:\Users\MANNHP\docker\labs\images> docker image ls --filter reference=ubuntu --filter reference=courselabs/base
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
courselabs/base:latest 2260313b31c8 160MB 45.3MB
ubuntu:20.04 8feb4d8ca535 111MB 29.3MB
ubuntu:latest 2260313b31c8 160MB 45.3MB
PS C:\Users\MANNHP\docker\labs\images>


Windows PowerShell
13b7e930469f: Pull complete
d49f7165cd13: Download complete
Digest: sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafe15388214
Status: Downloaded newer image for ubuntu:20.04
docker.io/library/ubuntu:20.04
PS C:\Users\MANNHP\docker\labs\images> docker image ls --filter reference=ubuntu --filter reference=courselabs/base
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
courselabs/base:latest 2260313b31c8 160MB 45.3MB
ubuntu:20.04 8feb4d8ca535 111MB 29.3MB
ubuntu:latest 2260313b31c8 160MB 45.3MB
PS C:\Users\MANNHP\docker\labs\images> docker build -t courselabs/curl ./curl
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
BuildKit is currently disabled; enable it by removing the DOCKER_BUILDKIT=0
environment-variable.
Sending build context to Docker daemon 3.072kB
Step 1/3 : FROM alpine
---> 28bd5fe8b56d
Step 2/3 : RUN apk add curl
---> Running in 1cf42620b24e
(1/9) Installing brotli-libs (1.2.0-r1)
(2/9) Installing c-ares (1.34.8-r0)
(3/9) Installing libunistring (1.4.2-r0)
(4/9) Installing libidn2 (2.3.8-r0)
(5/9) Installing nghttp2-libs (1.69.0-r0)
(6/9) Installing libpsl (0.21.5-r3)
(7/9) Installing zstd-libs (1.5.7-r2)
(8/9) Installing libcurl (8.22.0-r0)
(9/9) Installing curl (8.22.0-r0)
Executing busybox-1.37.0-r31.trigger
OK: 13.1 MiB in 25 packages
---> Removed intermediate container 1cf42620b24e
---> fe487b44bf30
Step 3/3 : CMD curl
---> Running in ff49c5a13416
---> Removed intermediate container ff49c5a13416
---> f4a815e51837
Successfully built f4a815e51837
Successfully tagged courselabs/curl:latest
PS C:\Users\MANNHP\docker\labs\images>


Windows PowerShell
ubuntu:latest 2260313b31c8 160MB 45.3MB
PS C:\Users\MANNHP\docker\labs\images> docker build -t courselabs/curl ./curl
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
BuildKit is currently disabled; enable it by removing the DOCKER_BUILDKIT=0
environment-variable.
Sending build context to Docker daemon 3.072kB
Step 1/3 : FROM alpine
---> 28bd5fe8b56d
Step 2/3 : RUN apk add curl
---> Running in 1cf42620b24e
(1/9) Installing brotli-libs (1.2.0-r1)
(2/9) Installing c-ares (1.34.8-r0)
(3/9) Installing libunistring (1.4.2-r0)
(4/9) Installing libidn2 (2.3.8-r0)
(5/9) Installing nghttp2-libs (1.69.0-r0)
(6/9) Installing libpsl (0.21.5-r3)
(7/9) Installing zstd-libs (1.5.7-r2)
(8/9) Installing libcurl (8.22.0-r0)
(9/9) Installing curl (8.22.0-r0)
Executing busybox-1.37.0-r31.trigger
OK: 13.1 MiB in 25 packages
---> Removed intermediate container 1cf42620b24e
---> fe487b44bf30
Step 3/3 : CMD curl
---> Running in ff49c5a13416
---> Removed intermediate container ff49c5a13416
---> f4a815e51837
Successfully built f4a815e51837
Successfully tagged courselabs/curl:latest
PS C:\Users\MANNHP\docker\labs\images> CMD curl
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.
C:\Users\MANNHP\docker\labs\images>docker run courselabs/curl
curl: try 'curl --help' or 'curl --manual' for more information
What's next:
Debug this container error with Gordon → docker ai "help me fix this container error"
C:\Users\MANNHP\docker\labs\images>


Windows PowerShell
Step 2/3 : RUN apk add curl
---> Running in 1cf42620b24e
(1/9) Installing brotli-libs (1.2.0-r1)
(2/9) Installing c-ares (1.34.8-r0)
(3/9) Installing libunistring (1.4.2-r0)
(4/9) Installing libidn2 (2.3.8-r0)
(5/9) Installing nghttp2-libs (1.69.0-r0)
(6/9) Installing libpsl (0.21.5-r3)
(7/9) Installing zstd-libs (1.5.7-r2)
(8/9) Installing libcurl (8.22.0-r0)
(9/9) Installing curl (8.22.0-r0)
Executing busybox-1.37.0-r31.trigger
OK: 13.1 MiB in 25 packages
---> Removed intermediate container 1cf42620b24e
---> fe487b44bf30
Step 3/3 : CMD curl
---> Running in ff49c5a13416
---> Removed intermediate container ff49c5a13416
---> f4a815e51837
Successfully built f4a815e51837
Successfully tagged courselabs/curl:latest
PS C:\Users\MANNHP\docker\labs\images> CMD curl
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.
C:\Users\MANNHP\docker\labs\images>docker run courselabs/curl
curl: try 'curl --help' or 'curl --manual' for more information
What's next:
Debug this container error with Gordon → docker ai "help me fix this container error"
C:\Users\MANNHP\docker\labs\images>docker run courselabs/curl docker.courselabs.co
What's next:
Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: failed to create task for container: failed to create shim task: OCI runtime create failed: runc create failed: unable t
o start container process: error during container init: exec: "docker.courselabs.co": executable file not found in $PATH
Run 'docker run --help' for more information
C:\Users\MANNHP\docker\labs\images>



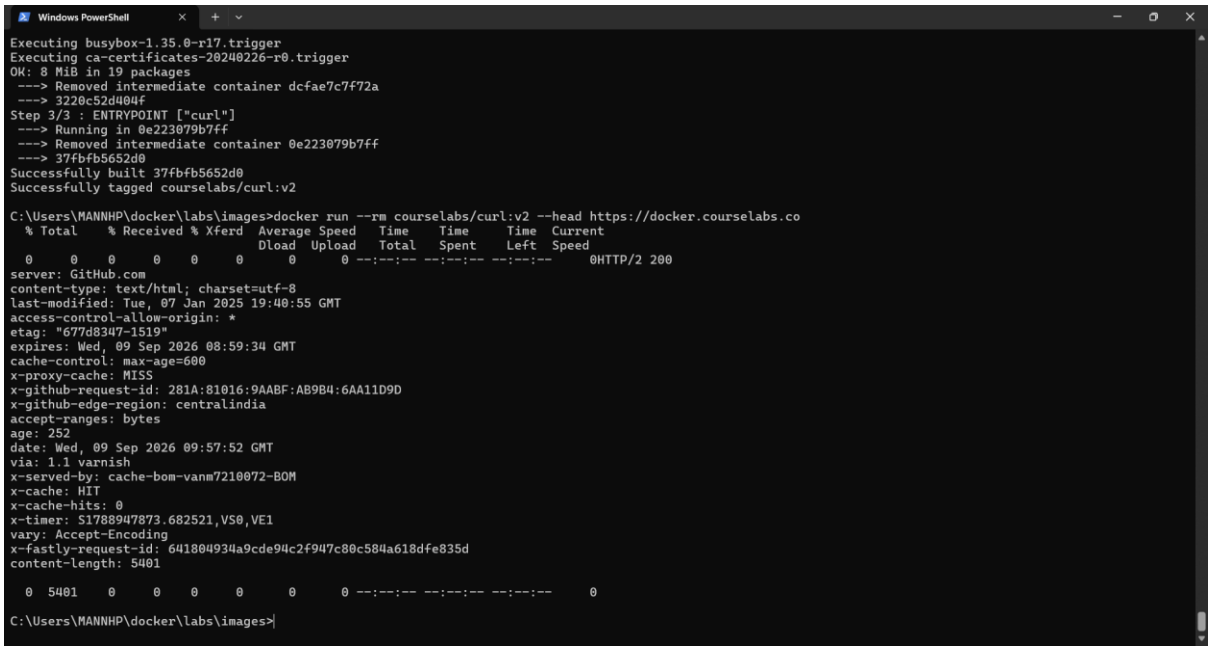
```
Windows PowerShell

C:\Users\MANNHP\docker\labs\images>notepad .\curl\Dockerfile.v2

C:\Users\MANNHP\docker\labs\images>docker build -t courselabs/curl:v2 -f .\curl\Dockerfile.v2 .\curl
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            BuildKit is currently disabled; enable it by removing the DOCKER_BUILDKIT=0
            environment-variable.

Sending build context to Docker daemon  3.072kB
Step 1/3 : FROM alpine:3.16
 ---> 452e7292acee
Step 2/3 : RUN apk add --no-cache curl
 ---> Running in dcfae7c7f72a
fetch https://dl-cdn.alpinelinux.org/alpine/v3.16/main/x86_64/APKINDEX.tar.gz
fetch https://dl-cdn.alpinelinux.org/alpine/v3.16/community/x86_64/APKINDEX.tar.gz
(1/5) Installing ca-certificates (20240226-r0)
(2/5) Installing brotli-libs (1.0.9-r6)
(3/5) Installing nghttp2-libs (1.47.0-r2)
(4/5) Installing libcurl (8.5.0-r0)
(5/5) Installing curl (8.5.0-r0)
Executing busybox-1.35.0-r17.trigger
Executing ca-certificates-20240226-r0.trigger
OK: 8 MiB in 19 packages
 ---> Removed intermediate container dcfae7c7f72a
 ---> 3220c52d404f
Step 3/3 : ENTRYPOINT ["curl"]
 ---> Running in 0e223079b7ff
 ---> Removed intermediate container 0e223079b7ff
 ---> 37fbfb5652d0
Successfully built 37fbfb5652d0
Successfully tagged courselabs/curl:v2

C:\Users\MANNHP\docker\labs\images>docker run --rm courselabs/curl:v2 --head https://docker.courselabs.co
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0HTTP/2 200
server: GitHub.com
content-type: text/html; charset=utf-8
last-modified: Tue, 07 Jan 2025 19:40:55 GMT
access-control-allow-origin: *
etag: "677d8347-1519"
expires: Wed, 09 Sep 2026 08:59:34 GMT
```



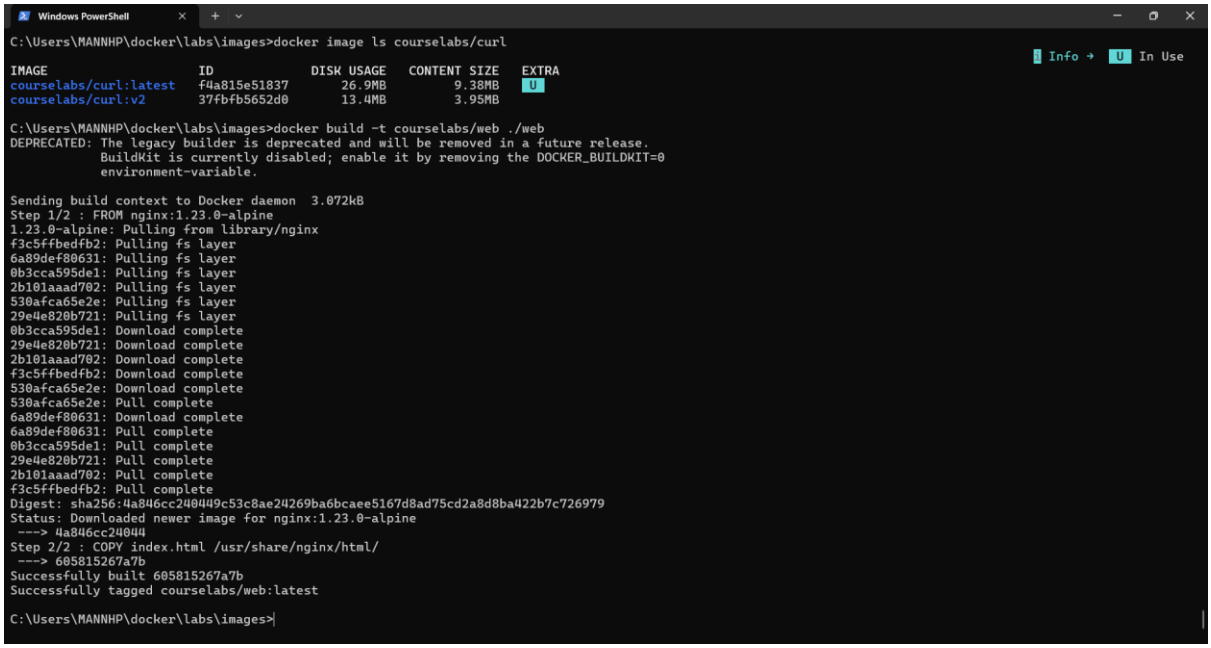
```
Windows PowerShell

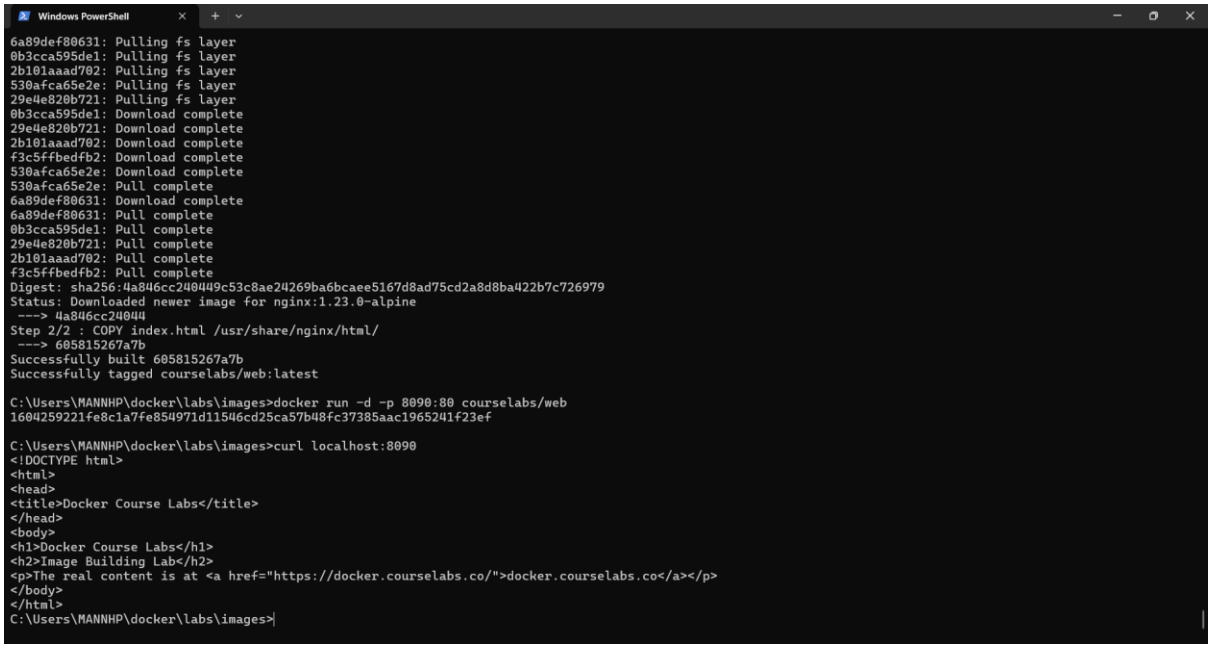
Executing busybox-1.35.0-r17.trigger
Executing ca-certificates-20240226-r0.trigger
OK: 8 MiB in 19 packages
 ---> Removed intermediate container dcfae7c7f72a
 ---> 3220c52d404f
Step 3/3 : ENTRYPOINT ["curl"]
 ---> Running in 0e223079b7ff
 ---> Removed intermediate container 0e223079b7ff
 ---> 37fbfb5652d0
Successfully built 37fbfb5652d0
Successfully tagged courselabs/curl:v2

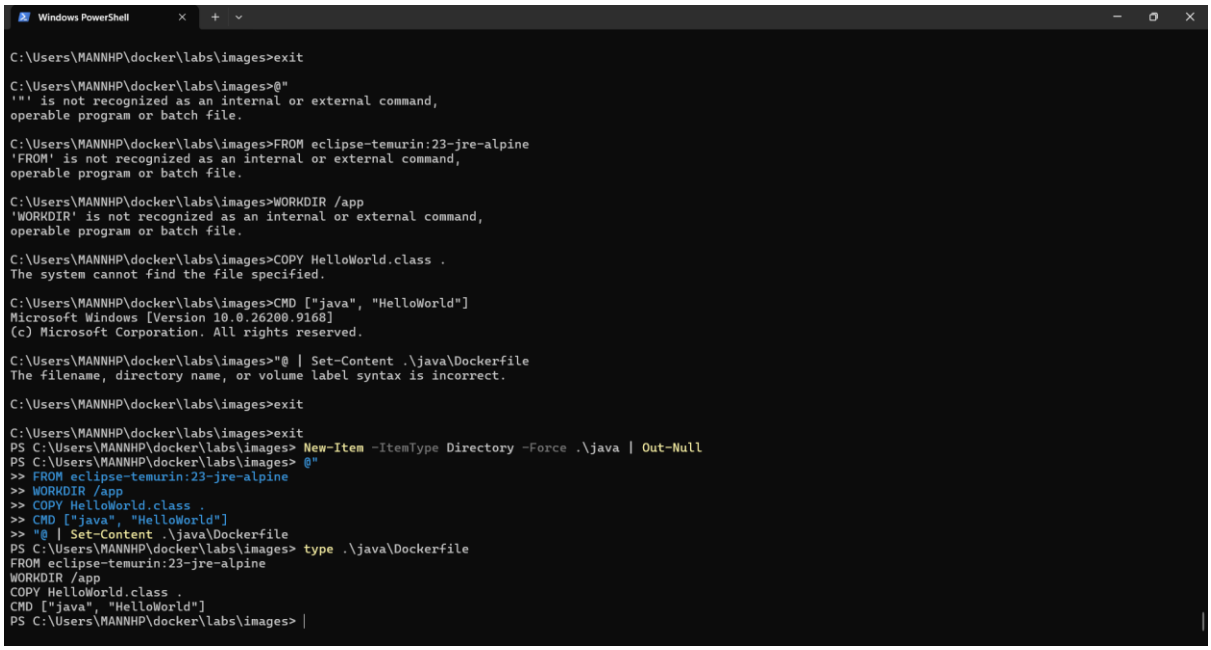
C:\Users\MANNHP\docker\labs\images>docker run --rm courselabs/curl:v2 --head https://docker.courselabs.co
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0HTTP/2 200
server: GitHub.com
content-type: text/html; charset=utf-8
last-modified: Tue, 07 Jan 2025 19:40:55 GMT
access-control-allow-origin: *
etag: "677d8347-1519"
expires: Wed, 09 Sep 2026 08:59:34 GMT
cache-control: max-age=600
x-proxy-cache: MISS
x-github-request-id: 281A:81016:9AABF:AB9B4:6AA11D9D
x-github-edge-region: centralindia
accept-ranges: bytes
age: 252
date: Wed, 09 Sep 2026 09:57:52 GMT
via: 1.1 varnish
x-served-by: cache-bom-vanm7210072-BOM
x-cache: HIT
x-cache-hits: 0
x-timer: S1788947873.682521,VS0,VE1
vary: Accept-Encoding
x-fastly-request-id: 641804934a9cde94c2f947c80c584a618dfe835d
content-length: 5401

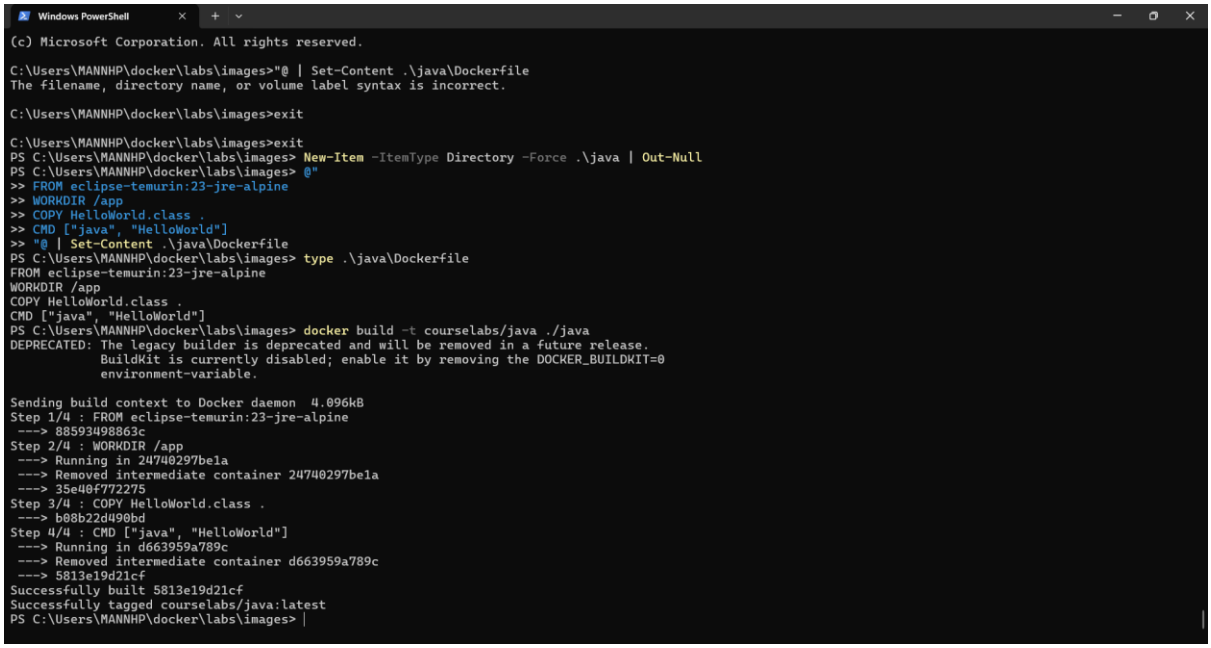
  0  5401    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0

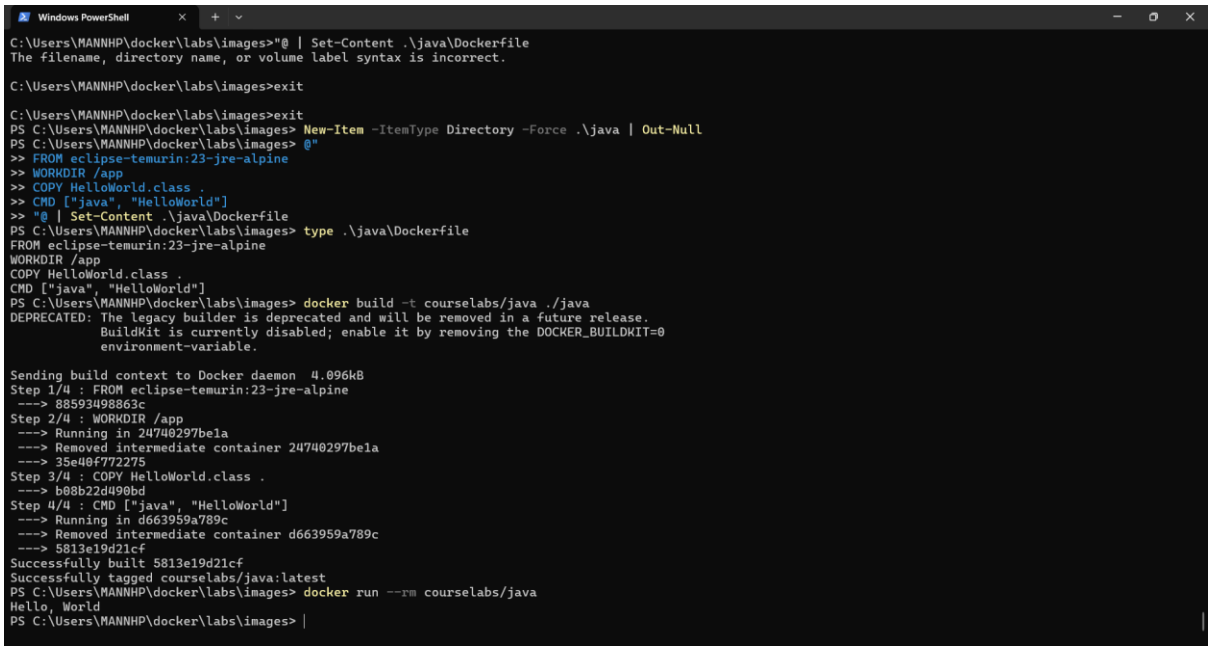
C:\Users\MANNHP\docker\labs\images>
```

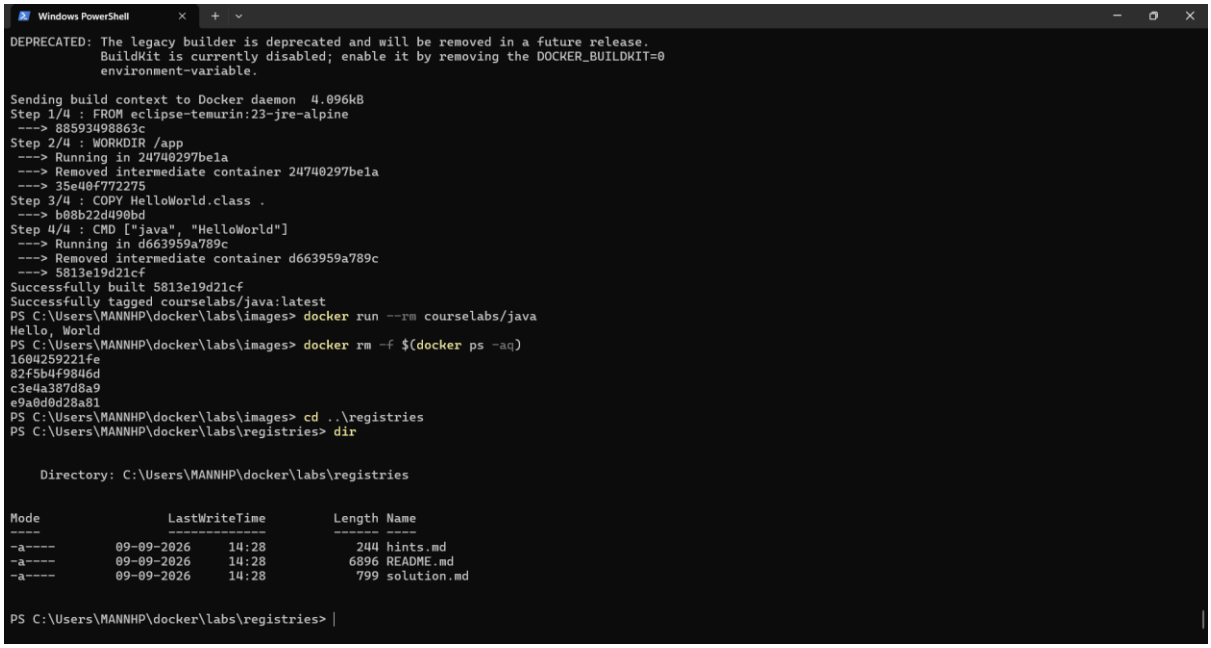

Windows PowerShell
C:\Users\MANNHP\docker\labs\images>docker image ls courselabs/curl
Info →
U
In Use
IMAGE
ID
DISK USAGE
CONTENT SIZE
EXTRA
courselabs/curl:latest
f4a815e51837
26.9MB
9.38MB
U
courselabs/curl:v2
37fbfb5652d0
13.4MB
3.95MB
C:\Users\MANNHP\docker\labs\images>docker build -t courselabs/web ./web
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
BuildKit is currently disabled; enable it by removing the DOCKER_BUILDKIT=0
environment-variable.
Sending build context to Docker daemon 3.072kB
Step 1/2 : FROM nginx:1.23.0-alpine
1.23.0-alpine: Pulling from library/nginx
f3c5ffbedfb2: Pulling fs layer
6a89def80631: Pulling fs layer
0b3cca595de1: Pulling fs layer
2b101aaad702: Pulling fs layer
530afca65e2e: Pulling fs layer
29e4e820b721: Pulling fs layer
0b3cca595de1: Download complete
29e4e820b721: Download complete
2b101aaad702: Download complete
f3c5ffbedfb2: Download complete
530afca65e2e: Download complete
530afca65e2e: Pull complete
6a89def80631: Download complete
6a89def80631: Pull complete
0b3cca595de1: Pull complete
29e4e820b721: Pull complete
2b101aaad702: Pull complete
f3c5ffbedfb2: Pull complete
Digest: sha256:4a846cc240449c53c8ae24269ba6bcaee5167d8ad75cd2a8d8ba422b7c726979
Status: Downloaded newer image for nginx:1.23.0-alpine
---> 4a846cc24044
Step 2/2 : COPY index.html /usr/share/nginx/html/
---> 605815267a7b
Successfully built 605815267a7b
Successfully tagged courselabs/web:latest
C:\Users\MANNHP\docker\labs\images>

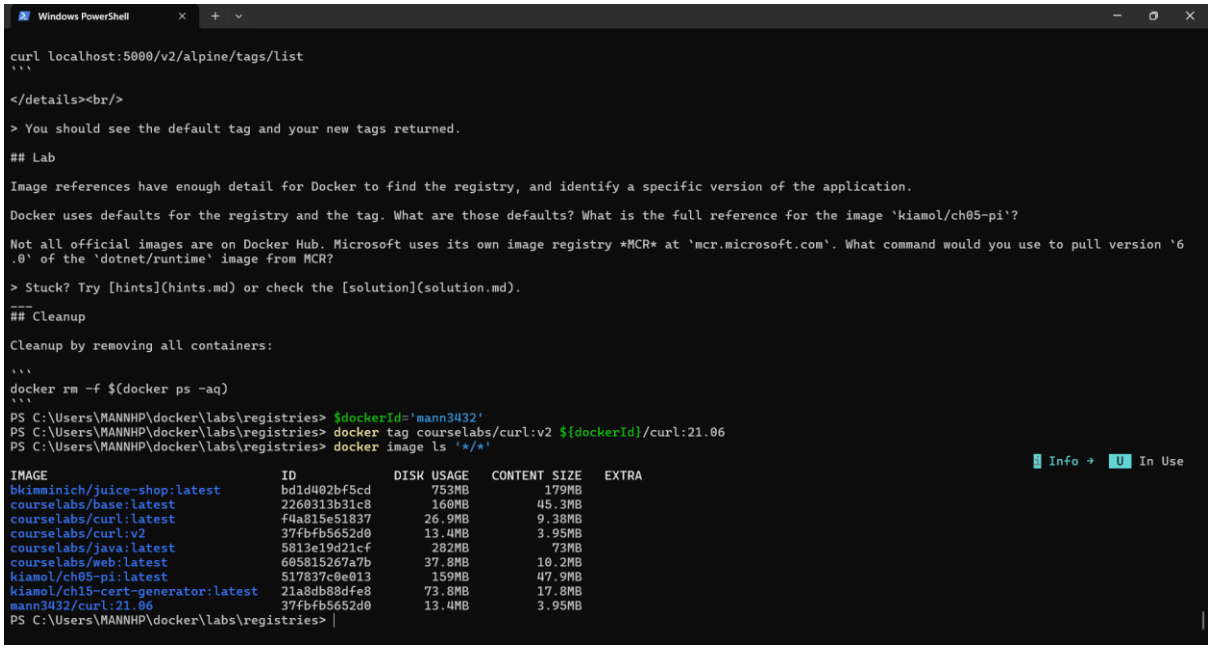

Windows PowerShell
6a89def80631: Pulling fs layer
0b3cca595de1: Pulling fs layer
2b101aaad702: Pulling fs layer
530afca65e2e: Pulling fs layer
29e4e820b721: Pulling fs layer
0b3cca595de1: Download complete
29e4e820b721: Download complete
2b101aaad702: Download complete
f3c5ffbedfb2: Download complete
530afca65e2e: Download complete
530afca65e2e: Pull complete
6a89def80631: Download complete
6a89def80631: Pull complete
0b3cca595de1: Pull complete
29e4e820b721: Pull complete
2b101aaad702: Pull complete
f3c5ffbedfb2: Pull complete
Digest: sha256:4a846cc240449c53c8ae24269ba6bcaee5167d8ad75cd2a8d8ba422b7c726979
Status: Downloaded newer image for nginx:1.23.0-alpine
---> 4a846cc24044
Step 2/2 : COPY index.html /usr/share/nginx/html/
---> 605815267a7b
Successfully built 605815267a7b
Successfully tagged courselabs/web:latest
C:\Users\MANNHP\docker\labs\images>docker run -d -p 8090:80 courselabs/web
1604259221fe8c1a7fe854971d11546cd25ca57b48fc37385aac1965241f23ef
C:\Users\MANNHP\docker\labs\images>curl localhost:8090
<!DOCTYPE html>
<html>
<head>
<title>Docker Course Labs</title>
</head>
<body>
<h1>Docker Course Labs</h1>
<h2>Image Building Lab</h2>
The real content is at docker.courselabs.co
</body>
</html>
C:\Users\MANNHP\docker\labs\images>

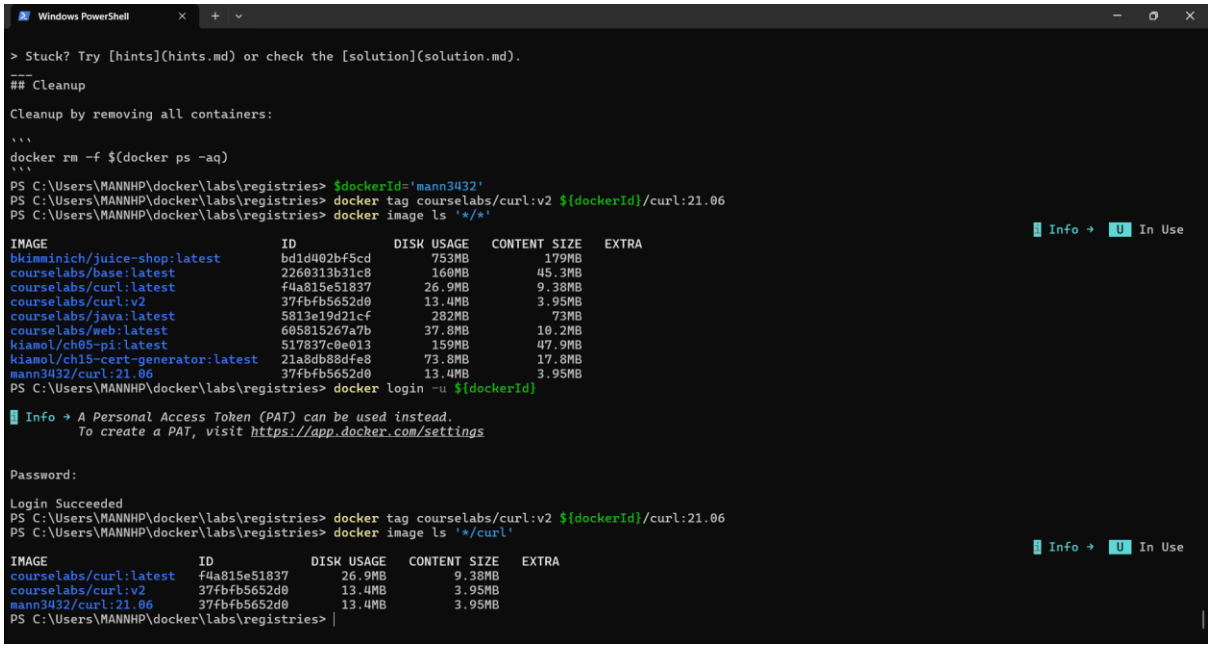

Windows PowerShell
C:\Users\MANNHP\docker\labs\images>exit
C:\Users\MANNHP\docker\labs\images>@"
'"' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\MANNHP\docker\labs\images>FROM eclipse-temurin:23-jre-alpine
'FROM' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\MANNHP\docker\labs\images>WORKDIR /app
'WORKDIR' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\MANNHP\docker\labs\images>COPY HelloWorld.class .
The system cannot find the file specified.
C:\Users\MANNHP\docker\labs\images>CMD ["java", "HelloWorld"]
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.
C:\Users\MANNHP\docker\labs\images>"@ | Set-Content .\java\Dockerfile
The filename, directory name, or volume label syntax is incorrect.
C:\Users\MANNHP\docker\labs\images>exit
C:\Users\MANNHP\docker\labs\images>exit
PS C:\Users\MANNHP\docker\labs\images> New-Item -ItemType Directory -Force .\java | Out-Null
PS C:\Users\MANNHP\docker\labs\images> @"
>> FROM eclipse-temurin:23-jre-alpine
>> WORKDIR /app
>> COPY HelloWorld.class .
>> CMD ["java", "HelloWorld"]
>> "@ | Set-Content .\java\Dockerfile
PS C:\Users\MANNHP\docker\labs\images> type .\java\Dockerfile
FROM eclipse-temurin:23-jre-alpine
WORKDIR /app
COPY HelloWorld.class .
CMD ["java", "HelloWorld"]
PS C:\Users\MANNHP\docker\labs\images>


Windows PowerShell
(c) Microsoft Corporation. All rights reserved.
C:\Users\MANNHP\docker\labs\images>"@ | Set-Content .\java\Dockerfile
The filename, directory name, or volume label syntax is incorrect.
C:\Users\MANNHP\docker\labs\images>exit
C:\Users\MANNHP\docker\labs\images>exit
PS C:\Users\MANNHP\docker\labs\images> New-Item -ItemType Directory -Force .\java | Out-Null
PS C:\Users\MANNHP\docker\labs\images> @"
>> FROM eclipse-temurin:23-jre-alpine
>> WORKDIR /app
>> COPY HelloWorld.class .
>> CMD ["java", "HelloWorld"]
>> "@ | Set-Content .\java\Dockerfile
PS C:\Users\MANNHP\docker\labs\images> type .\java\Dockerfile
FROM eclipse-temurin:23-jre-alpine
WORKDIR /app
COPY HelloWorld.class .
CMD ["java", "HelloWorld"]
PS C:\Users\MANNHP\docker\labs\images> docker build -t courselabs/java ./java
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
BuildKit is currently disabled; enable it by removing the DOCKER_BUILDKIT=0
environment-variable.
Sending build context to Docker daemon 4.096kB
Step 1/4 : FROM eclipse-temurin:23-jre-alpine
---> 88593498863c
Step 2/4 : WORKDIR /app
---> Running in 24740297be1a
---> Removed intermediate container 24740297be1a
---> 35e40f772275
Step 3/4 : COPY HelloWorld.class .
---> b08b22d490bd
Step 4/4 : CMD ["java", "HelloWorld"]
---> Running in d663959a789c
---> Removed intermediate container d663959a789c
---> 5813e19d21cf
Successfully built 5813e19d21cf
Successfully tagged courselabs/java:latest
PS C:\Users\MANNHP\docker\labs\images>

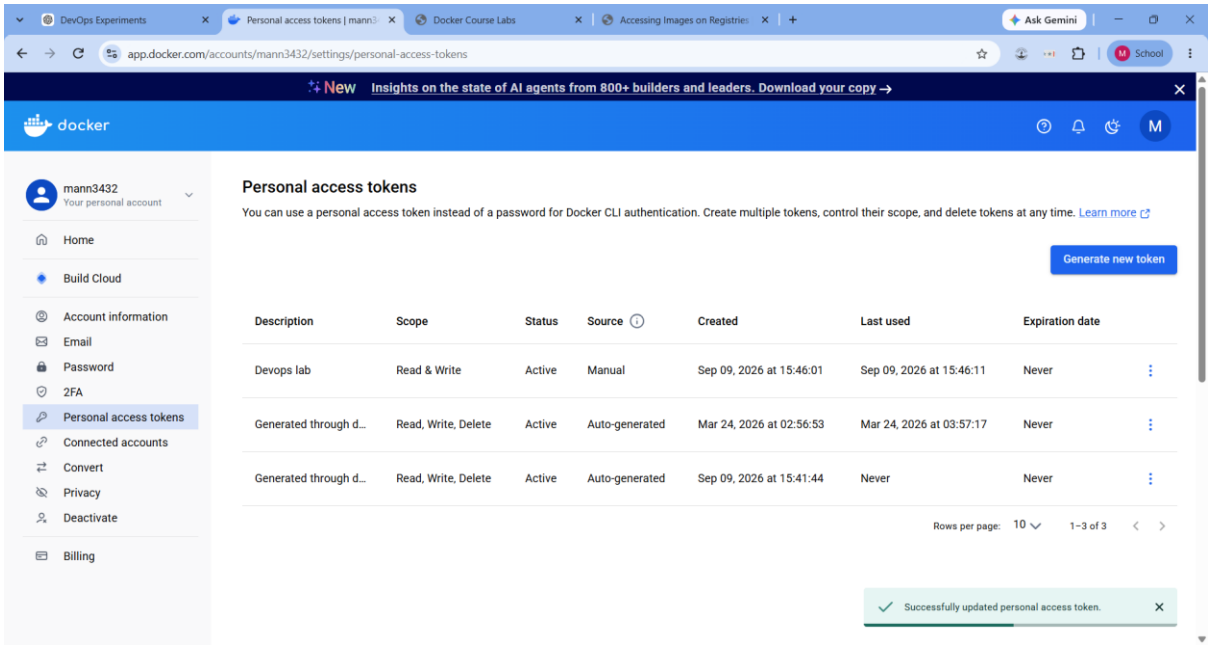

Windows PowerShell
C:\Users\MANNHP\docker\labs\images>"@ | Set-Content .\java\Dockerfile
The filename, directory name, or volume label syntax is incorrect.
C:\Users\MANNHP\docker\labs\images>exit
C:\Users\MANNHP\docker\labs\images>exit
PS C:\Users\MANNHP\docker\labs\images> New-Item -ItemType Directory -Force .\java | Out-Null
PS C:\Users\MANNHP\docker\labs\images> @"
>> FROM eclipse-temurin:23-jre-alpine
>> WORKDIR /app
>> COPY HelloWorld.class .
>> CMD ["java", "HelloWorld"]
>> "@ | Set-Content .\java\Dockerfile
PS C:\Users\MANNHP\docker\labs\images> type .\java\Dockerfile
FROM eclipse-temurin:23-jre-alpine
WORKDIR /app
COPY HelloWorld.class .
CMD ["java", "HelloWorld"]
PS C:\Users\MANNHP\docker\labs\images> docker build -t courselabs/java ./java
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
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Sending build context to Docker daemon 4.096kB
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---> Running in 24740297be1a
---> Removed intermediate container 24740297be1a
---> 35e40f772275
Step 3/4 : COPY HelloWorld.class .
---> b08b22d490bd
Step 4/4 : CMD ["java", "HelloWorld"]
---> Running in d663959a789c
---> Removed intermediate container d663959a789c
---> 5813e19d21cf
Successfully built 5813e19d21cf
Successfully tagged courselabs/java:latest
PS C:\Users\MANNHP\docker\labs\images> docker run --rm courselabs/java
Hello, World
PS C:\Users\MANNHP\docker\labs\images>

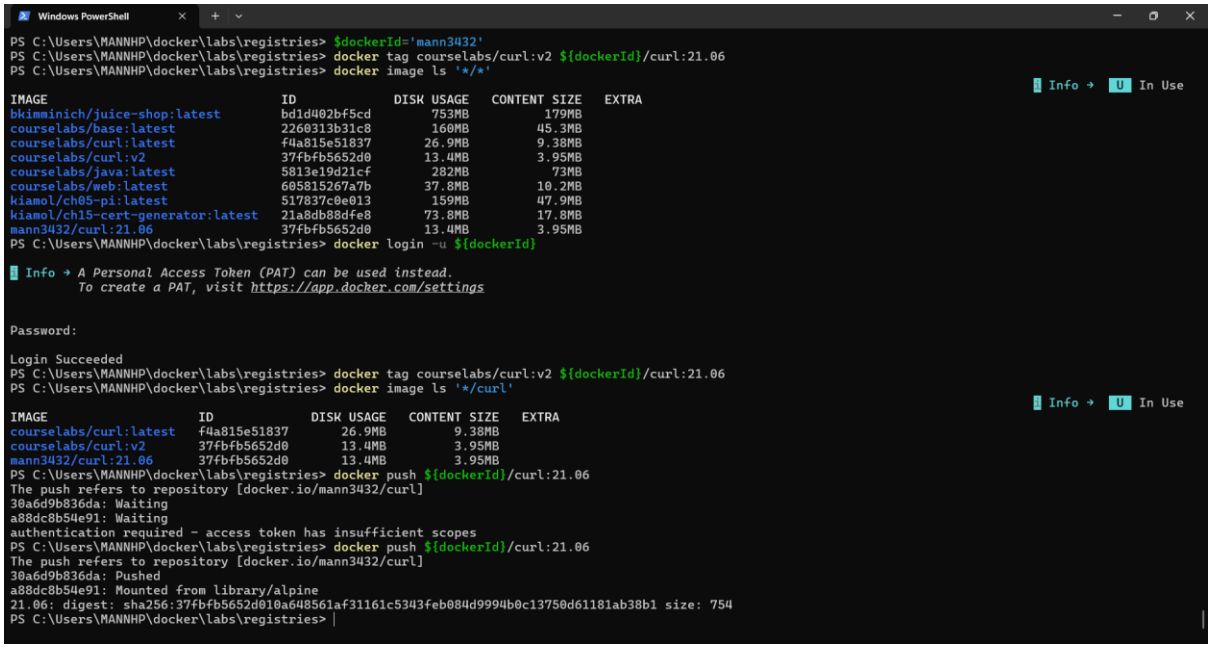

Windows PowerShell
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
BuildKit is currently disabled; enable it by removing the DOCKER_BUILDKIT=0
environment-variable.
Sending build context to Docker daemon 4.096kB
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---> 88593498863c
Step 2/4 : WORKDIR /app
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---> 35e40f772275
Step 3/4 : COPY HelloWorld.class .
---> b08b22d490bd
Step 4/4 : CMD ["java", "HelloWorld"]
---> Running in d663959a789c
---> Removed intermediate container d663959a789c
---> 5813e19d21cf
Successfully built 5813e19d21cf
Successfully tagged courselabs/java:latest
PS C:\Users\MANNHP\docker\labs\images> docker run --rm courselabs/java
Hello, World
PS C:\Users\MANNHP\docker\labs\images> docker rm -f $(docker ps -aq)
1604259221fe
82f5b4f9846d
c3e4a387d8a9
e9a0d0d28a81
PS C:\Users\MANNHP\docker\labs\images> cd ..\registries
PS C:\Users\MANNHP\docker\labs\registries> dir
Directory: C:\Users\MANNHP\docker\labs\registries
Mode LastWriteTime Length Name
-a---- 09-09-2026 14:28 244 hints.md
-a---- 09-09-2026 14:28 6896 README.md
-a---- 09-09-2026 14:28 799 solution.md
PS C:\Users\MANNHP\docker\labs\registries>

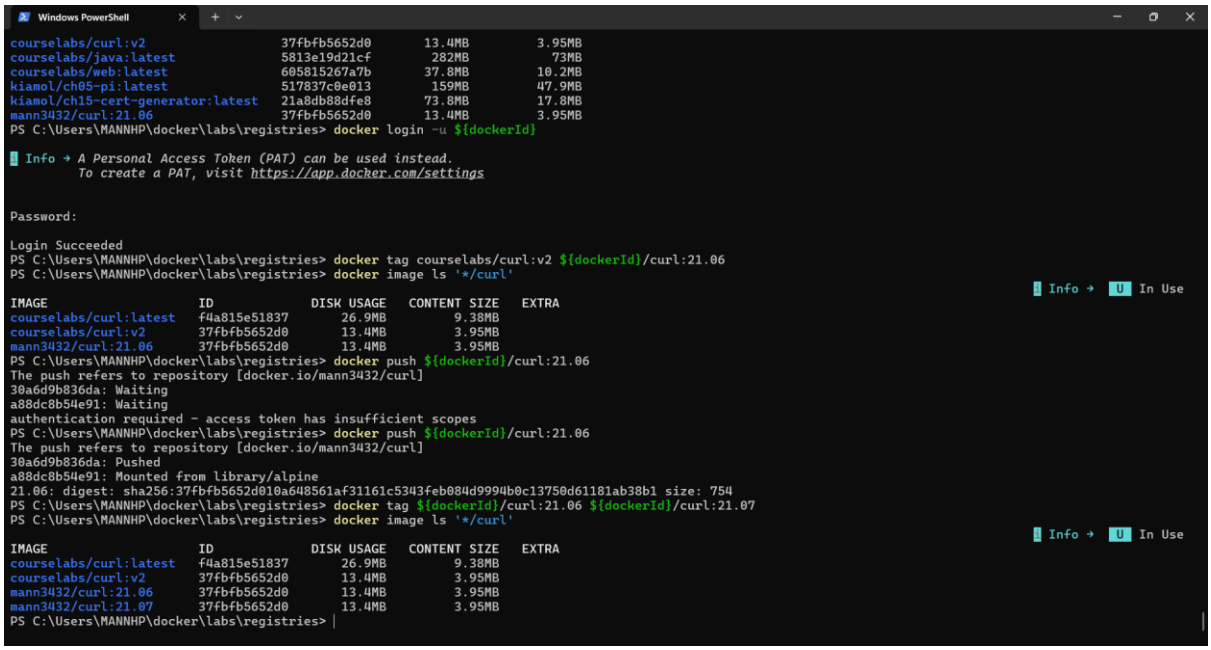

Windows PowerShell
curl localhost:5000/v2/alpine/tags/list
```
</details><br/>
> You should see the default tag and your new tags returned.
## Lab
Image references have enough detail for Docker to find the registry, and identify a specific version of the application.
Docker uses defaults for the registry and the tag. What are those defaults? What is the full reference for the image `kiamol/ch05-pi`?
Not all official images are on Docker Hub. Microsoft uses its own image registry *MCR* at `mcr.microsoft.com`. What command would you use to pull version `6.0` of the `dotnet/runtime` image from MCR?
> Stuck? Try [hints](hints.md) or check the [solution](solution.md).
---
## Cleanup
Cleanup by removing all containers:
```
docker rm -f $(docker ps -aq)
```
PS C:\Users\MANNHP\docker\labs\registries> $dockerId='mann3432'
PS C:\Users\MANNHP\docker\labs\registries> docker tag courselabs/curl:v2 ${dockerId}/curl:21.06
PS C:\Users\MANNHP\docker\labs\registries> docker image ls '*/*'
I Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
bkimminich/juice-shop:latest bd1d402bf5cd 753MB 179MB
courselabs/base:latest 2260313b31c8 160MB 45.3MB
courselabs/curl:latest f4a815e51837 26.9MB 9.38MB
courselabs/curl:v2 37fbfb5652d0 13.4MB 3.95MB
courselabs/java:latest 5813e19d21cf 282MB 73MB
courselabs/web:latest 605815267a7b 37.8MB 10.2MB
kiamol/ch05-pi:latest 517837c0e013 159MB 47.9MB
kiamol/ch15-cert-generator:latest 21a8db88dfe8 73.8MB 17.8MB
mann3432/curl:21.06 37fbfb5652d0 13.4MB 3.95MB
PS C:\Users\MANNHP\docker\labs\registries>
```

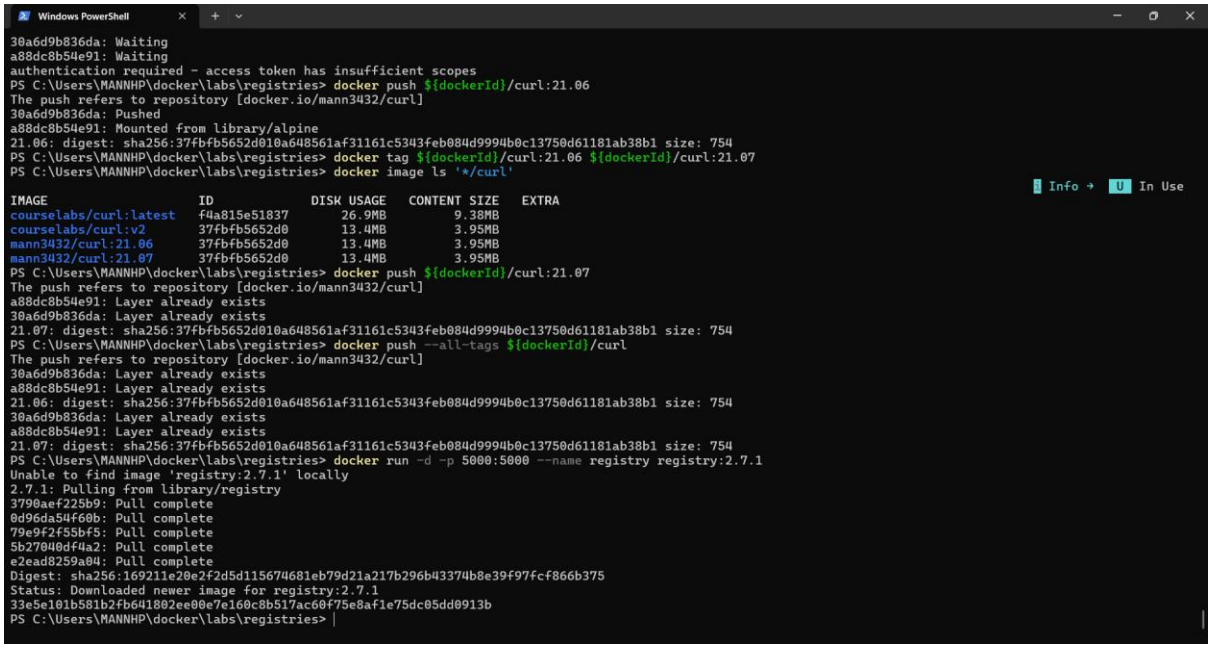

Windows PowerShell
> Stuck? Try [hints](hints.md) or check the [solution](solution.md).

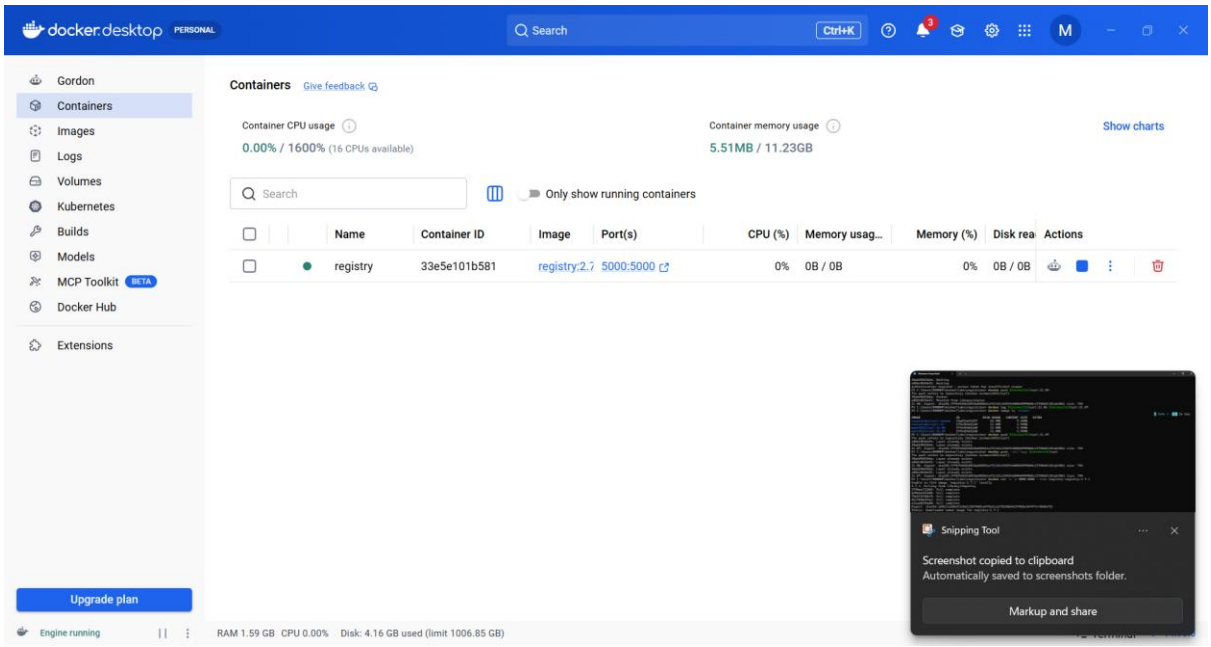
Cleanup
Cleanup by removing all containers:
```
docker rm -f $(docker ps -aq)
```
PS C:\Users\MANNHP\docker\labs\registries> $dockerId='mann3432'
PS C:\Users\MANNHP\docker\labs\registries> docker tag courselabs/curl:v2 ${dockerId}/curl:21.06
PS C:\Users\MANNHP\docker\labs\registries> docker image ls '*/*'
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
bkimminich/juice-shop:latest bd1d402bf5cd 753MB 179MB
courselabs/base:latest 2260313b31c8 160MB 45.3MB
courselabs/curl:latest f4a815e51837 26.9MB 9.38MB
courselabs/curl:v2 37fbfb5652d0 13.4MB 3.95MB
courselabs/java:latest 5813e19d21cf 282MB 73MB
courselabs/web:latest 605815267a7b 37.8MB 10.2MB
kiamol/ch05-pi:latest 517837c0e013 159MB 47.9MB
kiamol/ch15-cert-generator:latest 21a8db88dfe8 73.8MB 17.8MB
mann3432/curl:21.06 37fbfb5652d0 13.4MB 3.95MB
PS C:\Users\MANNHP\docker\labs\registries> docker login -u ${dockerId}
Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings
Password:
Login Succeeded
PS C:\Users\MANNHP\docker\labs\registries> docker tag courselabs/curl:v2 ${dockerId}/curl:21.06
PS C:\Users\MANNHP\docker\labs\registries> docker image ls '*/curl'
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
courselabs/curl:latest f4a815e51837 26.9MB 9.38MB
courselabs/curl:v2 37fbfb5652d0 13.4MB 3.95MB
mann3432/curl:21.06 37fbfb5652d0 13.4MB 3.95MB
PS C:\Users\MANNHP\docker\labs\registries>

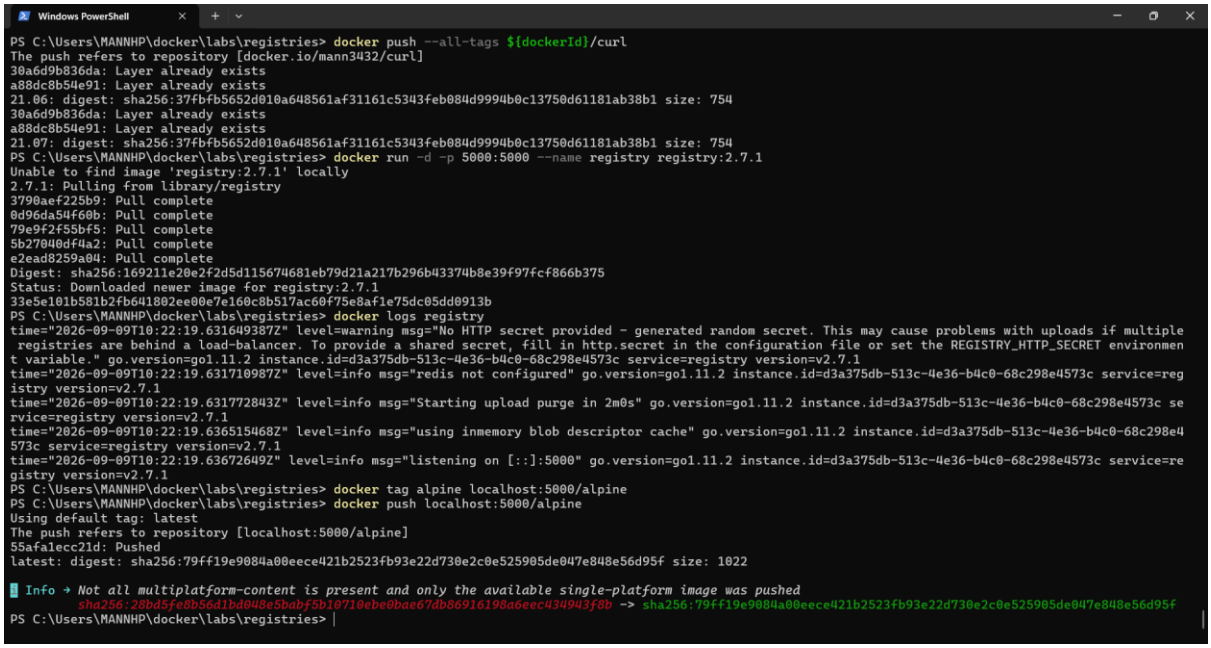

DevOps Experiments
Personal access tokens | mann3
Docker Course Labs
Accessing Images on Registries
Ask Gemini
app.docker.com/accounts/mann3432/settings/personal-access-tokens
School
New Insights on the state of AI agents from 800+ builders and leaders. Download your copy →
docker
M
mann3432
Your personal account
Home
Build Cloud
Account information
Email
Password
2FA
Personal access tokens
Connected accounts
Convert
Privacy
Deactivate
Billing
Personal access tokens
You can use a personal access token instead of a password for Docker CLI authentication. Create multiple tokens, control their scope, and delete tokens at any time. Learn more
Generate new token
Description Scope Status Source Created Last used Expiration date
Devops lab Read & Write Active Manual Sep 09, 2026 at 15:46:01 Sep 09, 2026 at 15:46:11 Never
Generated through d... Read, Write, Delete Active Auto-generated Mar 24, 2026 at 02:56:53 Mar 24, 2026 at 03:57:17 Never
Generated through d... Read, Write, Delete Active Auto-generated Sep 09, 2026 at 15:41:44 Never Never
Rows per page: 10
1–3 of 3
Successfully updated personal access token.

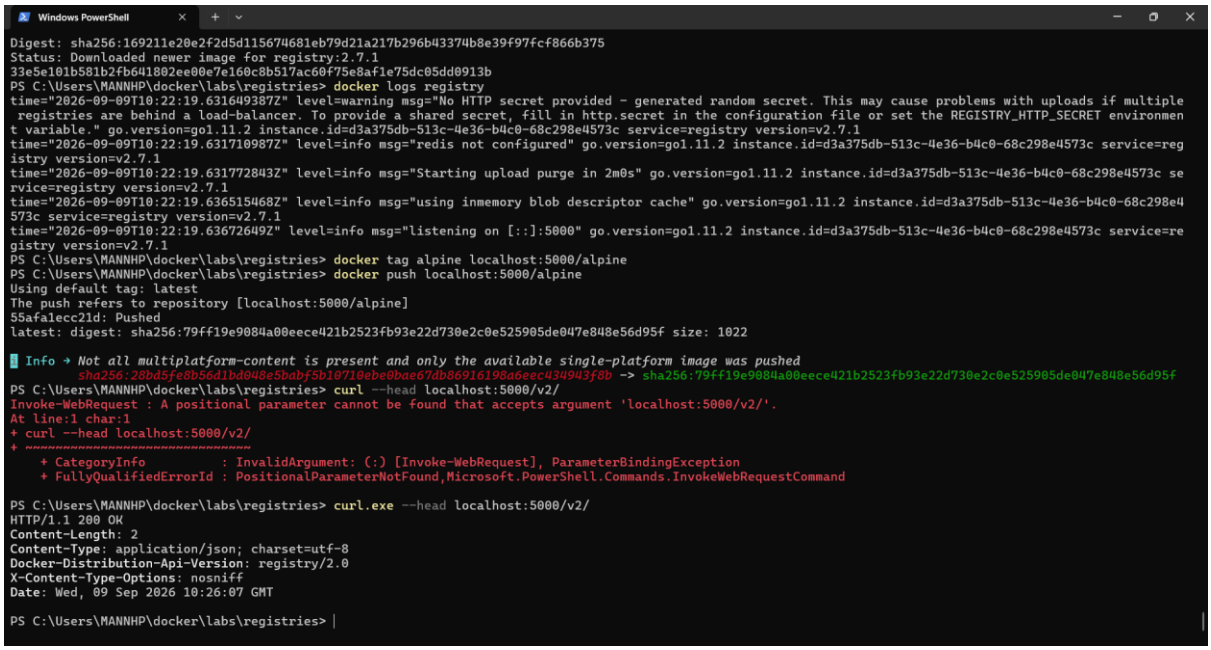

Windows PowerShell
PS C:\Users\MANNHP\docker\labs\registries> $dockerId='mann3432'
PS C:\Users\MANNHP\docker\labs\registries> docker tag courselabs/curl:v2 ${dockerId}/curl:21.06
PS C:\Users\MANNHP\docker\labs\registries> docker image ls '*/*'
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
bkimminich/juice-shop:latest bd1d402bf5cd 753MB 179MB
courselabs/base:latest 2260313b31c8 160MB 45.3MB
courselabs/curl:latest f4a815e51837 26.9MB 9.38MB
courselabs/curl:v2 37fbfb5652d0 13.4MB 3.95MB
courselabs/java:latest 5813e19d21cf 282MB 73MB
courselabs/web:latest 605815267a7b 37.8MB 10.2MB
kiamol/ch05-pi:latest 517837c0e013 159MB 47.9MB
kiamol/ch15-cert-generator:latest 21a8db88dfe8 73.8MB 17.8MB
mann3432/curl:21.06 37fbfb5652d0 13.4MB 3.95MB
PS C:\Users\MANNHP\docker\labs\registries> docker login -u ${dockerId}
Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings
Password:
Login Succeeded
PS C:\Users\MANNHP\docker\labs\registries> docker tag courselabs/curl:v2 ${dockerId}/curl:21.06
PS C:\Users\MANNHP\docker\labs\registries> docker image ls '*/curl'
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
courselabs/curl:latest f4a815e51837 26.9MB 9.38MB
courselabs/curl:v2 37fbfb5652d0 13.4MB 3.95MB
mann3432/curl:21.06 37fbfb5652d0 13.4MB 3.95MB
PS C:\Users\MANNHP\docker\labs\registries> docker push ${dockerId}/curl:21.06
The push refers to repository [docker.io/mann3432/curl]
30a6d9b836da: Waiting
a88dc8b54e91: Waiting
authentication required - access token has insufficient scopes
PS C:\Users\MANNHP\docker\labs\registries> docker push ${dockerId}/curl:21.06
The push refers to repository [docker.io/mann3432/curl]
30a6d9b836da: Pushed
a88dc8b54e91: Mounted from library/alpine
21.06: digest: sha256:37fbfb5652d010a648561af31161c5343feb084d9994b0c13750d61181ab38b1 size: 754
PS C:\Users\MANNHP\docker\labs\registries>

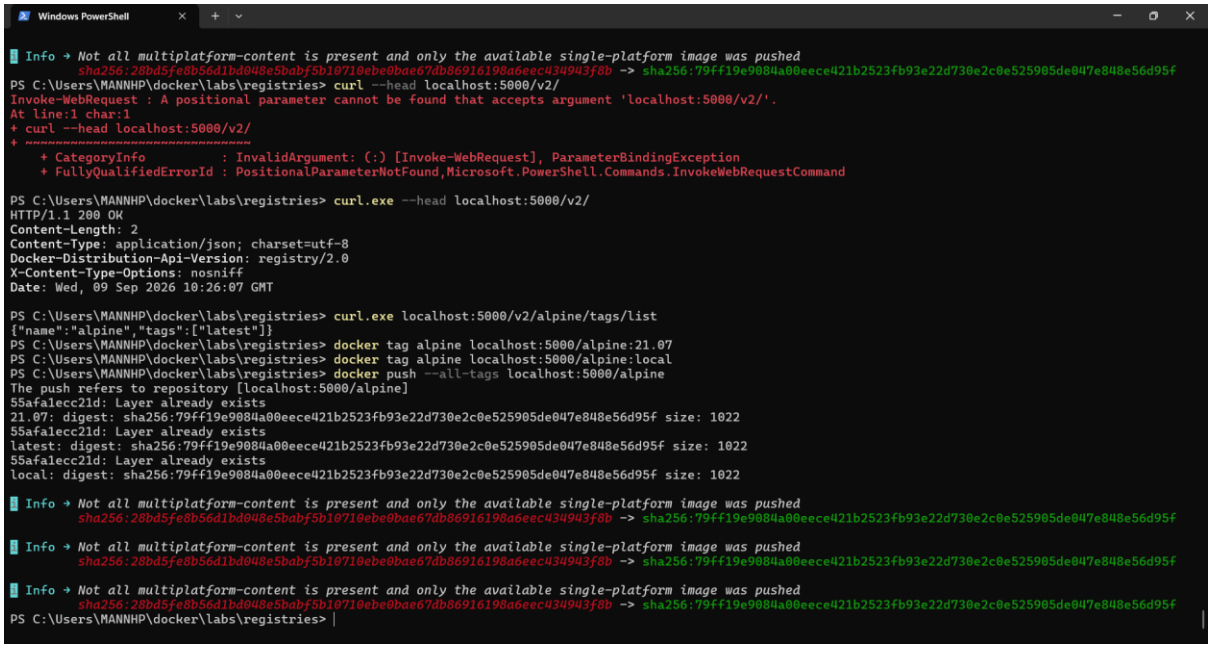

Windows PowerShell
courselabs/curl:v2 37fbfb5652d0 13.4MB 3.95MB
courselabs/java:latest 5813e19d21cf 282MB 73MB
courselabs/web:latest 605815267a7b 37.8MB 10.2MB
kiamol/ch05-pi:latest 517837c0e013 159MB 47.9MB
kiamol/ch15-cert-generator:latest 21a8db88dfe8 73.8MB 17.8MB
mann3432/curl:21.06 37fbfb5652d0 13.4MB 3.95MB
PS C:\Users\MANNHP\docker\labs\registries> docker login -u ${dockerId}
Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings
Password:
Login Succeeded
PS C:\Users\MANNHP\docker\labs\registries> docker tag courselabs/curl:v2 ${dockerId}/curl:21.06
PS C:\Users\MANNHP\docker\labs\registries> docker image ls '*/curl'
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
courselabs/curl:latest f4a815e51837 26.9MB 9.38MB
courselabs/curl:v2 37fbfb5652d0 13.4MB 3.95MB
mann3432/curl:21.06 37fbfb5652d0 13.4MB 3.95MB
PS C:\Users\MANNHP\docker\labs\registries> docker push ${dockerId}/curl:21.06
The push refers to repository [docker.io/mann3432/curl]
30a6d9b836da: Waiting
a88dc8b54e91: Waiting
authentication required - access token has insufficient scopes
PS C:\Users\MANNHP\docker\labs\registries> docker push ${dockerId}/curl:21.06
The push refers to repository [docker.io/mann3432/curl]
30a6d9b836da: Pushed
a88dc8b54e91: Mounted from library/alpine
21.06: digest: sha256:37fbfb5652d010a648561af31161c5343feb084d9994b0c13750d61181ab38b1 size: 754
PS C:\Users\MANNHP\docker\labs\registries> docker tag ${dockerId}/curl:21.06 ${dockerId}/curl:21.07
PS C:\Users\MANNHP\docker\labs\registries> docker image ls '*/curl'
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
courselabs/curl:latest f4a815e51837 26.9MB 9.38MB
courselabs/curl:v2 37fbfb5652d0 13.4MB 3.95MB
mann3432/curl:21.06 37fbfb5652d0 13.4MB 3.95MB
mann3432/curl:21.07 37fbfb5652d0 13.4MB 3.95MB
PS C:\Users\MANNHP\docker\labs\registries>

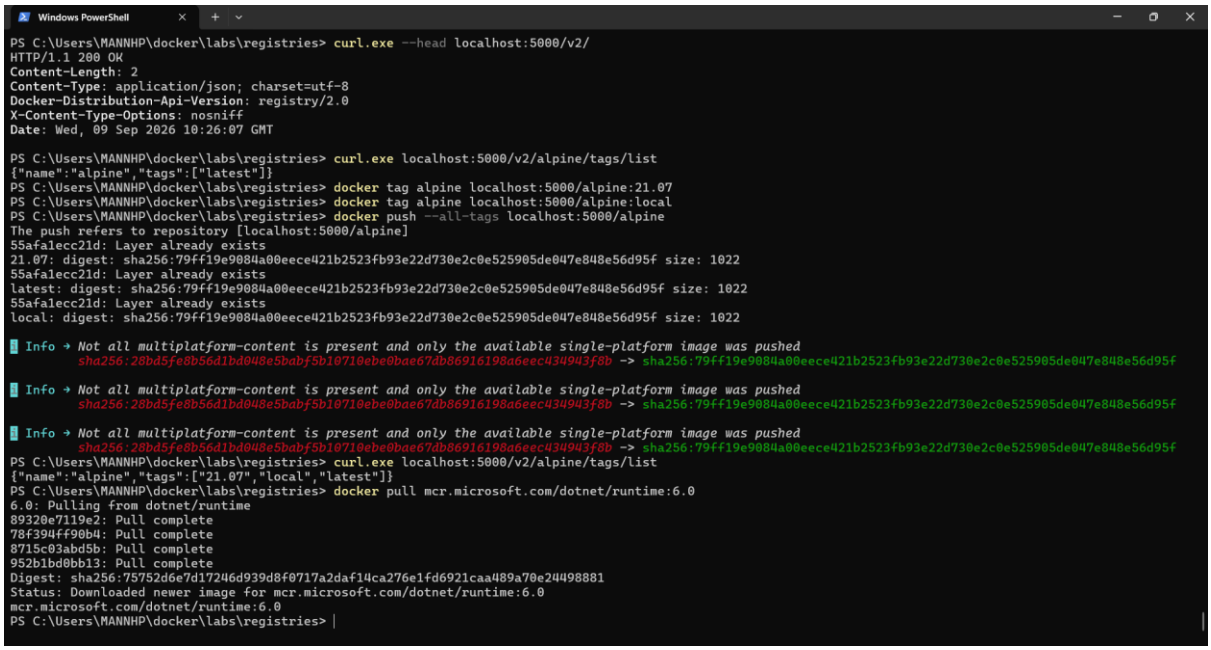

Windows PowerShell
30a6d9b836da: Waiting
a88dc8b54e91: Waiting
authentication required - access token has insufficient scopes
PS C:\Users\MANNHP\docker\labs\registries> docker push ${dockerId}/curl:21.06
The push refers to repository [docker.io/mann3432/curl]
30a6d9b836da: Pushed
a88dc8b54e91: Mounted from library/alpine
21.06: digest: sha256:37fbfb5652d010a648561af31161c5343feb084d9994b0c13750d61181ab38b1 size: 754
PS C:\Users\MANNHP\docker\labs\registries> docker tag ${dockerId}/curl:21.06 ${dockerId}/curl:21.07
PS C:\Users\MANNHP\docker\labs\registries> docker image ls '*/curl'
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
courselabs/curl:latest f4a815e51837 26.9MB 9.38MB
courselabs/curl:v2 37fbfb5652d0 13.4MB 3.95MB
mann3432/curl:21.06 37fbfb5652d0 13.4MB 3.95MB
mann3432/curl:21.07 37fbfb5652d0 13.4MB 3.95MB
PS C:\Users\MANNHP\docker\labs\registries> docker push ${dockerId}/curl:21.07
The push refers to repository [docker.io/mann3432/curl]
a88dc8b54e91: Layer already exists
30a6d9b836da: Layer already exists
21.07: digest: sha256:37fbfb5652d010a648561af31161c5343feb084d9994b0c13750d61181ab38b1 size: 754
PS C:\Users\MANNHP\docker\labs\registries> docker push --all-tags ${dockerId}/curl
The push refers to repository [docker.io/mann3432/curl]
30a6d9b836da: Layer already exists
a88dc8b54e91: Layer already exists
21.06: digest: sha256:37fbfb5652d010a648561af31161c5343feb084d9994b0c13750d61181ab38b1 size: 754
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21.07: digest: sha256:37fbfb5652d010a648561af31161c5343feb084d9994b0c13750d61181ab38b1 size: 754
PS C:\Users\MANNHP\docker\labs\registries> docker run -d -p 5000:5000 --name registry registry:2.7.1
Unable to find image 'registry:2.7.1' locally
2.7.1: Pulling from library/registry
3790aef225b9: Pull complete
0d96da54f60b: Pull complete
79e9f2f55bf5: Pull complete
5b27040df4a2: Pull complete
e2ead8259a04: Pull complete
Digest: sha256:169211e20e2f2d5d115674681eb79d21a217b296b43374b8e39f97fcf866b375
Status: Downloaded newer image for registry:2.7.1
33e5e101b581b2fb641802ee00e7e160c8b517ac60f75e8af1e75dc05dd0913b
PS C:\Users\MANNHP\docker\labs\registries>

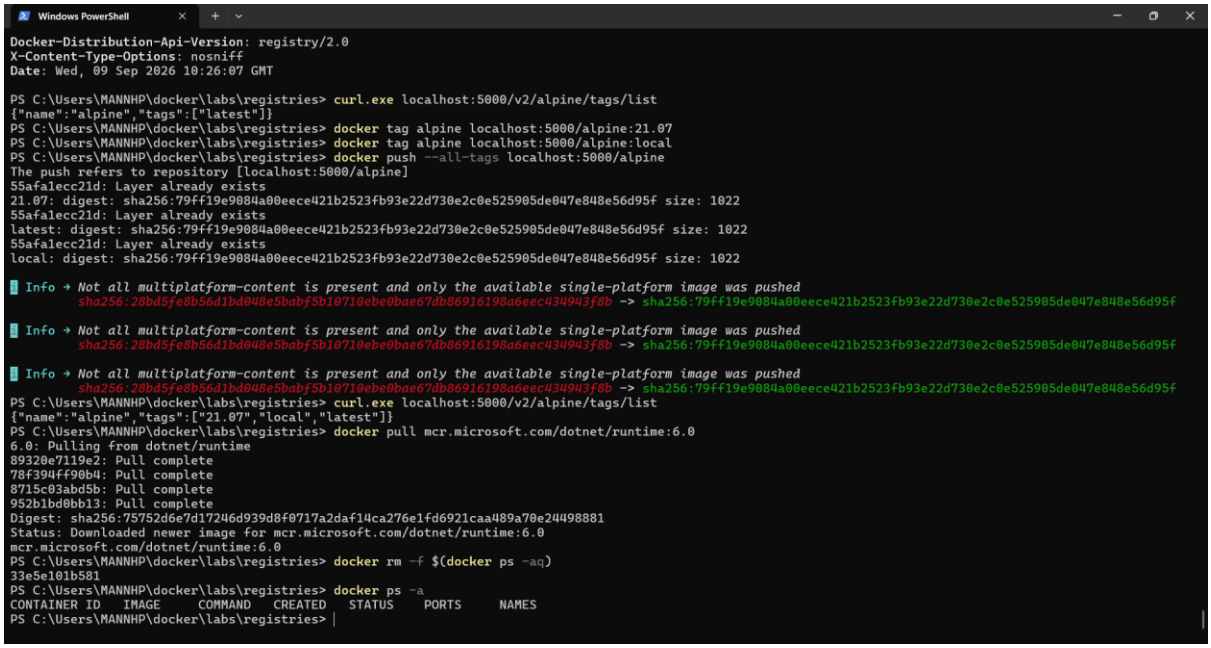

docker desktop PERSONAL
Search Ctrl+K
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Extensions
Upgrade plan
Containers
Give feedback
Container CPU usage
0.00% / 1600% (16 CPUs available)
Container memory usage
5.51MB / 11.23GB
Show charts
Search
Only show running containers
Name Container ID Image Port(s) CPU (%) Memory usag... Memory (%) Disk rea Actions
registry 33e5e101b581 registry:2.7 5000:5000 0% 0B / 0B 0% 0B / 0B
Engine running
RAM 1.59 GB CPU 0.00% Disk: 4.16 GB used (limit 1006.85 GB)
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Markup and share


Windows PowerShell
PS C:\Users\MANNHP\docker\labs\registries> docker push --all-tags ${dockerId}/curl
The push refers to repository [docker.io/mann3432/curl]
30a6d9b836da: Layer already exists
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e2ead8259a04: Pull complete
Digest: sha256:169211e20e2f2d5d115674681eb79d21a217b296b43374b8e39f97fcf866b375
Status: Downloaded newer image for registry:2.7.1
33e5e101b581b2fb641802ee00e7e160c8b517ac60f75e8af1e75dc05dd0913b
PS C:\Users\MANNHP\docker\labs\registries> docker logs registry
time="2026-09-09T10:22:19.631649387Z" level=warning msg="No HTTP secret provided - generated random secret. This may cause problems with uploads if multiple registries are behind a load-balancer. To provide a shared secret, fill in http.secret in the configuration file or set the REGISTRY_HTTP_SECRET environment variable." go.version=go1.11.2 instance.id=d3a375db-513c-4e36-b4c0-68c298e4573c service=registry version=v2.7.1
time="2026-09-09T10:22:19.631710987Z" level=info msg="redis not configured" go.version=go1.11.2 instance.id=d3a375db-513c-4e36-b4c0-68c298e4573c service=registry version=v2.7.1
time="2026-09-09T10:22:19.631772843Z" level=info msg="Starting upload purge in 2m0s" go.version=go1.11.2 instance.id=d3a375db-513c-4e36-b4c0-68c298e4573c service=registry version=v2.7.1
time="2026-09-09T10:22:19.636515468Z" level=info msg="using inmemory blob descriptor cache" go.version=go1.11.2 instance.id=d3a375db-513c-4e36-b4c0-68c298e4573c service=registry version=v2.7.1
time="2026-09-09T10:22:19.636726449Z" level=info msg="listening on [::]:5000" go.version=go1.11.2 instance.id=d3a375db-513c-4e36-b4c0-68c298e4573c service=registry version=v2.7.1
PS C:\Users\MANNHP\docker\labs\registries> docker tag alpine localhost:5000/alpine
PS C:\Users\MANNHP\docker\labs\registries> docker push localhost:5000/alpine
Using default tag: latest
The push refers to repository [localhost:5000/alpine]
55afa1ecc21d: Pushed
latest: digest: sha256:79ff19e9084a00eece421b2523fb93e22d730e2c0e525905de047e848e56d95f size: 1022
Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b -> sha256:79ff19e9084a00eece421b2523fb93e22d730e2c0e525905de047e848e56d95f
PS C:\Users\MANNHP\docker\labs\registries>


Windows PowerShell
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time="2026-09-09T10:22:19.636726449Z" level=info msg="listening on [::]:5000" go.version=go1.11.2 instance.id=d3a375db-513c-4e36-b4c0-68c298e4573c service=registry version=v2.7.1
PS C:\Users\MANNHP\docker\labs\registries> docker tag alpine localhost:5000/alpine
PS C:\Users\MANNHP\docker\labs\registries> docker push localhost:5000/alpine
Using default tag: latest
The push refers to repository [localhost:5000/alpine]
55afa1ecc21d: Pushed
latest: digest: sha256:79ff19e9084a00eece421b2523fb93e22d730e2c0e525905de047e848e56d95f size: 1022
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sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b -> sha256:79ff19e9084a00eece421b2523fb93e22d730e2c0e525905de047e848e56d95f
PS C:\Users\MANNHP\docker\labs\registries> curl --head localhost:5000/v2/
Invoke-WebRequest : A positional parameter cannot be found that accepts argument 'localhost:5000/v2/'.
At line:1 char:1
+ curl --head localhost:5000/v2/
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
+ CategoryInfo : InvalidArgument: (:) [Invoke-WebRequest], ParameterBindingException
+ FullyQualifiedErrorId : PositionalParameterNotFound,Microsoft.PowerShell.Commands.InvokeWebRequestCommand
PS C:\Users\MANNHP\docker\labs\registries> curl.exe --head localhost:5000/v2/
HTTP/1.1 200 OK
Content-Length: 2
Content-Type: application/json; charset=utf-8
Docker-Distribution-Api-Version: registry/2.0
X-Content-Type-Options: nosniff
Date: Wed, 09 Sep 2026 10:26:07 GMT
PS C:\Users\MANNHP\docker\labs\registries>


Windows PowerShell
[Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b -> sha256:79ff19e9084a00eece421b2523fb93e22d730e2c0e525905de047e848e56d95f
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Docker-Distribution-Api-Version: registry/2.0
X-Content-Type-Options: nosniff
Date: Wed, 09 Sep 2026 10:26:07 GMT
PS C:\Users\MANNHP\docker\labs\registries> curl.exe localhost:5000/v2/alpine/tags/list
{"name":"alpine","tags":["latest"]}
PS C:\Users\MANNHP\docker\labs\registries> docker tag alpine localhost:5000/alpine:21.07
PS C:\Users\MANNHP\docker\labs\registries> docker tag alpine localhost:5000/alpine:local
PS C:\Users\MANNHP\docker\labs\registries> docker push --all-tags localhost:5000/alpine
The push refers to repository [localhost:5000/alpine]
55afa1ecc21d: Layer already exists
21.07: digest: sha256:79ff19e9084a00eece421b2523fb93e22d730e2c0e525905de047e848e56d95f size: 1022
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local: digest: sha256:79ff19e9084a00eece421b2523fb93e22d730e2c0e525905de047e848e56d95f size: 1022
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PS C:\Users\MANNHP\docker\labs\registries>


Windows PowerShell
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PS C:\Users\MANNHP\docker\labs\registries> curl.exe localhost:5000/v2/alpine/tags/list
{"name":"alpine","tags":["21.07","local","latest"]}
PS C:\Users\MANNHP\docker\labs\registries> docker pull mcr.microsoft.com/dotnet/runtime:6.0
6.0: Pulling from dotnet/runtime
89320e7119e2: Pull complete
78f394ff90b4: Pull complete
8715c03abd5b: Pull complete
952b1bd0bb13: Pull complete
Digest: sha256:75752d6e7d17246d939d8f0717a2daf14ca276e1fd6921caa489a70e24498881
Status: Downloaded newer image for mcr.microsoft.com/dotnet/runtime:6.0
mcr.microsoft.com/dotnet/runtime:6.0
PS C:\Users\MANNHP\docker\labs\registries>


Windows PowerShell
Docker-Distribution-Api-Version: registry/2.0
X-Content-Type-Options: nosniff
Date: Wed, 09 Sep 2026 10:26:07 GMT
PS C:\Users\MANNHP\docker\labs\registries> curl.exe localhost:5000/v2/alpine/tags/list
{"name":"alpine","tags":["latest"]}
PS C:\Users\MANNHP\docker\labs\registries> docker tag alpine localhost:5000/alpine:21.07
PS C:\Users\MANNHP\docker\labs\registries> docker tag alpine localhost:5000/alpine:local
PS C:\Users\MANNHP\docker\labs\registries> docker push --all-tags localhost:5000/alpine
The push refers to repository [localhost:5000/alpine]
55afa1ecc21d: Layer already exists
21.07: digest: sha256:79ff19e9084a00eece421b2523fb93e22d730e2c0e525905de047e848e56d95f size: 1022
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Digest: sha256:75752d6e7d17246d939d8f0717a2daf14ca276e1fd6921caa489a70e24498881
Status: Downloaded newer image for mcr.microsoft.com/dotnet/runtime:6.0
mcr.microsoft.com/dotnet/runtime:6.0
PS C:\Users\MANNHP\docker\labs\registries> docker rm -f $(docker ps -aq)
33e5e101b581
PS C:\Users\MANNHP\docker\labs\registries> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
PS C:\Users\MANNHP\docker\labs\registries>